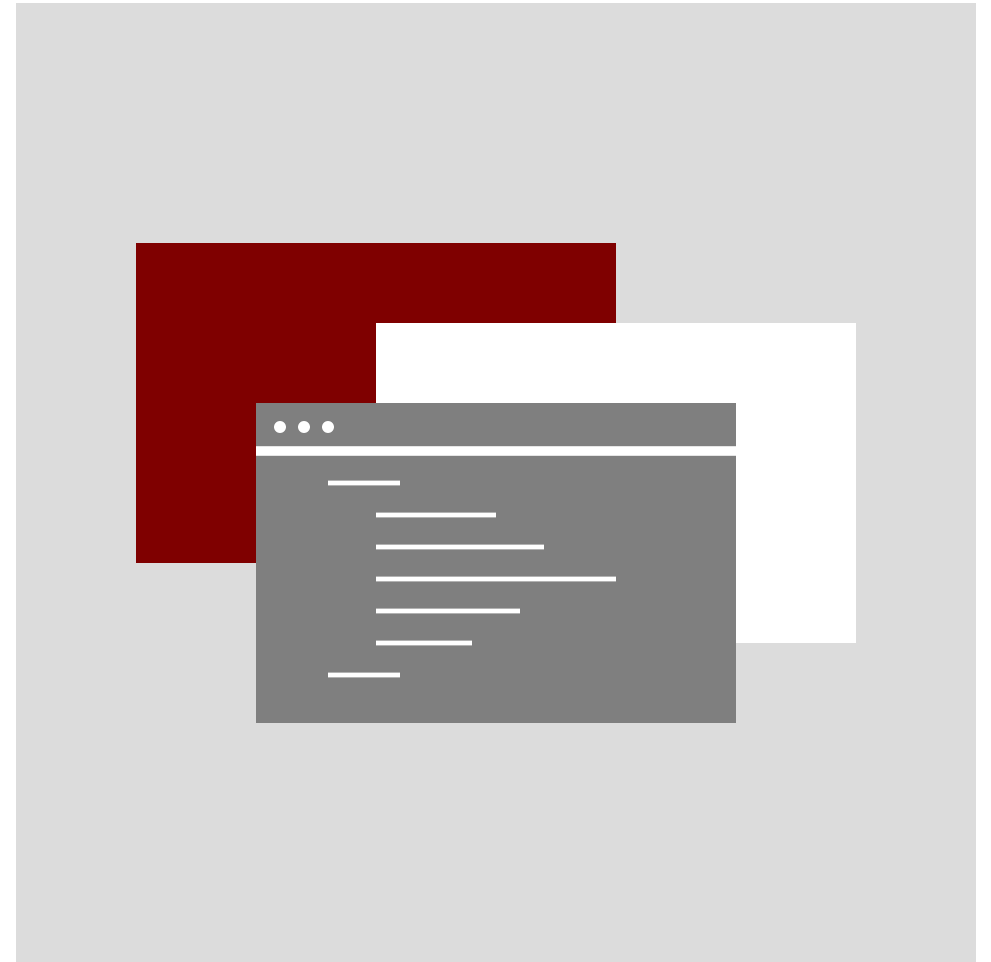


decksh reference



Version
2026-06-22-1.1.0

Introduction

decksh is a Domain-Specific Language (DSL) for making presentations, visualizations, and information displays.

This reference describes the keywords and elements of the language, how to structure decksh code, along with how to use variables, assignments, and binary operations.

Also included is a color reference and a detailed description, with examples, for all decksh elements.

Keywords

Structure Text

deck/edeck
slide/eslide
canvas
def/edef
for/efor
func
grid
import
include
if/else/eif
text
btext
ctext
etext
para/epara
rtext
arctext
textblock
textblockfile
textfile
textcode

Graphics

acircle
arc
circle
curve
ellipse
dline
hline
line
pill
polygon
polyline
rect
rrect
ruler
square
star
vline

Lists

list
blist
nlist
clist
li
elist

Braces

lbrace
rbrace
ubrace
dbrace
lbracket
rbracket
dbracket
ubracket

Arrows

arrow
rcarrow
lcarrow
ucarrow
dcarrow

Maps

geoarc
geoborder
geobound
geocanvas
geoimage
geolabel
geoloc
geomark
geopath
geopathfile
georegion
georuler

Images

image
cimage

Charts

chartbbox
dchart
legend
area
barchart
hbar
wbar
linechart
scatter
slopechart
donut
pie
waffle
pgrid
pmap
bowtie
fanchart

Assign

polar
polarx
polary
random
area
format
substr
vmap
dump

Math

geocoord
cosine
sine
sqrt
tangent

Data

data
edata
content

Keywords and arguments

keyword

arguments

mandatory

optional

text `"..string...." x y n`

`"font" "color" op`

text "hello, world" 80 50 2

hello, world

text "hello, world" 80 40 2

"serif"

hello, world

text "hello, world" 80 30 2

"serif" "red"

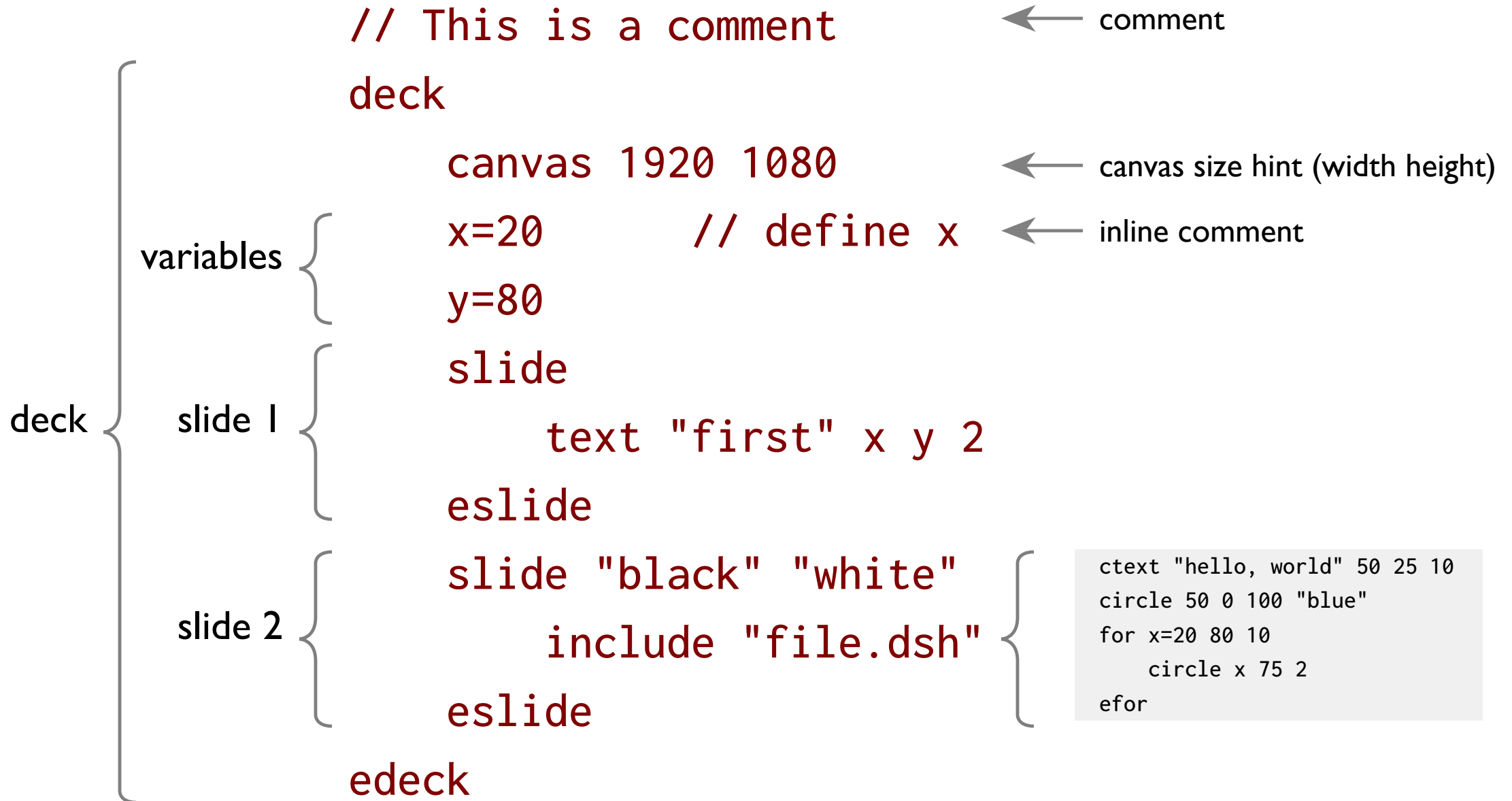
hello, world

text "hello, world" 80 20 2

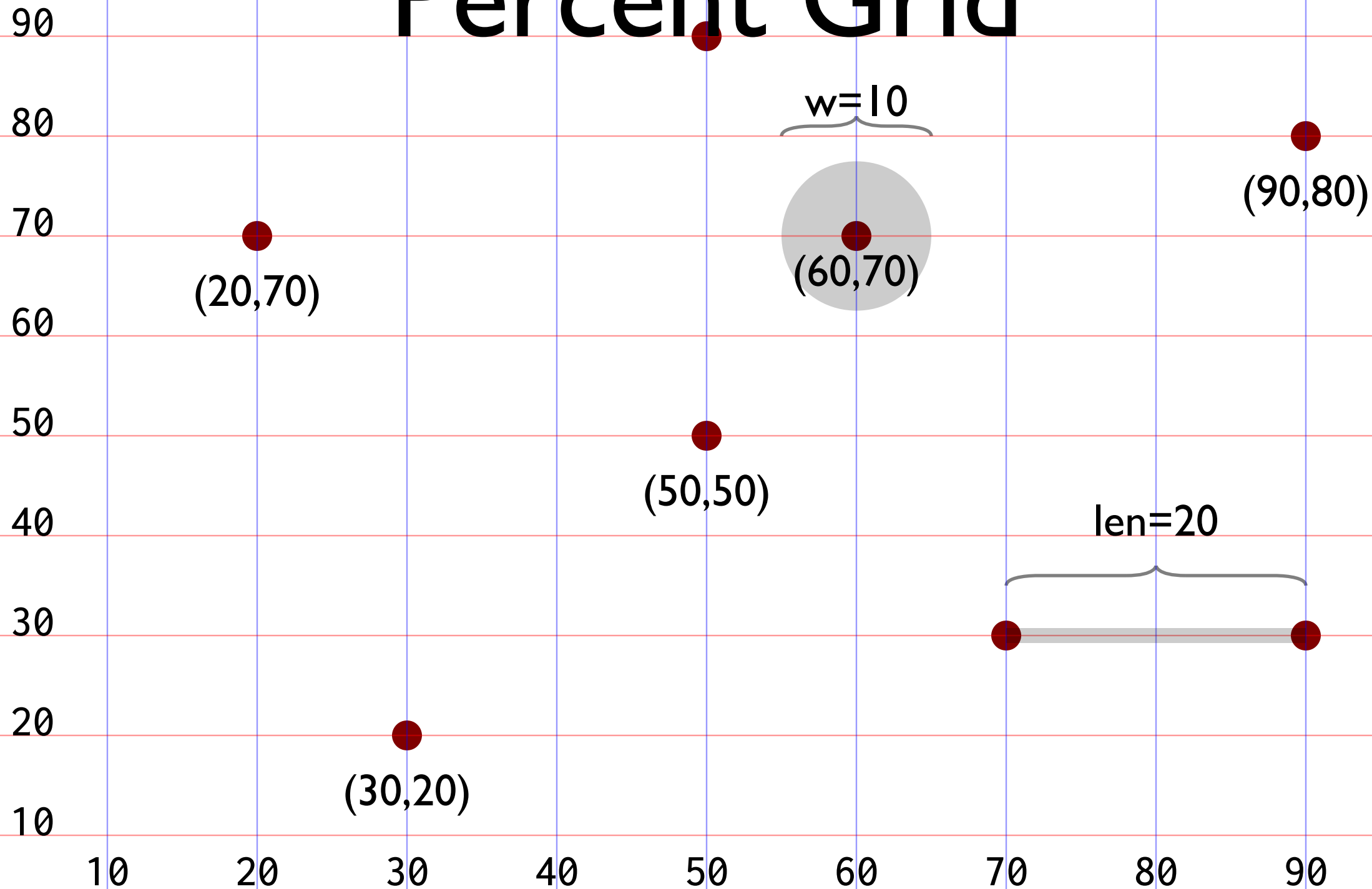
"serif" "red" 50

hello, world

Structure



Percent Grid



Defining and using variables

name x	is =	thing 3.14159265	number
s	=	"hello"	string
y	=	x	another variable
x	=	a + b	binary operation
x	*=	10	assignment operator
text	s	x y 2	variable use
dump	[var list]		dump the value of variables

Binary operators

$x = a + b$

addition

$x = a - b$

subtraction

$x = a * b$

multiplication

$x = a / b$

division

$x = a \% b$

modulo

Assignment operators

`x += 10`

increase x by 10

`x -= 10`

decrease x by 10

`x *= 10`

multiply x by 10

`x /= 10`

divide x by 10

Special Assignments

`p=(expr, expr)`

`p=geocoord "loc" or geocoord lat long`

`p=polar cx cy r theta`

`x=polarx cx cy r theta`

`y=polary cx cy r theta`

`v=format string expr`

`v=substr string begin end`

`v=random v1 v2`

`v=vmap data v1 v2 v3 v4`

`v=area expr`

`v=cosine expr`

`v=sine expr`

`v=sqrt expr`

`v=tangent expr`

coordinates (p_x, p_y)

geo mapped coordinates (p_x, p_y)

polar coordinate (p_x, p_y)

polar coordinate (x)

polar coordinate (y)

number formatting

substring

random number

range map

area

cosine

sine

square root

tangent

Colors, fonts, opacity, gradient

Colors	Fonts		Opacity (0-100)	
"steelblue"	"sans"	Sans Serif	100	
"#4682b4"	"serif"	Serif	50	
"rgb(70,130,180)"	"mono"	Monospace	25	
"hsv(207,61,71)"	"symbol"	❀❇❁❄❅	10	

"red/blue/90" 

(applies to rect and square only)

Color Index

name	hex	RGB
 aliceblue	#f0f8ff	rgb(240,248,255)
 antiquewhite	#faebd7	rgb(250,235,215)
 aqua	#00ffff	rgb(0,255,255)
 aquamarine	#7fffd4	rgb(127,255,212)
 azure	#f0ffff	rgb(240,255,255)
 beige	#f5f5dc	rgb(245,245,220)
 bisque	#ffe4c4	rgb(255,228,196)
 black	#000000	rgb(0,0,0)
 blanchedalmond	#ffebcd	rgb(255,235,205)
 blue	#0000ff	rgb(0,0,255)
 blueviolet	#8a2be2	rgb(138,43,226)
 brown	#a52a2a	rgb(165,42,42)
 burlywood	#deb887	rgb(222,184,135)
 cadetblue	#5f9ea0	rgb(95,158,160)
 chartreuse	#7fff00	rgb(127,255,0)
 chocolate	#d2691e	rgb(210,105,30)

name	hex	RGB
 coral	#ff7f50	rgb(255,127,80)
 cornflowerblue	#6495ed	rgb(100,149,237)
 cornsilk	#fff8dc	rgb(255,248,220)
 crimson	#dc143c	rgb(220,20,60)
 cyan	#00ffff	rgb(0,255,255)
 darkblue	#00008b	rgb(0,0,139)
 darkcyan	#008b8b	rgb(0,139,139)
 darkgoldenrod	#b8860b	rgb(184,134,11)
 darkgray	#a9a9a9	rgb(169,169,169)
 darkgreen	#006400	rgb(0,100,0)
 darkgrey	#a9a9a9	rgb(169,169,169)
 darkkhaki	#bdb76b	rgb(189,183,107)
 darkmagenta	#8b008b	rgb(139,0,139)
 darkolivegreen	#556b2f	rgb(85,107,47)
 darkorange	#ff8c00	rgb(255,140,0)
 darkorchid	#9932cc	rgb(153,50,204)

Color Index (2)

name	hex	RGB	name	hex	RGB
 darkred	#8b0000	rgb(139,0,0)	 fuchsia	#ff00ff	rgb(255,0,255)
 darksalmon	#e9967a	rgb(233,150,122)	 gainsboro	#dcdcdc	rgb(220,220,220)
 darkseagreen	#8fbc8f	rgb(143,188,143)	 ghostwhite	#f8f8ff	rgb(248,248,255)
 darkslateblue	#483d8b	rgb(72,61,139)	 gold	#ffd700	rgb(255,215,0)
 darkslategray	#2f4f4f	rgb(47,79,79)	 goldenrod	#daa520	rgb(218,165,32)
 darkslategrey	#2f4f4f	rgb(47,79,79)	 gray	#808080	rgb(128,128,128)
 darkturquoise	#00ced1	rgb(0,206,209)	 green	#008000	rgb(0,128,0)
 darkviolet	#9400d3	rgb(148,0,211)	 greenyellow	#adff2f	rgb(173,255,47)
 deeppink	#ff1493	rgb(255,20,147)	 grey	#808080	rgb(128,128,128)
 deepskyblue	#00bfff	rgb(0,191,255)	 honeydew	#f0fff0	rgb(240,255,240)
 dimgray	#696969	rgb(105,105,105)	 hotpink	#ff69b4	rgb(255,105,180)
 dimgrey	#696969	rgb(105,105,105)	 indianred	#cd5c5c	rgb(205,92,92)
 dodgerblue	#1e90ff	rgb(30,144,255)	 indigo	#4b0082	rgb(75,0,130)
 firebrick	#b22222	rgb(178,34,34)	 ivory	#fffff0	rgb(255,255,240)
 floralwhite	#fffaf0	rgb(255,250,240)	 khaki	#f0e68c	rgb(240,230,140)
 forestgreen	#228b22	rgb(34,139,34)	 lavender	#e6e6fa	rgb(230,230,250)

Color Index (3)

name	hex	RGB
 lavenderblush	#fff0f5	rgb(255,240,245)
 lawngreen	#7cfc00	rgb(124,252,0)
 lemonchiffon	#ffffac	rgb(255,250,205)
 lightblue	#add8e6	rgb(173,216,230)
 lightcoral	#f08080	rgb(240,128,128)
 lightcyan	#e0ffff	rgb(224,255,255)
 lightgoldenrodyellow	#fafad2	rgb(250,250,210)
 lightgray	#d3d3d3	rgb(211,211,211)
 lightgreen	#90ee90	rgb(144,238,144)
 lightgrey	#d3d3d3	rgb(211,211,211)
 lightpink	#ffb6c1	rgb(255,182,193)
 lightsalmon	#ffa07a	rgb(255,160,122)
 lightseagreen	#20b2aa	rgb(32,178,170)
 lightskyblue	#87cefa	rgb(135,206,250)
 lightslategray	#778899	rgb(119,136,153)
 lightslategrey	#778899	rgb(119,136,153)

name	hex	RGB
 lightsteelblue	#b0c4de	rgb(176,196,222)
 lightyellow	#ffffe0	rgb(255,255,224)
 lime	#00ff00	rgb(0,255,0)
 limegreen	#32cd32	rgb(50,205,50)
 linen	#faf0e6	rgb(250,240,230)
 magenta	#ff00ff	rgb(255,0,255)
 maroon	#800000	rgb(128,0,0)
 mediumaquamarine	#66cdaa	rgb(102,205,170)
 mediumblue	#0000cd	rgb(0,0,205)
 mediumorchid	#ba55d3	rgb(186,85,211)
 mediumpurple	#9370db	rgb(147,112,219)
 mediumseagreen	#3cb371	rgb(60,179,113)
 mediumslateblue	#7b68ee	rgb(123,104,238)
 mediumspringgreen	#00fa9a	rgb(0,250,154)
 mediumturquoise	#48d1cc	rgb(72,209,204)
 mediumvioletred	#c71585	rgb(199,21,133)

Color Index (4)

name	hex	RGB	name	hex	RGB
 midnightblue	#191970	rgb(25,25,112)	 papayawhip	#ffefd5	rgb(255,239,213)
 mintcream	#f5fffa	rgb(245,255,250)	 peachpuff	#ffdab9	rgb(255,218,185)
 mistyrose	#ffe4e1	rgb(255,228,225)	 peru	#cd853f	rgb(205,133,63)
 moccasin	#ffe4b5	rgb(255,228,181)	 pink	#ffc0cb	rgb(255,192,203)
 navajowhite	#ffdead	rgb(255,222,173)	 plum	#dda0dd	rgb(221,160,221)
 navy	#000080	rgb(0,0,128)	 powderblue	#b0e0e6	rgb(176,224,230)
 oldlace	#fdf5e6	rgb(253,245,230)	 purple	#800080	rgb(128,0,128)
 olive	#808000	rgb(128,128,0)	 red	#ff0000	rgb(255,0,0)
 olivedrab	#6b8e23	rgb(107,142,35)	 rosybrown	#bc8f8f	rgb(188,143,143)
 orange	#ffa500	rgb(255,165,0)	 royalblue	#4169e1	rgb(65,105,225)
 orangered	#ff4500	rgb(255,69,0)	 saddlebrown	#8b4513	rgb(139,69,19)
 orchid	#da70d6	rgb(218,112,214)	 salmon	#fa8072	rgb(250,128,114)
 palegoldenrod	#eee8aa	rgb(238,232,170)	 sandybrown	#f4a460	rgb(244,164,96)
 palegreen	#98fb98	rgb(152,251,152)	 seagreen	#2e8b57	rgb(46,139,87)
 paleturquoise	#afeeee	rgb(175,238,238)	 seashell	#fff5ee	rgb(255,245,238)
 palevioletred	#db7093	rgb(219,112,147)	 sienna	#a0522d	rgb(160,82,45)

Color Index (5)

name	hex	RGB
 silver	#c0c0c0	rgb(192,192,192)
 skyblue	#87ceeb	rgb(135,206,235)
 slateblue	#6a5acd	rgb(106,90,205)
 slategray	#708090	rgb(112,128,144)
 slategrey	#708090	rgb(112,128,144)
 snow	#ffaafa	rgb(255,250,250)
 springgreen	#00ff7f	rgb(0,255,127)
 steelblue	#4682b4	rgb(70,130,180)
 tan	#d2b48c	rgb(210,180,140)
 teal	#008080	rgb(0,128,128)
 thistle	#d8bfd8	rgb(216,191,216)
 tomato	#ff6347	rgb(255,99,71)
 turquoise	#40e0d0	rgb(64,224,208)
 violet	#ee82ee	rgb(238,130,238)
 wheat	#f5deb3	rgb(245,222,179)
 white	#ffffff	rgb(255,255,255)

name	hex	RGB
 whitesmoke	#f5f5f5	rgb(245,245,245)
 yellow	#ffff00	rgb(255,255,0)
 yellowgreen	#9acd32	rgb(154,205,50)

Neutrals

name	hex	RGB
 aliceblue	#f0f8ff	rgb(240,248,255)
 antiquewhite	#faebd7	rgb(250,235,215)
 azure	#f0ffff	rgb(240,255,255)
 beige	#f5f5dc	rgb(245,245,220)
 bisque	#ffe4c4	rgb(255,228,196)
 black	#000000	rgb(0,0,0)
 blanchdalmond	#ffeabcd	rgb(255,235,205)
 brown	#a52a2a	rgb(165,42,42)
 burlywood	#deb887	rgb(222,184,135)
 chocolate	#d2691e	rgb(210,105,30)
 cornsilk	#fff8dc	rgb(255,248,220)
 darkgray	#a9a9a9	rgb(169,169,169)
 darkgrey	#a9a9a9	rgb(169,169,169)
 darksalmon	#e9967a	rgb(233,150,122)
 darkslategray	#2f4f4f	rgb(47,79,79)
 darkslategrey	#2f4f4f	rgb(47,79,79)

name	hex	RGB
 dimgray	#696969	rgb(105,105,105)
 dimgrey	#696969	rgb(105,105,105)
 floralwhite	#fffaf0	rgb(255,250,240)
 gainsboro	#dcdcdc	rgb(220,220,220)
 ghostwhite	#f8f8ff	rgb(248,248,255)
 gray	#808080	rgb(128,128,128)
 grey	#808080	rgb(128,128,128)
 honeydew	#f0fff0	rgb(240,255,240)
 ivory	#fffff0	rgb(255,255,240)
 lavender	#e6e6fa	rgb(230,230,250)
 lavenderblush	#fff0f5	rgb(255,240,245)
 lightgray	#d3d3d3	rgb(211,211,211)
 lightgrey	#d3d3d3	rgb(211,211,211)
 lightslategray	#778899	rgb(119,136,153)
 lightslategrey	#778899	rgb(119,136,153)
 linen	#faf0e6	rgb(250,240,230)

Neutrals (2)

name	hex	RGB
 mintcream	#f5fffa	rgb(245,255,250)
 mistyrose	#ffe4e1	rgb(255,228,225)
 moccasin	#ffe4b5	rgb(255,228,181)
 navajowhite	#ffdead	rgb(255,222,173)
 oldlace	#fdf5e6	rgb(253,245,230)
 papayawhip	#ffefd5	rgb(255,239,213)
 peachpuff	#ffdab9	rgb(255,218,185)
 peru	#cd853f	rgb(205,133,63)
 rosybrown	#bc8f8f	rgb(188,143,143)
 saddlebrown	#8b4513	rgb(139,69,19)
 salmon	#fa8072	rgb(250,128,114)
 sandybrown	#f4a460	rgb(244,164,96)
 seashell	#fff5ee	rgb(255,245,238)
 sienna	#a0522d	rgb(160,82,45)
 silver	#c0c0c0	rgb(192,192,192)
 slategray	#708090	rgb(112,128,144)

name	hex	RGB
 slategrey	#708090	rgb(112,128,144)
 snow	#fffafa	rgb(255,250,250)
 tan	#d2b48c	rgb(210,180,140)
 wheat	#f5deb3	rgb(245,222,179)
 white	#ffffff	rgb(255,255,255)
 whitesmoke	#f5f5f5	rgb(245,245,245)

Reds

name	hex	RGB
 coral	#ff7f50	rgb(255,127,80)
 crimson	#dc143c	rgb(220,20,60)
 darkmagenta	#8b008b	rgb(139,0,139)
 darkred	#8b0000	rgb(139,0,0)
 deeppink	#ff1493	rgb(255,20,147)
 firebrick	#b22222	rgb(178,34,34)
 fuchsia	#ff00ff	rgb(255,0,255)
 hotpink	#ff69b4	rgb(255,105,180)
 indianred	#cd5c5c	rgb(205,92,92)
 lightcoral	#f08080	rgb(240,128,128)
 lightpink	#ffb6c1	rgb(255,182,193)
 lightsalmon	#ffa07a	rgb(255,160,122)
 magenta	#ff00ff	rgb(255,0,255)
 maroon	#800000	rgb(128,0,0)
 orangered	#ff4500	rgb(255,69,0)
 orchid	#da70d6	rgb(218,112,214)

name	hex	RGB
 palevioletred	#db7093	rgb(219,112,147)
 pink	#ffc0cb	rgb(255,192,203)
 plum	#dda0dd	rgb(221,160,221)
 red	#ff0000	rgb(255,0,0)
 thistle	#d8bfd8	rgb(216,191,216)
 tomato	#ff6347	rgb(255,99,71)

Greens

name	hex	RGB
 aquamarine	#7fffd4	rgb(127,255,212)
 chartreuse	#7fff00	rgb(127,255,0)
 darkgreen	#006400	rgb(0,100,0)
 darkkhaki	#bdb76b	rgb(189,183,107)
 darkolivegreen	#556b2f	rgb(85,107,47)
 darkseagreen	#8fbc8f	rgb(143,188,143)
 forestgreen	#228b22	rgb(34,139,34)
 green	#008000	rgb(0,128,0)
 greenyellow	#adff2f	rgb(173,255,47)
 lawngreen	#7cfc00	rgb(124,252,0)
 lightgreen	#90ee90	rgb(144,238,144)
 lightseagreen	#20b2aa	rgb(32,178,170)
 lime	#00ff00	rgb(0,255,0)
 limegreen	#32cd32	rgb(50,205,50)
 mediumseagreen	#3cb371	rgb(60,179,113)
 mediumspringgreen	#00fa9a	rgb(0,250,154)

name	hex	RGB
 olive	#808000	rgb(128,128,0)
 olivedrab	#6b8e23	rgb(107,142,35)
 palegreen	#98fb98	rgb(152,251,152)
 seagreen	#2e8b57	rgb(46,139,87)
 springgreen	#00ff7f	rgb(0,255,127)
 teal	#008080	rgb(0,128,128)
 yellowgreen	#9acd32	rgb(154,205,50)

Blues

name	hex	RGB
 aqua	#00ffff	rgb(0,255,255)
 blue	#0000ff	rgb(0,0,255)
 cadetblue	#5f9ea0	rgb(95,158,160)
 cornflowerblue	#6495ed	rgb(100,149,237)
 cyan	#00ffff	rgb(0,255,255)
 darkblue	#00008b	rgb(0,0,139)
 darkcyan	#008b8b	rgb(0,139,139)
 darkslateblue	#483d8b	rgb(72,61,139)
 darkturquoise	#00ced1	rgb(0,206,209)
 deepskyblue	#00bfff	rgb(0,191,255)
 dodgerblue	#1e90ff	rgb(30,144,255)
 lightblue	#add8e6	rgb(173,216,230)
 lightcyan	#e0ffff	rgb(224,255,255)
 lightskyblue	#87cefa	rgb(135,206,250)
 lightsteelblue	#b0c4de	rgb(176,196,222)
 mediumaquamarine	#66cdaa	rgb(102,205,170)

name	hex	RGB
 mediumblue	#0000cd	rgb(0,0,205)
 mediumslateblue	#7b68ee	rgb(123,104,238)
 mediumturquoise	#48d1cc	rgb(72,209,204)
 midnightblue	#191970	rgb(25,25,112)
 navy	#000080	rgb(0,0,128)
 paleturquoise	#afeeee	rgb(175,238,238)
 powderblue	#b0e0e6	rgb(176,224,230)
 royalblue	#4169e1	rgb(65,105,225)
 skyblue	#87ceeb	rgb(135,206,235)
 slateblue	#6a5acd	rgb(106,90,205)
 steelblue	#4682b4	rgb(70,130,180)
 turquoise	#40e0d0	rgb(64,224,208)

Violets

	name	hex	RGB
	blueviolet	#8a2be2	rgb(138,43,226)
	darkorchid	#9932cc	rgb(153,50,204)
	darkviolet	#9400d3	rgb(148,0,211)
	indigo	#4b0082	rgb(75,0,130)
	mediumorchid	#ba55d3	rgb(186,85,211)
	mediumpurple	#9370db	rgb(147,112,219)
	mediumvioletred	#c71585	rgb(199,21,133)
	purple	#800080	rgb(128,0,128)
	violet	#ee82ee	rgb(238,130,238)

Yellows

name	hex	RGB
 darkgoldenrod	#b8860b	rgb(184,134,11)
 darkorange	#ff8c00	rgb(255,140,0)
 gold	#ffd700	rgb(255,215,0)
 goldenrod	#daa520	rgb(218,165,32)
 khaki	#f0e68c	rgb(240,230,140)
 lemonchiffon	#ffffac	rgb(255,250,205)
 lightgoldenrodyellow	#fafad2	rgb(250,250,210)
 lightyellow	#ffffe0	rgb(255,255,224)
 orange	#ffa500	rgb(255,165,0)
 palegoldenrod	#eee8aa	rgb(238,232,170)
 yellow	#ffff00	rgb(255,255,0)

(b)text

begin

c**text**

center

e**text**

end

r**text**

rotate

a**rc****text**

hello, world

t**ext****box**

Now is the time
for all good men
to come

t**ext****file**

This is the contents
of a file. it has lines of text.
Reading is fundamental.

t**ext****code**

import "fmt"
func main() {
 fmt.Println("Go")
}

l**ine**

h**line**

v**line**

a**rc**

c**urve**

p**oly****line**

c**ircle**

a**circle**

e**llipse**

s**quare**

r**ect**

r**rect**

p**ill**

p**olygon**

s**tar**

i**mage**

c**image**

l**brace**

r**brace**

u**brace**

d**brace**

l**bracket**

r**bracket**

u**bracket**

d**bracket**

s**ky**
a**rr****ow**

l**car****row**

r**car****row**

d**car****row**

u**car****row**

l**ist**

one
two
three

b**list**

● one
● two
● three

n**list**

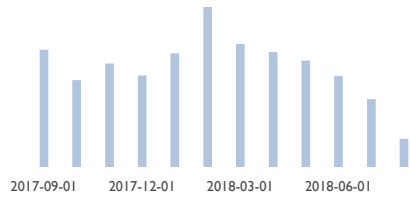
1. one
2. two
3. three

c**list**

first
second item
thrid

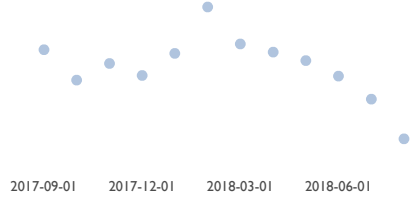
barchart

AAPL Volume (Millions)



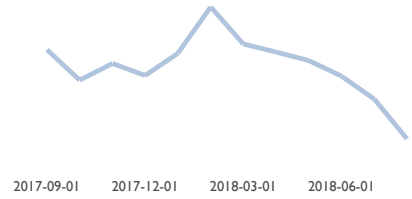
scatter

AAPL Volume (Millions)



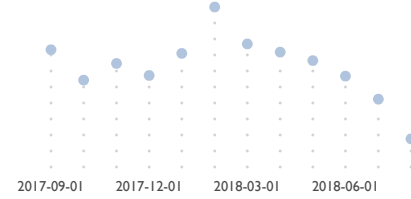
linechart

AAPL Volume (Millions)



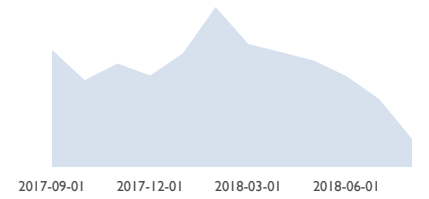
dotchart

AAPL Volume (Millions)

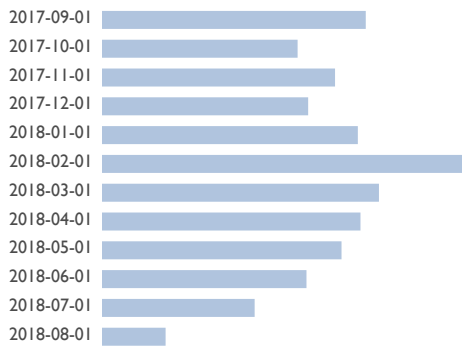


areachart

AAPL Volume (Millions)

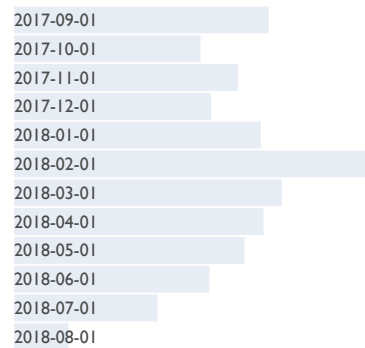


hbar



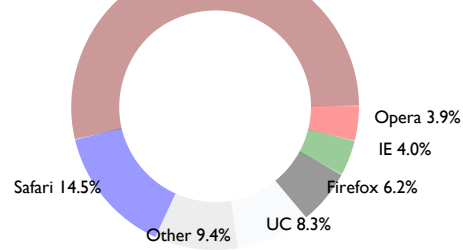
wbar

AAPL Volume (Millions)

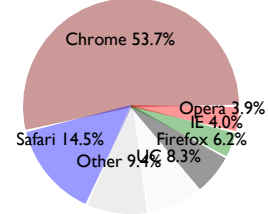


donut

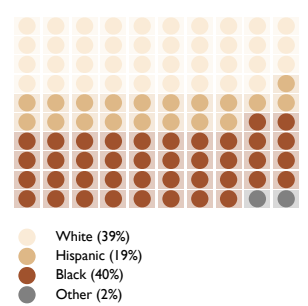
AAPL Volume (Millions)



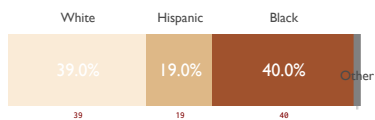
pie



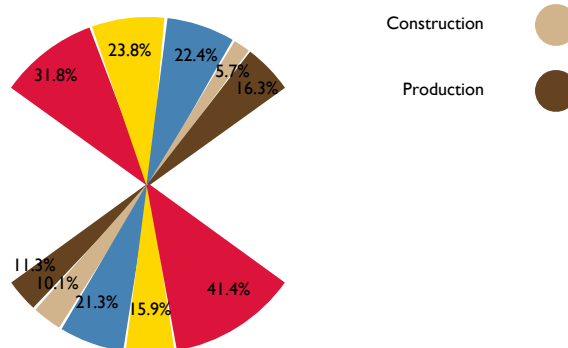
waffle



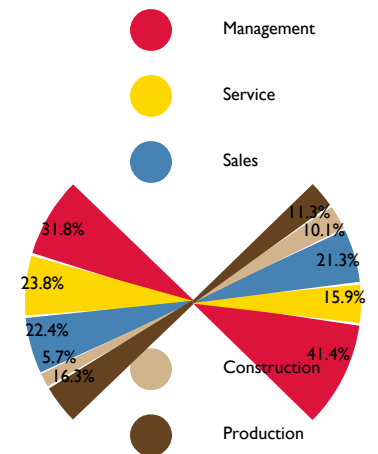
pmap



fanchart



bowtie



georegion



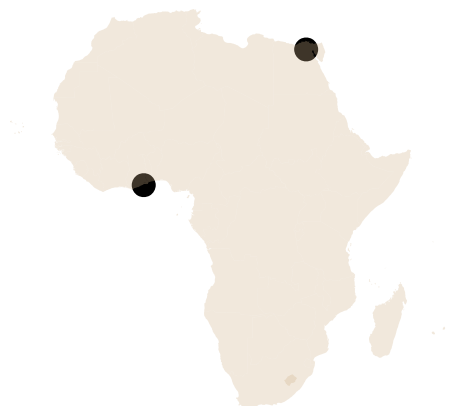
geoborder



geolabel



geomark



geoloc



geoimage



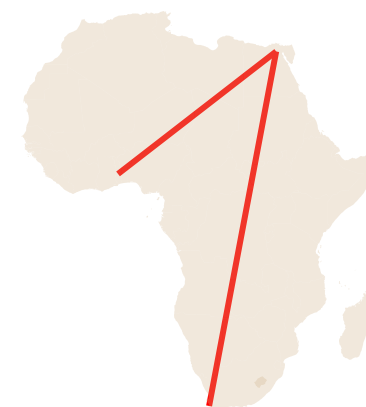
geopath



geoarc



geopathfile



Textual Elements

description	keyword	mandatory	optional
Left-aligned	text	"..." x y fontsize	font color op link
Centered	ctext	"..." x y fontsize	font color op link
End-aligned	etext	"..." x y fontsize	font color op link
Rotated	rtext	"..." x y angle fontsize	font color op link
Text on an arc	arctext	"..." x y rad a1 a2 fontsize	font color op link
Block text	textblock	"..." x y w fontsize	font color op link
Paragraphs	para...epara	x y w fontsize	font color op link
Block text from file	textblockfile	"file" x y w fontsize	font color op link
File contents	textfile	"file" x y fontsize	font color op spacing
Code listing	textcode	"file" x y w fontsize	font color

hello, world
(x,y)

text "... " x y fontsize font color op link

abc

text "abc" 20 20 4

abc

text "abc" 75 20 7 "mono" "maroon"

hello, world

(x,y)

`ctext "..."` `x y` `fontsize` `font` `color` `op` `link`

abc

`ctext "abc" 20 20 4`

abc

`ctext "abc" 80 20 7 "mono" "maroon"`

hello, world.

(x,y)

`etext "..."` `x y` `fontsize` `font` `color` `op` `link`

abc

`etext "abc" 20 20 4`

abc

`etext "abc" 80 20 7 "mono" "maroon"`

hello, world

(x,y)

`rtext "..."` `x` `y` `angle` `fontsize` `font` `color` `op` `link`

abc

`rtext 20 20 30 3`

abc

`rtext 50 20 90 5`

abc

`rtext 80 20 270 4 "sans" "maroon"`

a1 hello there world a2
(x,y)

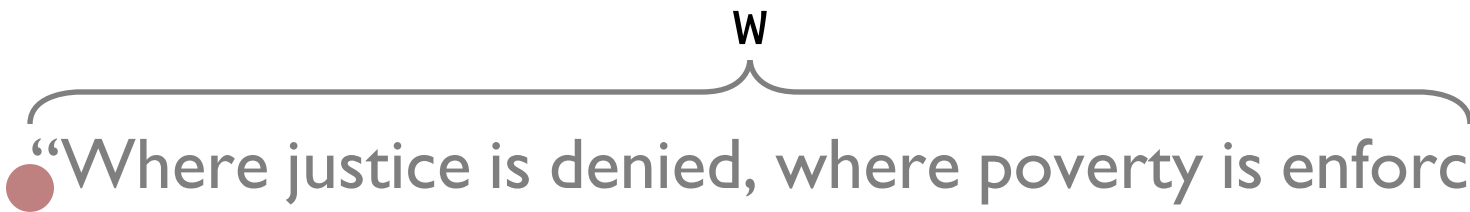
```
arctext "... " x y radius a1 a2 fontsize font color op link
```

What is up

This is curvy

```
arctext "What is up" 25 20 10 180 90 3 "mono"
```

```
arctext "This is curvy" 75 30 10 180 360 3 "mono"
```


(x, y)  “Where justice is denied, where poverty is enforced,
where ignorance prevails, and where any one class
is made to feel that society is an organized conspiracy
to oppress, rob and degrade them, neither persons
nor property will be safe.”

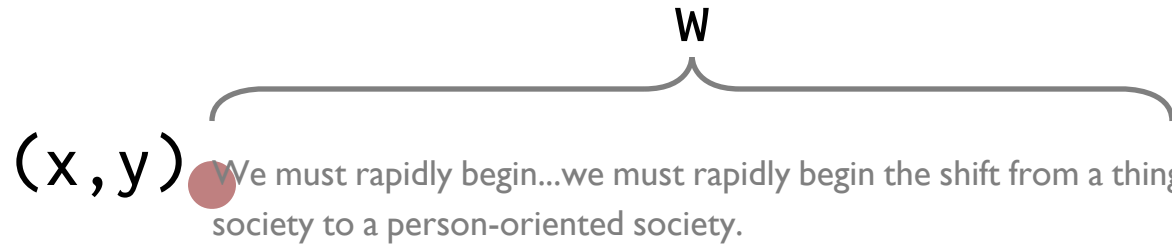
textblock "... " x y w fontsize font color op link

“Where justice is denied, where poverty is enforced,
where ignorance prevails, and where any one class
is made to feel that society is an organized conspiracy
to oppress, rob and degrade them, neither persons
nor property will be safe.”

textblock "... " 10 35 30 2

“Where justice is denied,
where poverty is enforced,
where ignorance prevails,
and where any one class is
made to feel that society
is an organized conspiracy
to oppress, rob and degrade
them, neither persons nor
property will be safe.”

textblock "... " 50 35 10 1 "sans" "maroon"

(x,y)  We must rapidly begin...we must rapidly begin the shift from a thing-oriented society to a person-oriented society.

When machines and computers, profit motives and property rights, are considered more important than people, the giant triplets of racism, extreme materialism, and militarism are incapable of being conquered.

A true revolution of values will soon cause us to question the fairness and justice of many of our past and present policies. On the one hand, we are called to play the Good Samaritan on life's roadside, but that will be only an initial act.

para x y w fontsize font color op link ... **epara**

For, lo, the winter is passed,
the rain is over and gone.

The flowers appear on the
earth, the time for the singing
of birds is come, And the voice
of the turtle is heard on the
land.

```
para 5 35 20 1.8 2.5 "serif" "gray"
```

```
For, lo, the winter is passed, the rain is over and gone.
```

```
The flowers appear on the earth, the time  
for the singing of birds is come,  
And the voice of the turtle is heard on the land.
```

```
epara
```

(x,y) This is the contents
of a file. it has lines of text.
Reading is fundamental.

`textfile "..."` x y fontsize font color op link

This is the contents
of a file. it has lines of text.
Reading is fundamental.

```
import "fmt"
func main() {
    fmt.Println("Go")
}
```

`textfile "example.txt" 10 35 2`

`textfile "hw-go" 55 35 1.6 "mono" "maroon"`

(x,y)

W

```
import "fmt"
func main() {
    fmt.Println("Go")
}
```

textcode "... " x y w fontsize font color

```
import "fmt"
func main() {
    fmt.Println("Go")
}
```

textcode "hw-go" 10 35 25 1.0

```
import "fmt"
func main() {
    fmt.Println("Go")
}
```

textcode "hw-go" 55 35 40 1.6 "maroon"

Graphical Elements

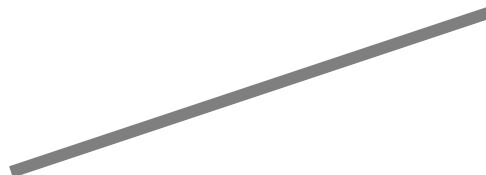
description	keyword	mandatory	optional
Line	line	x1 y1 x2 y2 lw	color op
Dotted Line	dline	x1 y1 x2 y2	lw gap color op
Horizontal line	hline	x y w	lw color op
Vertical line	vline	x y h	lw color op
Elliptical arc	arc	x y w h a1 a2	lw color op
Quadratic Bezier	curve	bx by cx cy ex ey	lw color op
Circle	circle	x y w	color op
Area circle	acircle	x y area	color op
Ellipse	ellipse	x y w h	color op
Square	square	x y w	color op
Rectangle	rect	x y w h	color op
Rounded rectangle	rrect	x y w h radius	color
Pill shape	pill	x y w h	color
Polygon	polygon	"x1 x2...xn" "y1 y2...yn"	color op
Polyline	polyline	"x1 x2...xn" "y1 y2...yn"	lw color op
N-sided star	star	x y sides inner outer	color op

`lw {`  `}`
`(x1,y1)` `(x2,y2)`

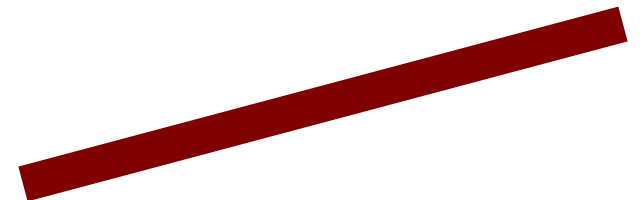
`line x1 y1 x2 y2 lw color op`



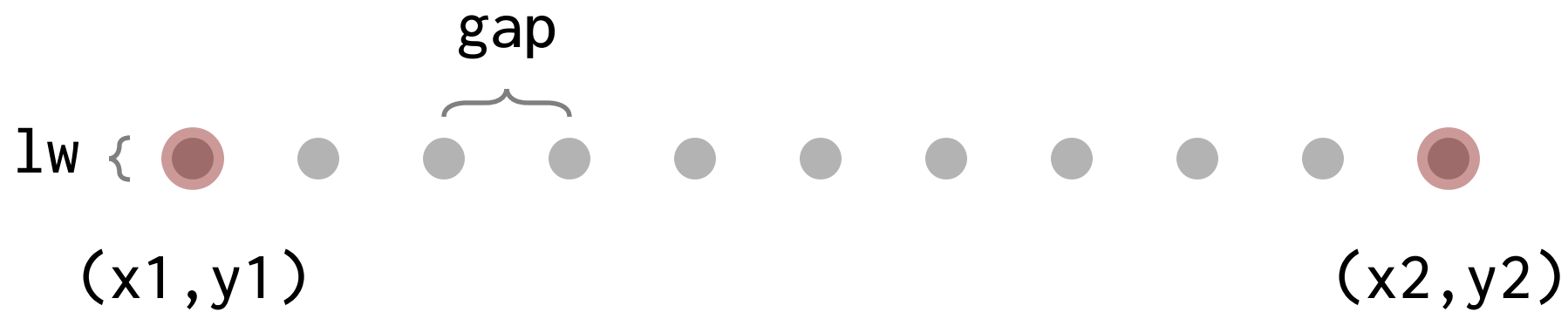
`line 10 20 30 20`



`line 40 20 60 30 0.5`



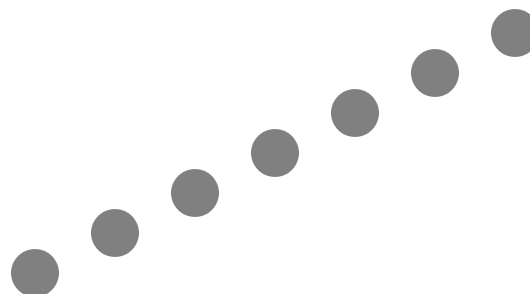
`line 70 20 95 30 1.5 "maroon"`



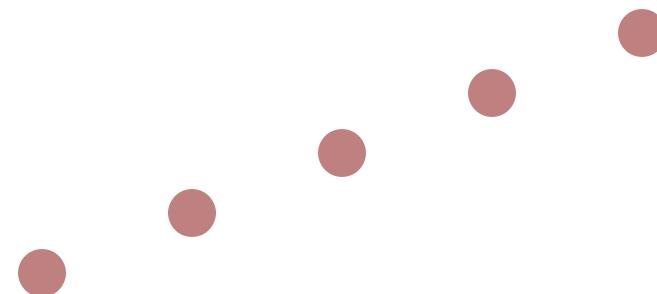
`dline x1 y1 x2 y2 lw gap color op`

.....

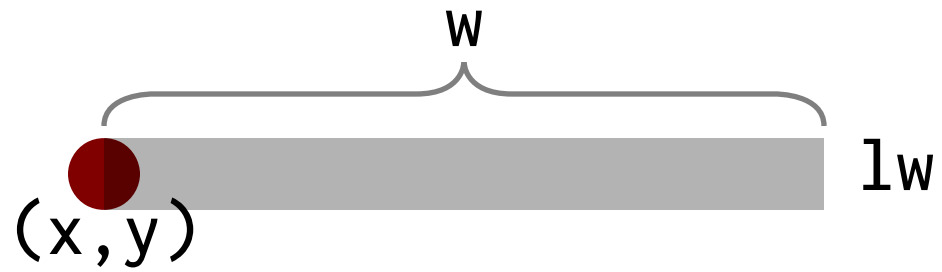
`dline 10 20 30 20`



`dline 40 20 60 35 2 4`



`dline 70 20 95 35 2 6 "maroon" 50`



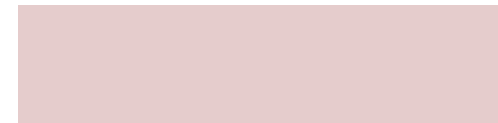
```
hline x y w lw color op
```



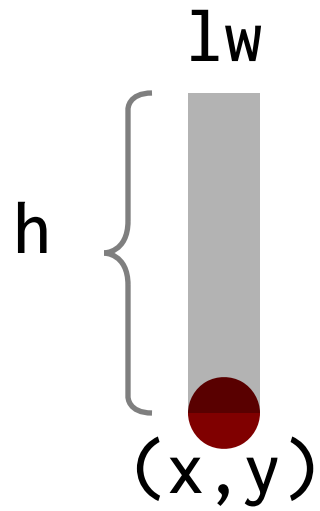
```
hline 15 20 10
```



```
hline 40 20 20 1
```



```
hline 70 20 20 5 "maroon" 20
```

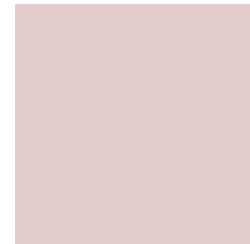
`vline x y w lw color op`



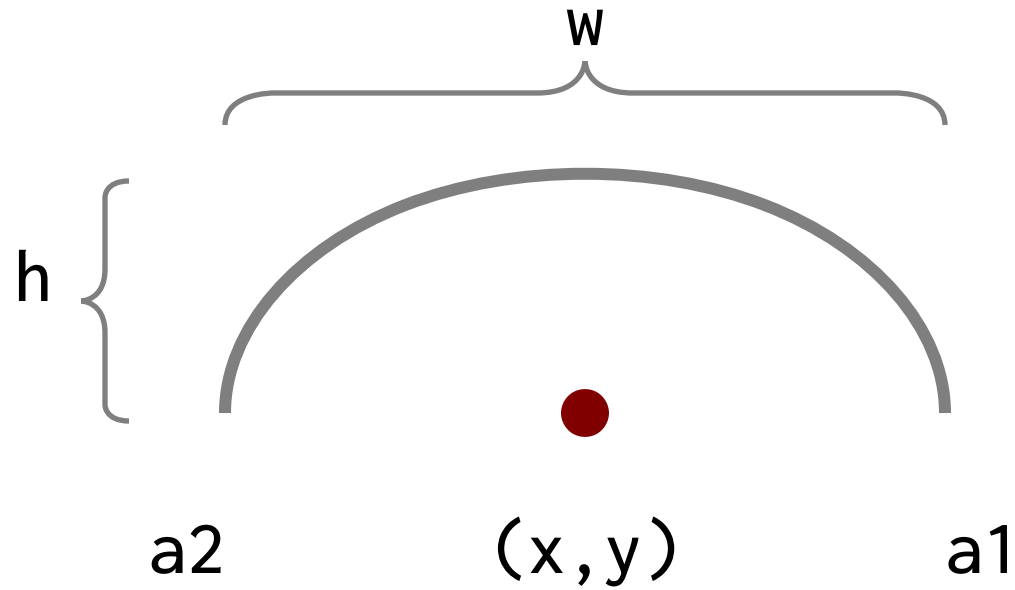
`vline 20 20 15`



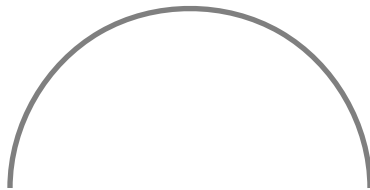
`vline 50 20 15 2`



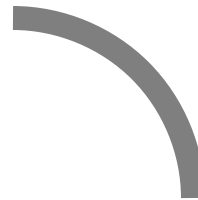
`vline 80 20 15 10 "maroon" 20`



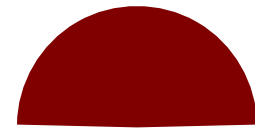
`arc x y w h a1 a2 lw color op`



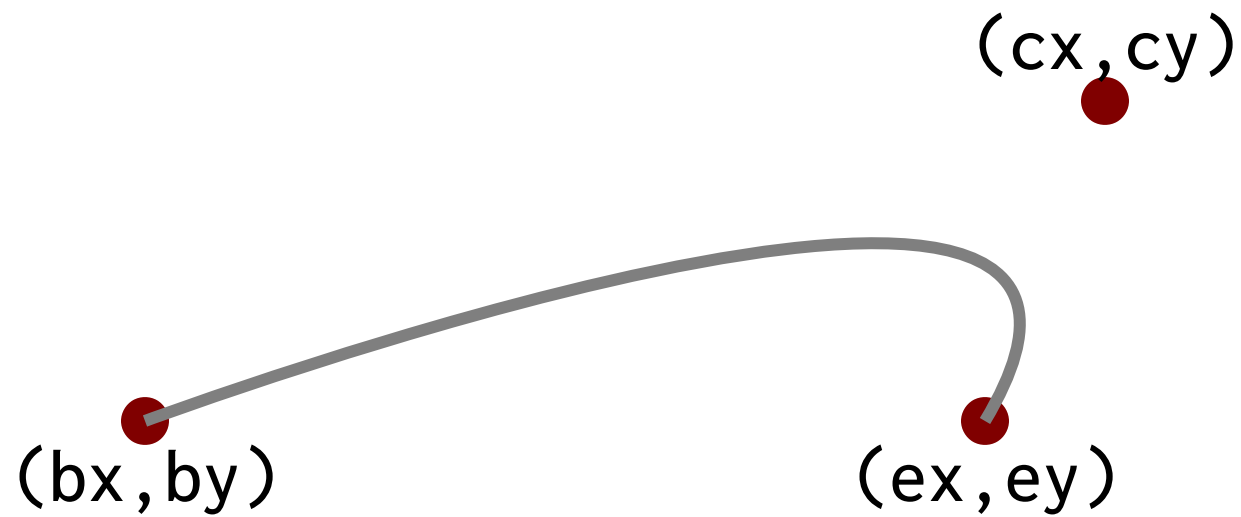
`arc 20 20 15 15 0 180`



`arc 50 20 15 15 0 90 1`



`arc 80 20 5 5 0 180 5 "maroon"`



`curve` `bx` `by` `cx` `cy` `ex` `ey` `lw` `color` `op`



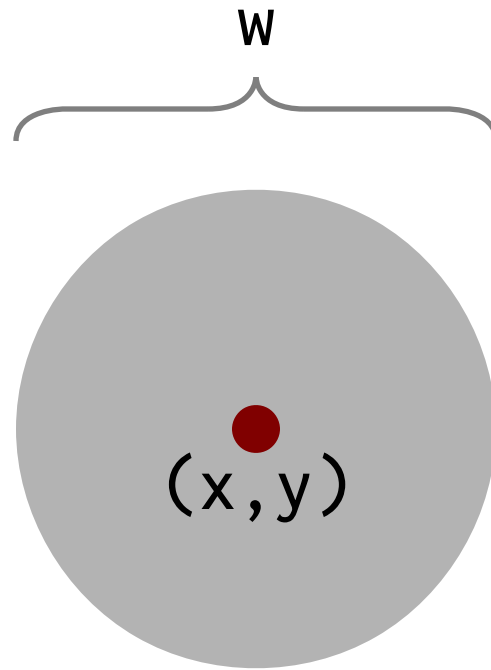
`curve 15 20 25 30 30 25`



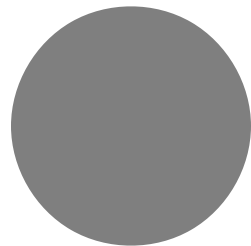
`curve 15 20 25 30 30 25`



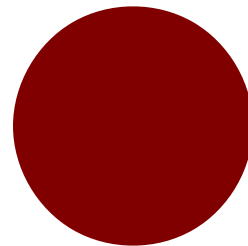
`curve 70 20 70 30 90 25 0.5 "maroon"`



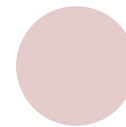
`circle x y w color op`



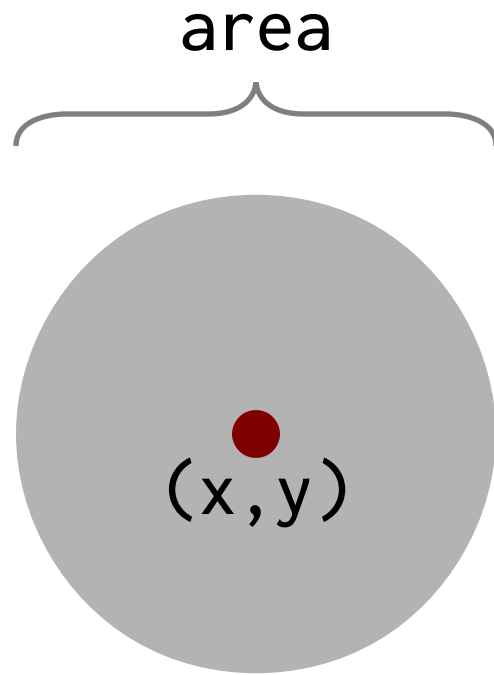
`circle 20 20 10`



`circle 50 20 10 "maroon"`



`circle 80 20 5 "maroon" 20`



`circle x y area color op`



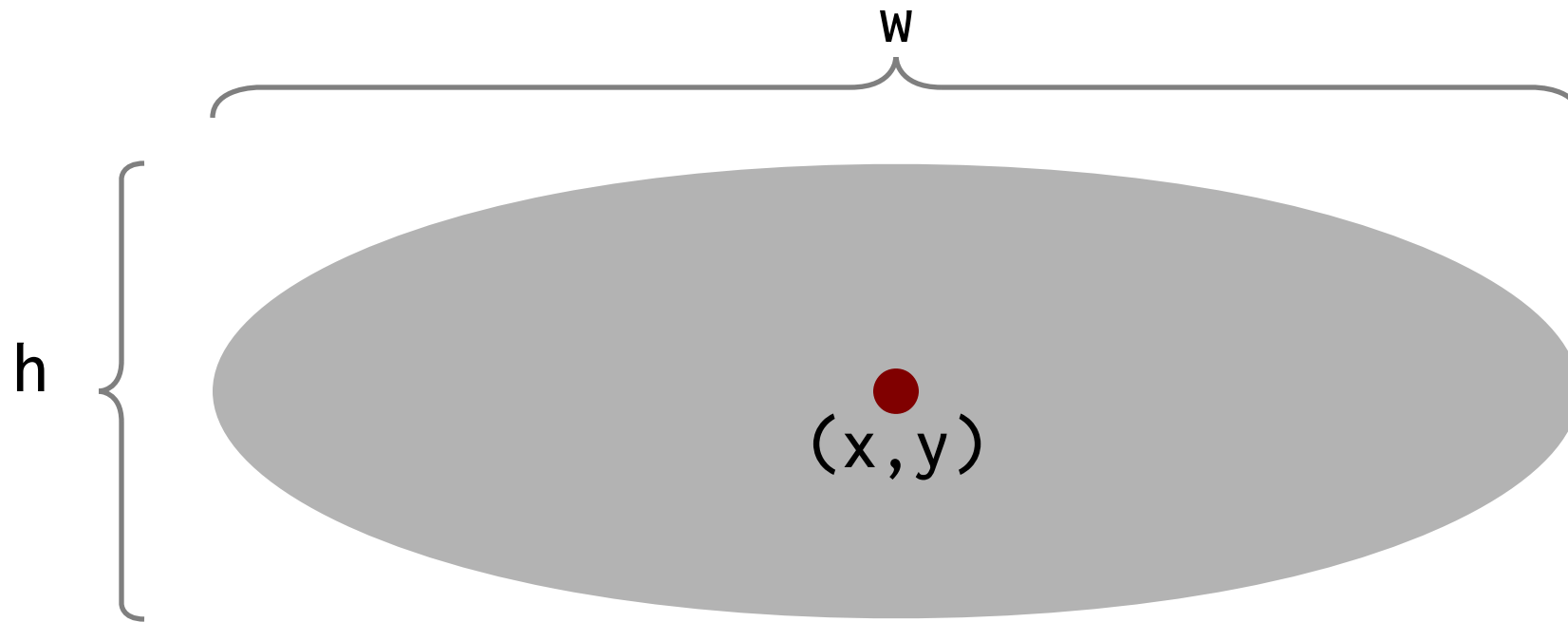
`acircle 20 20 10`



`acircle 50 20 10 "maroon"`



`acircle 80 20 5 "maroon" 20`



`ellipse x y w h color op`



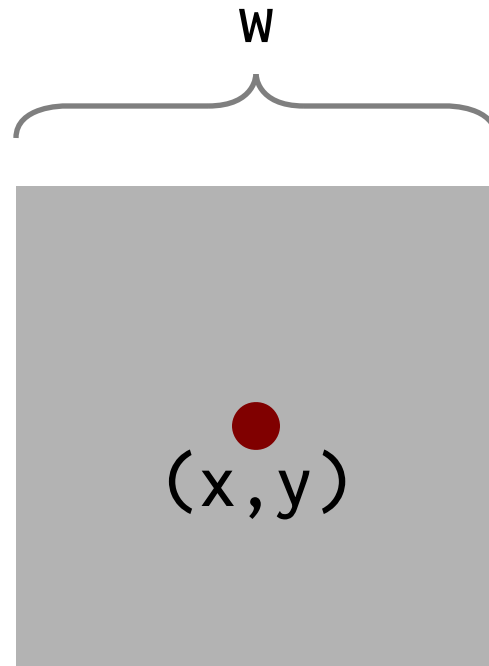
`ellipse 20 20 10 5`



`ellipse 50 20 10 5 "maroon"`



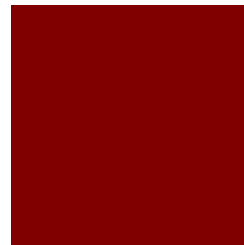
`ellipse 80 20 5 10 "maroon" 20`



`square x y w color op`



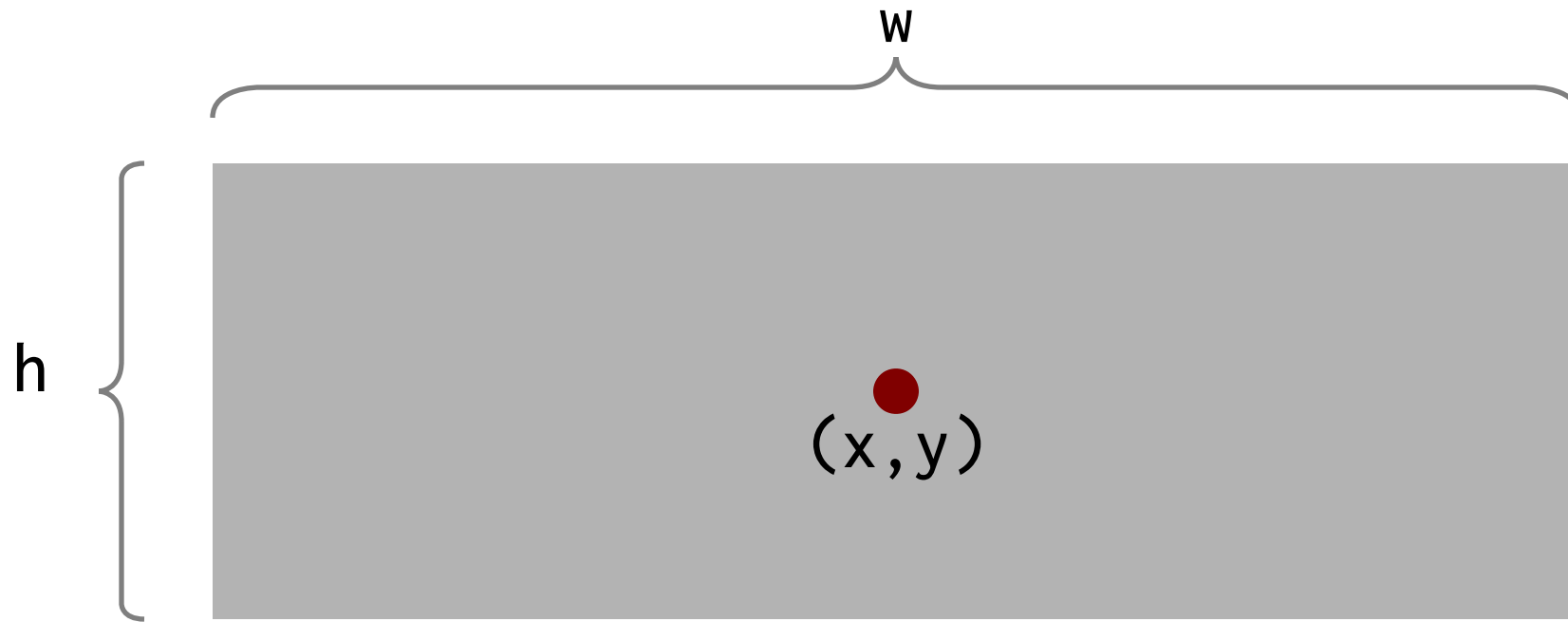
`square 20 20 10`



`square 50 20 10 "maroon"`



`square 80 20 5 "maroon" 20`



`rect x y w h color op`



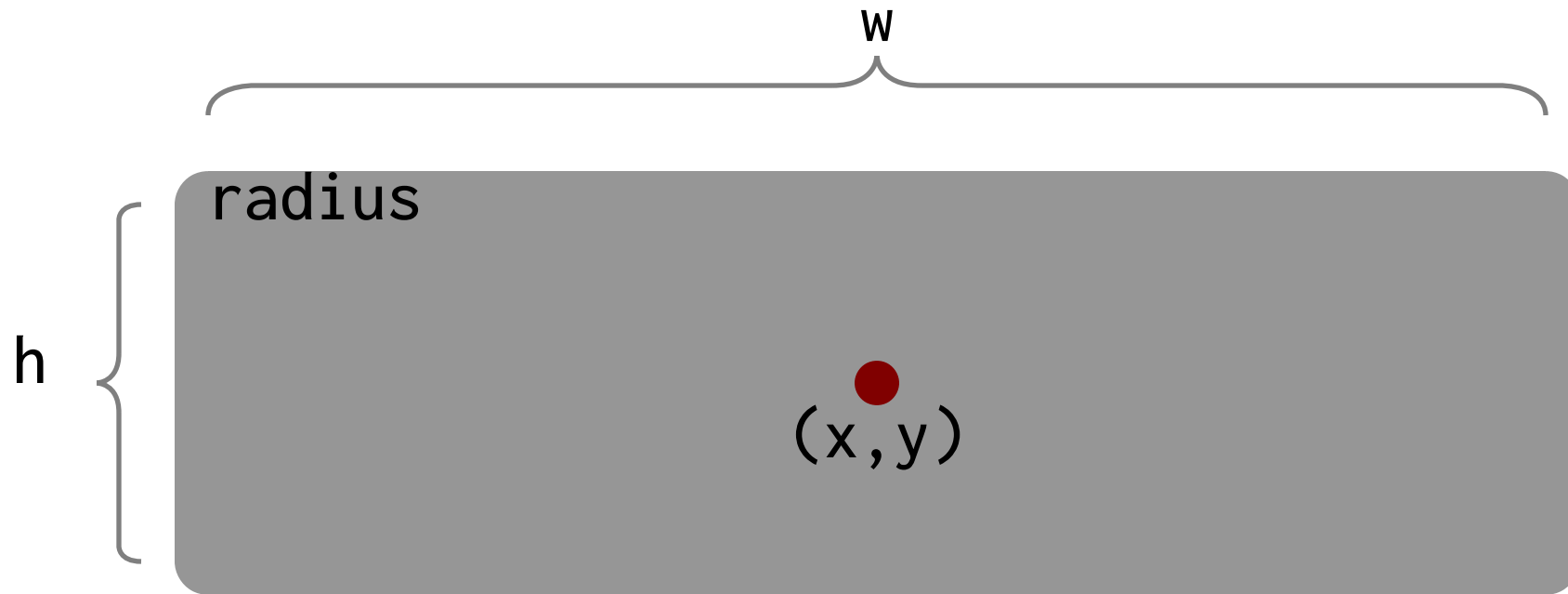
`rect 20 20 10 5`



`rect 50 20 10 5 "maroon"`



`rect 80 20 5 10 "maroon" 20`



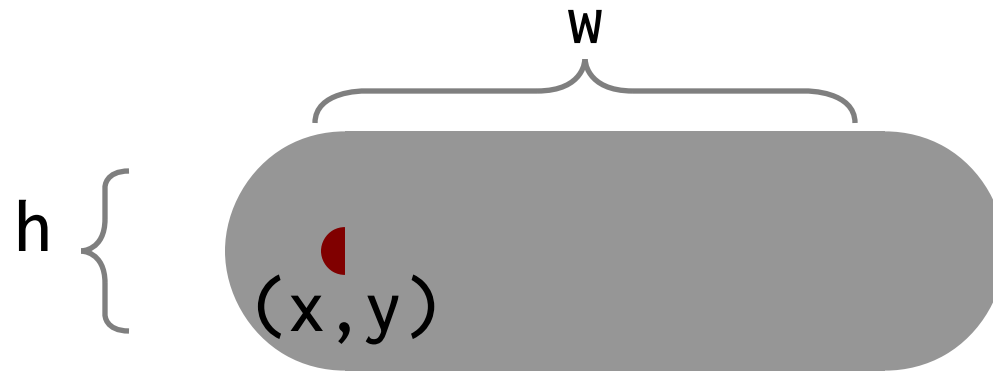
`rrect x y w h radius color op`



`rrect 20 20 10 5 1`



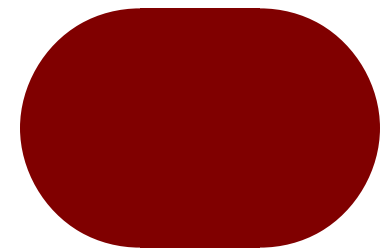
`rrect 80 20 5 10 1 "maroon"`



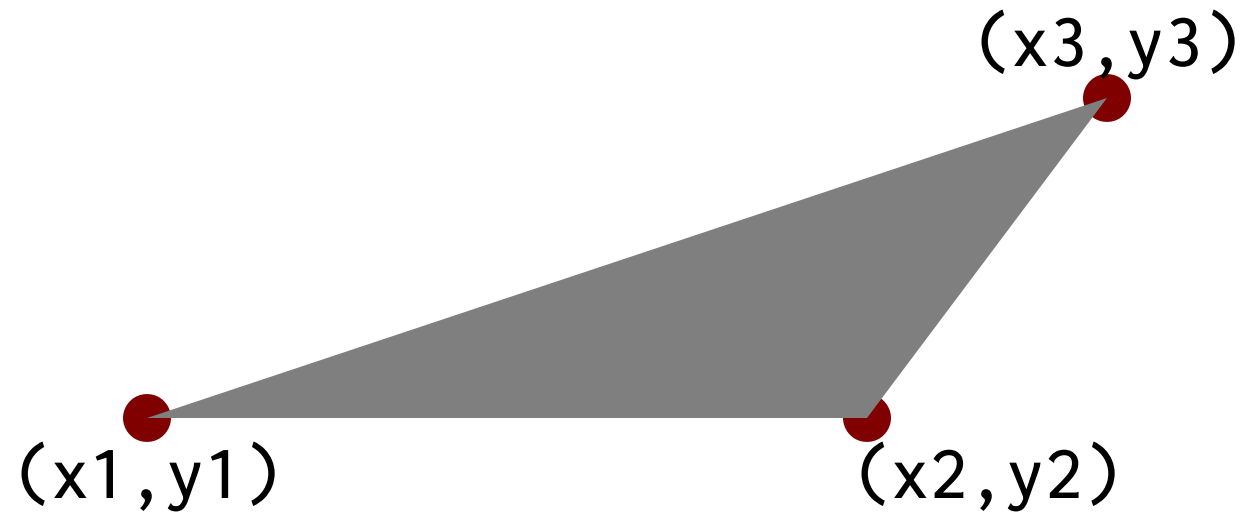
`pill x y w h color`



`pill 20 20 10 5`



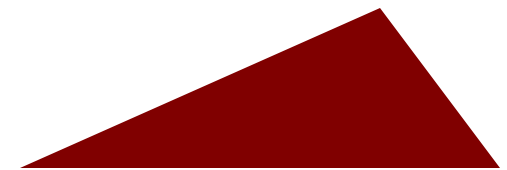
`pill 80 20 5 10 "maroon"`



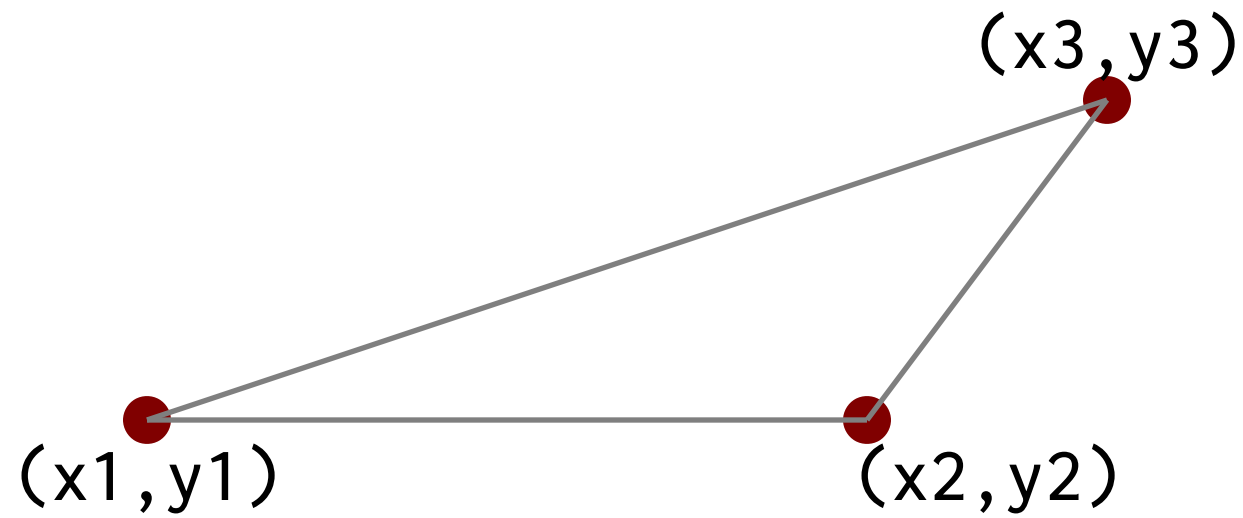
`polygon "x1 x2...xn" "y1 y2...yn" color op`



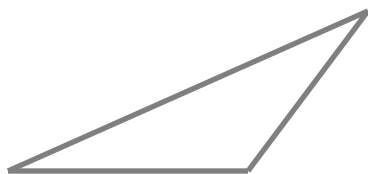
`polygon "10 25 20" "20 30 20"`



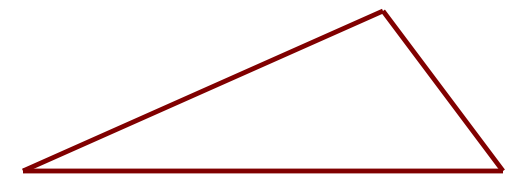
`polygon "70 85 90" "20 30 20" "maroon"`



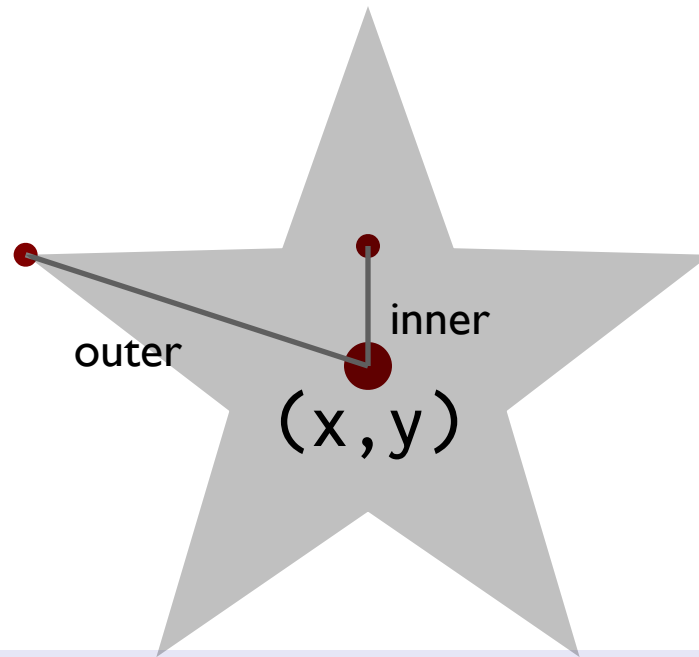
`polyline "x1 x2...xn" "y1 y2...yn" lw color op`



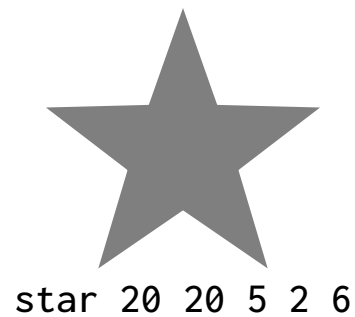
`polyline "10 25 20" "20 30 20"`



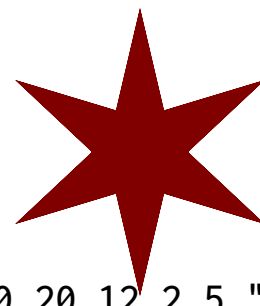
`polyline "70 85 90" "20 30 20" 0.2 "maroon"`



`star x y sides inner outer color op`



`star 20 20 5 2 6`



`star 50 20 12 2 5 "maroon"`



`star 80 ey 24 2 8 "maroon" 20`

Images

description	keyword	mandatory	optional
Image	<code>image</code>	<code>"file" x y w h</code>	<code>scale "link"</code>
Captioned image	<code>cimage</code>	<code>"file" "caption" x y w h</code>	<code>scale "link" capsize</code>

If `h = 0`, `w` specifies the image width in terms of canvas width.

The scale value is a percentage from 1-100, and link is a URL. capsize is the text size of the caption

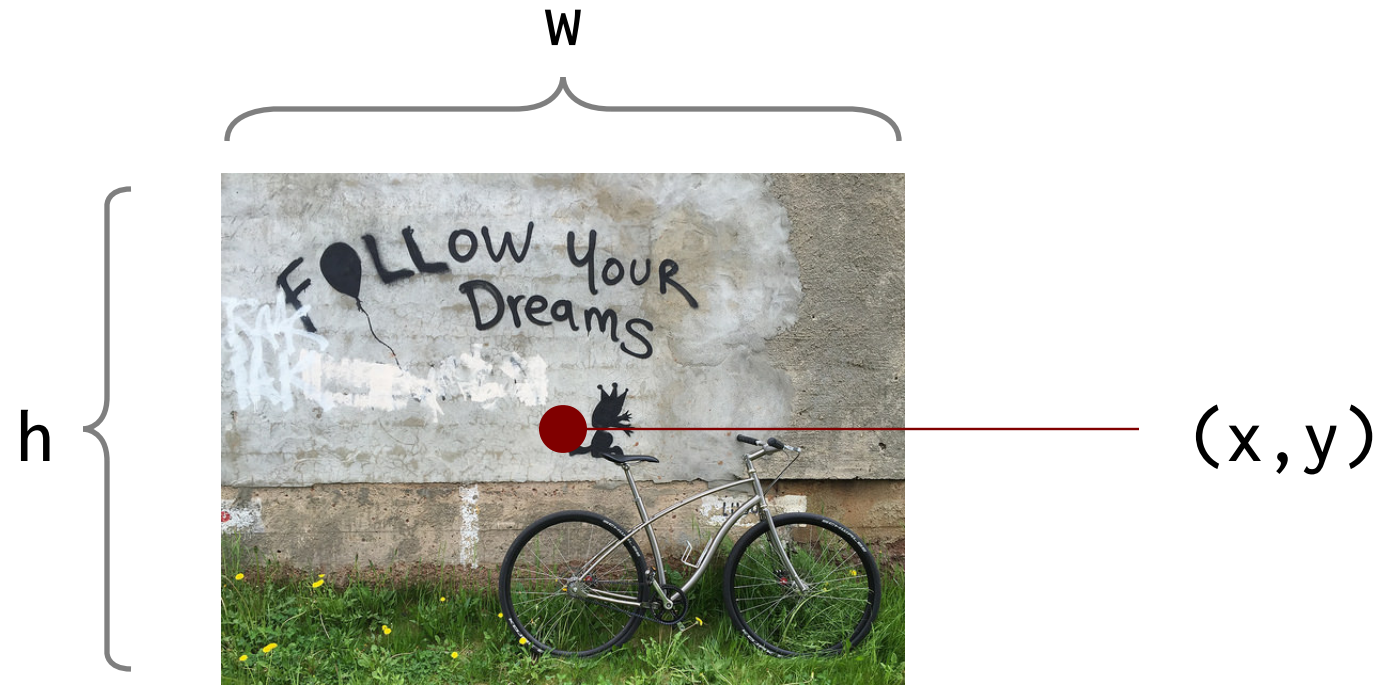


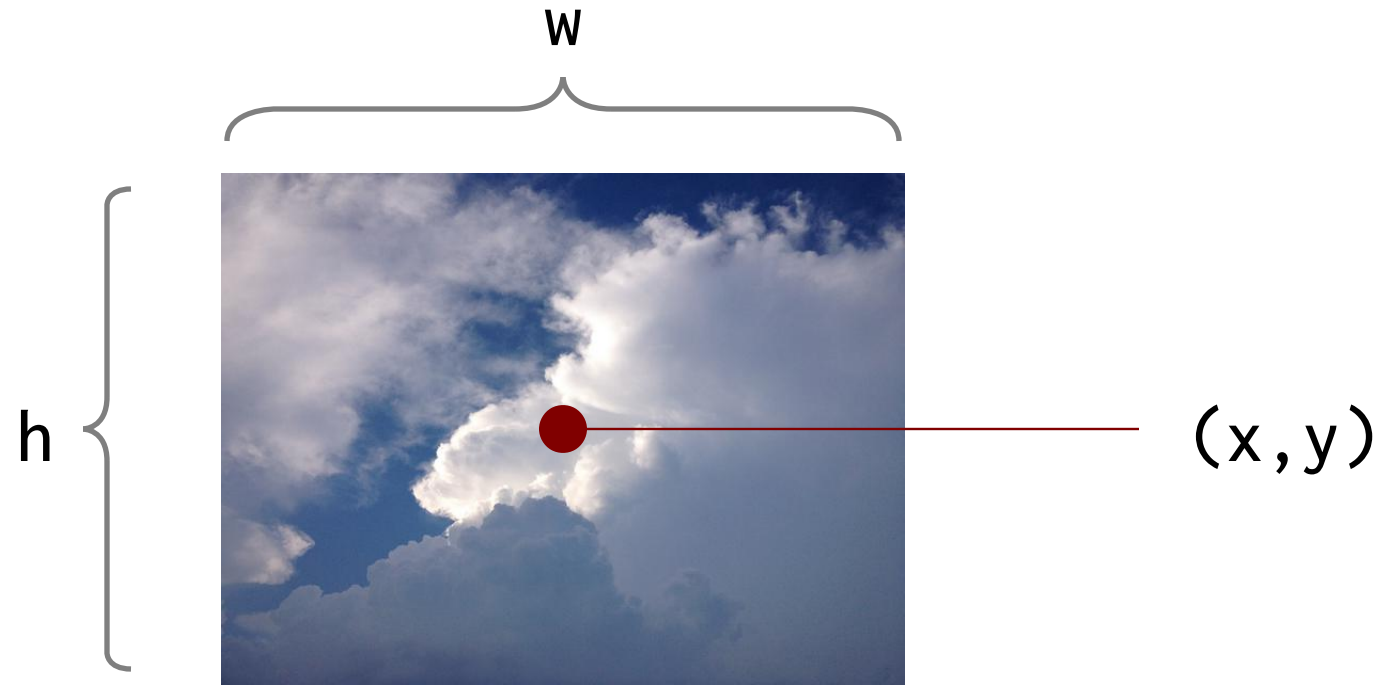
image "file" x y w h scale link



image "follow.jpg" 20 25 640 480 10

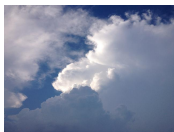


image "follow.jpg" 75 25 640 480 30



sky

`cimage "file" x y w h scale link`



sky



sky

`cimage "cloudy.jpg" "sky" 20 25 640 480 10`

`cimage "cloudy.jpg" "sky" 75 25 640 480 30 "" 1.5`

Lists

description	keyword	mandatory	optional
Plain list	<code>list</code>	<code>x y fontsize</code>	<code>font color op spacing</code>
Bullet list	<code>blist</code>	<code>x y fontsize</code>	<code>font color op spacing</code>
Numbered list	<code>nlist</code>	<code>x y fontsize</code>	<code>font color op spacing</code>
Centered list	<code>clist</code>	<code>x y fontsize</code>	<code>font color op spacing</code>

```
list x y fs
(x,y) li "first"
      li "second"
      li "third"
elist
```

list x y fontsize font color op spacing

```
list 20 30 2.5 one
      li "one"
      li "two" two
      li "three" three
elist
```

```
list 85 30 2.5 "serif" "maroon" 100 1.0 one
      li "one" two
      li "two" three
      li "three"
elist
```

```
    blist x y fs
(x,y)  li "first"
        li "second"
        li "third"
    elist
```

blist x y fontsize font color op spacing

```
blist 20 30 2.5 ● one
    li "one"
    li "two" ● two
    li "three" ● three
elist
```

```
blist 85 30 2.5 "serif" "maroon" 100 1.0 ● one
    li "one" ● two
    li "two" ● three
    li "three"
elist
```

```
nlist x y fs
(x,y) li "first"
      li "second"
      li "third"
elist
```

nlist x y fontsize font color op spacing

```
nlist 20 30 2.5 1. one
      li "one"
      li "two" 2. two
      li "three" 3. three
elist
```

```
nlist 85 30 2.5 "serif" "maroon" 100 1.0
      li "one"
      li "two"
      li "three"
elist
```

1. one
2. two
3. three

```
clist x y fs
(x,y) li "first"
      li "second"
      li "third"
elist
```

clist x y fontsize font color op spacing

```
clist 30 30 2.5      first one
  li "first one"
  li "next"          next
  li "and last"      and last
elist
```

```
clist 90 30 2.5 "serif" "maroon" 100 1.0 first
  li "first"        next
  li "next"         and last
  li "and last"
elist
```

Arrows

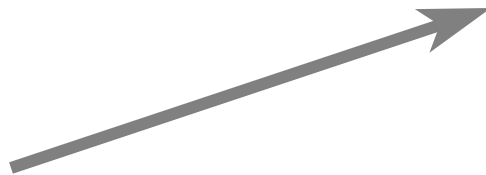
description	keyword	mandatory	optional
Straight	arrow	x1 y1 x2 y2	lw aw ah color op
Left curved	lcarrow	bx by cx cy ex ey	lw aw ah color op
Right curved	rcarrow	bx by cx cy ex ey	lw aw ah color op
Up curved	ucarrow	bx by cx cy ex ey	lw aw ah color op
Down curved	dcarrow	bx by cx cy ex ey	lw aw ah color op



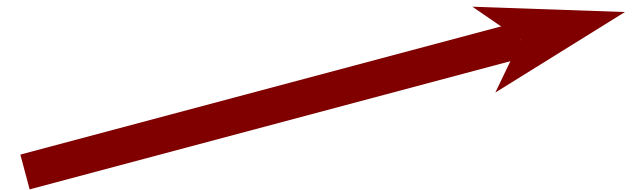
`arrow x1 y1 x2 y2 lw aw ah color op`



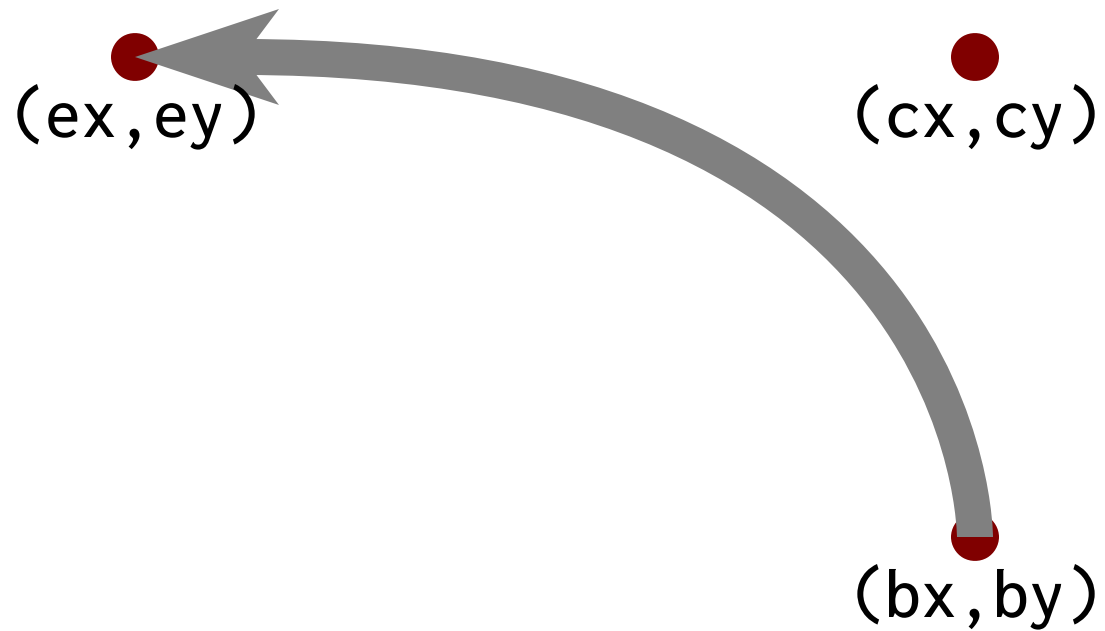
`arrow 10 20 30 20`



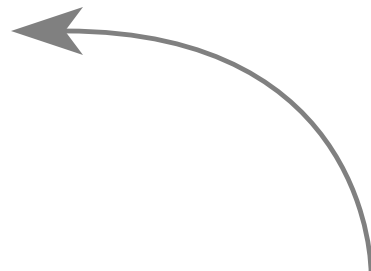
`arrow 40 20 60 30 0.5`



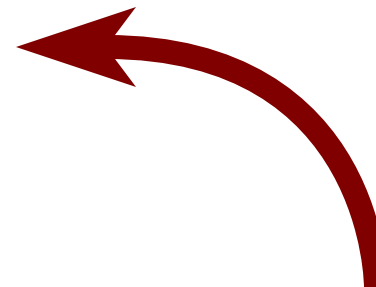
`arrow 70 20 95 30 1.5 6 6 "maroon"`



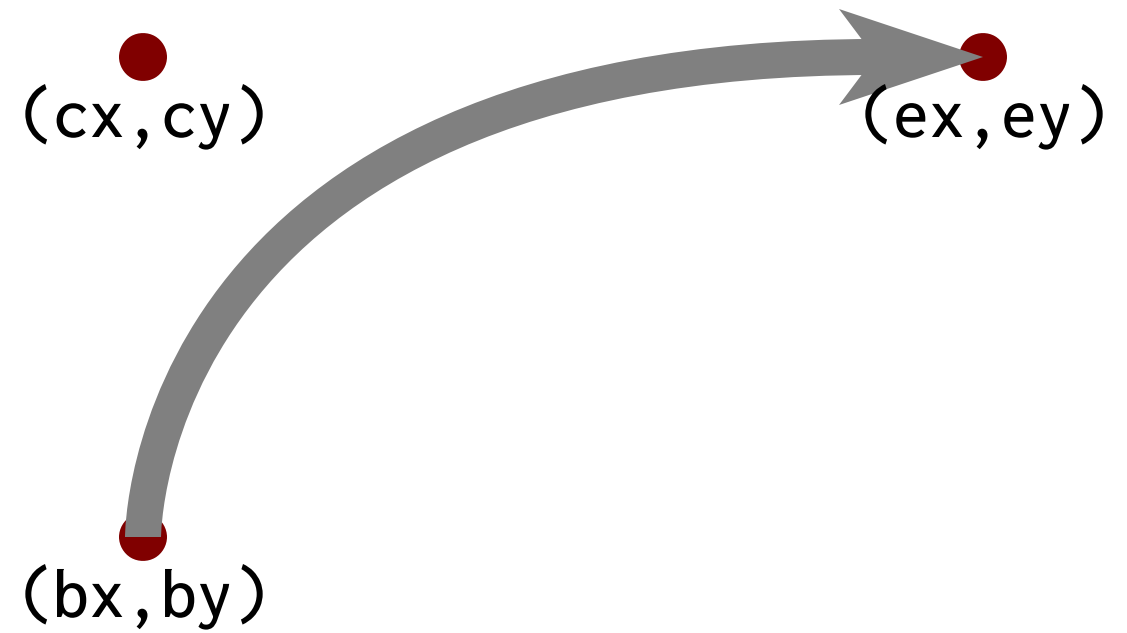
`lcarrow` `bx by cx cy ex ey lw aw ah color op`



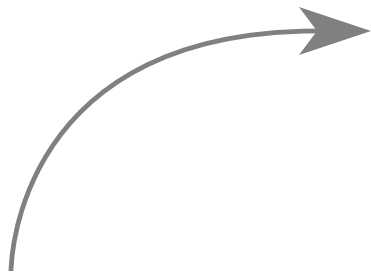
`lcarrow 30 20 30 35 15 35`



`lcarrow 70 20 70 35 55 35 1 5 5 "maroon"`



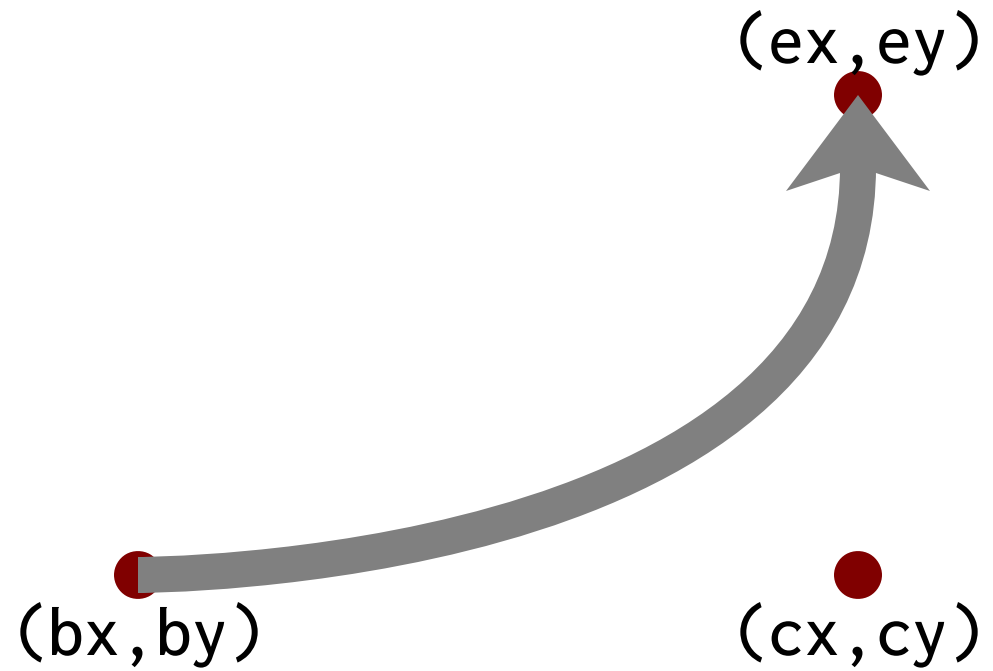
`rcarrow bx by cx cy ex ey lw aw ah color op`



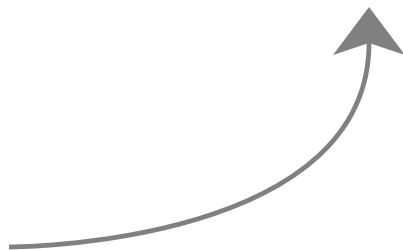
`rcarrow 15 20 15 35 30 35`



`rcarrow 50 20 50 35 70 35 1 5 5 "maroon"`



`ucarrow` `bx` `by` `cx` `cy` `ex` `ey` `lw` `aw` `ah` `color` `op`



`ucarrow 15 20 30 20 30 35`



`rcarrow 50 20 70 20 70 35 1 5 5 "maroon"`

(bx, by)

(cx, cy)

(ex, ey)

`dcarrow bx by cx cy ex ey lw aw ah color op`



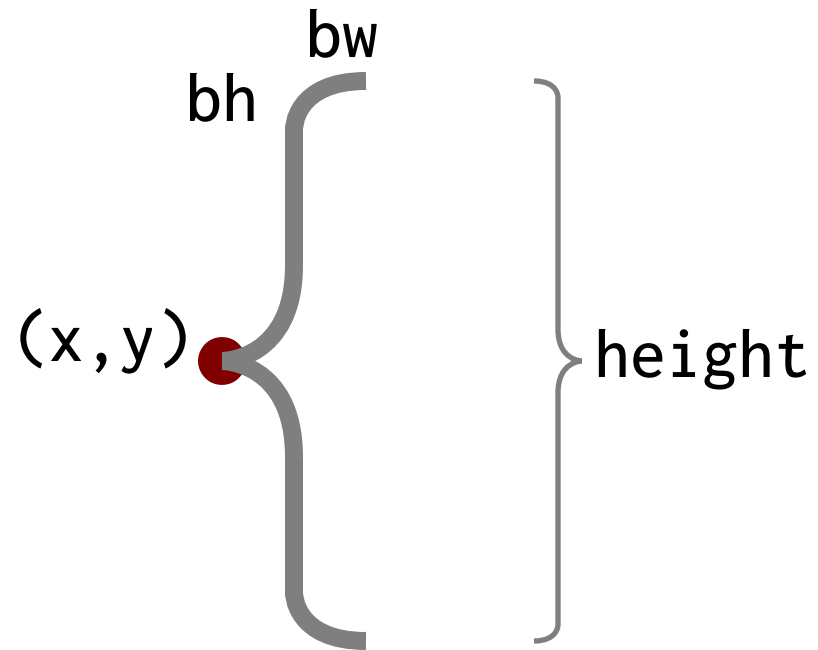
`dcarrow 15 35 30 30 20`



`dcarrow 50 35 70 35 20 1 5 5 "maroon"`

Braces and Brackets

description	keyword	mandatory	optional
Left brace	lbrace	x y height bw bh	lw color op
Right brace	rbrace	x y height bw bh	lw color op
Up brace	ubrace	x y width bw bh	lw color op
Down brace	dbrace	x y width bw bh	lw color op
Left bracket	lbracket	x y width height	lw color op
Right bracket	rbracket	x y width height	lw color op
Up bracket	ubracket	x y width height	lw color op
Down bracket	dbacket	x y width height	lw color op



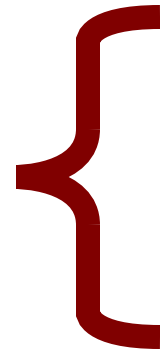
`lbrace` `x` `y` `height` `bw` `bh` `lw` `color` `op`



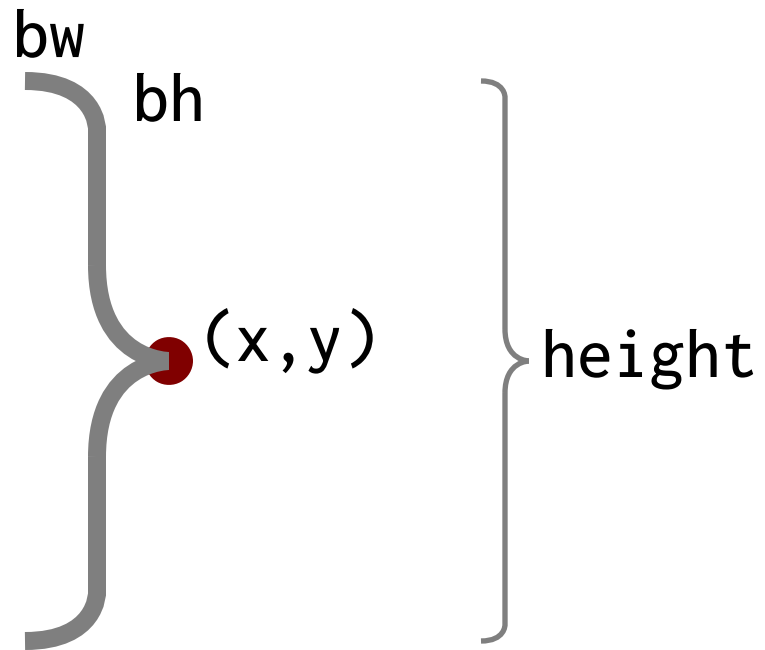
`lbrace 20 25 20 2 2`



`lbrace 50 25 20 4 4 1`



`lbrace 80 25 20 6 3 1 "maroon"`



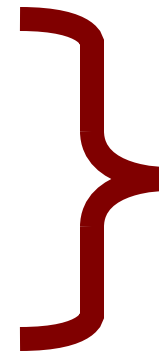
`rbrace` `x` `y` `height` `bw` `bh` `lw` `color` `op`



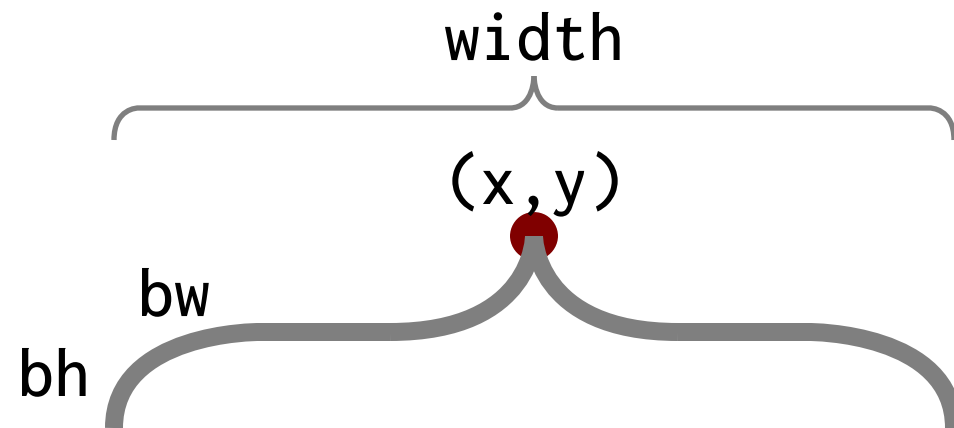
`rbrace 20 25 20 2 2`



`rbrace 50 25 20 4 4 1`



`rbrace 80 25 20 6 3 1 "maroon"`



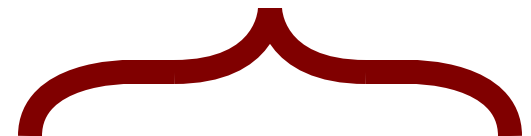
`ubrace` `x y width bw bh` `lw color op`



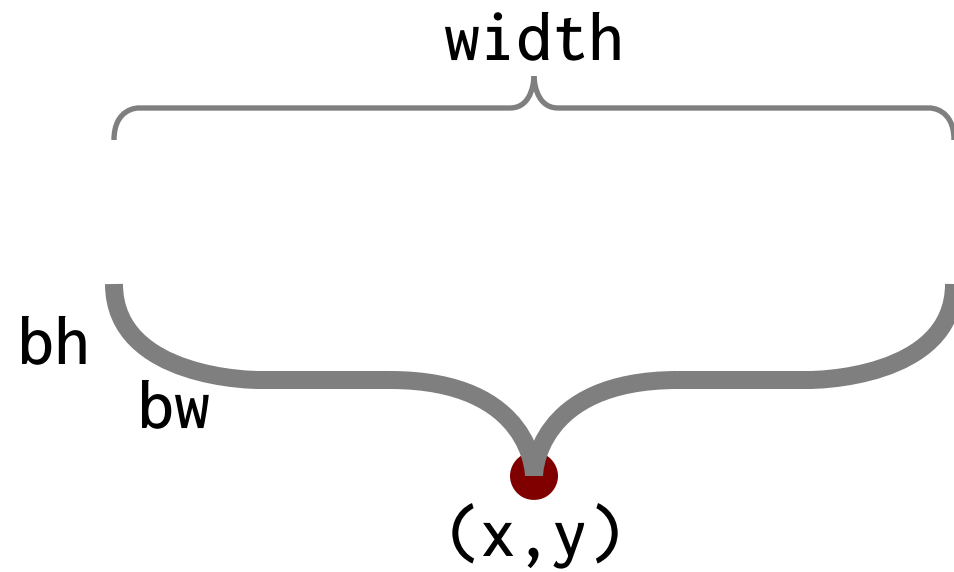
`ubrace 20 25 20 2 4`



`ubrace 50 25 20 4 8 1`



`ubrace 80 25 20 4 8 1 "maroon"`



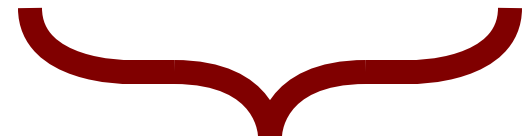
dbrace **x y width bw bh** lw color op



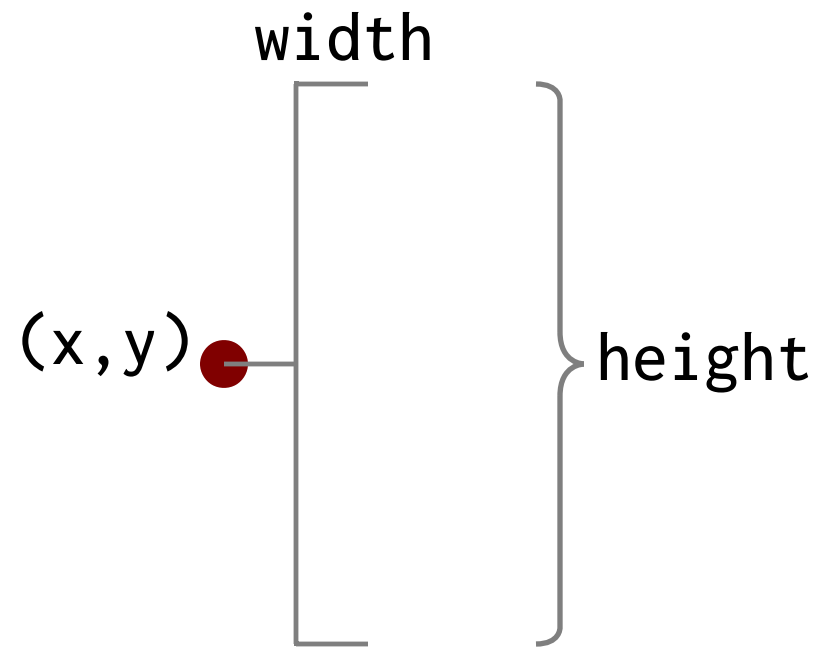
dbrace 20 25 20 2 4



dbrace 50 25 20 4 8 1



dbrace 80 25 20 4 8 1 "maroon"



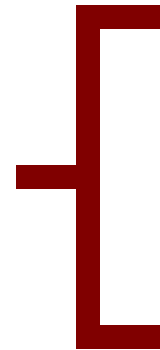
`lbracket` `x` `y` `width` `height` `lw` `color` `op`



`lbrace 20 25 2 20`



`lbracket 50 25 4 20 1`



`lbracket 80 25 6 20 1 "maroon"`

width



(x,y)



height

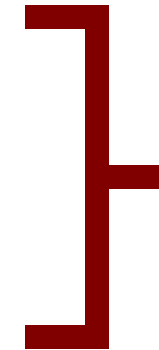
rbracket x y width height lw color op



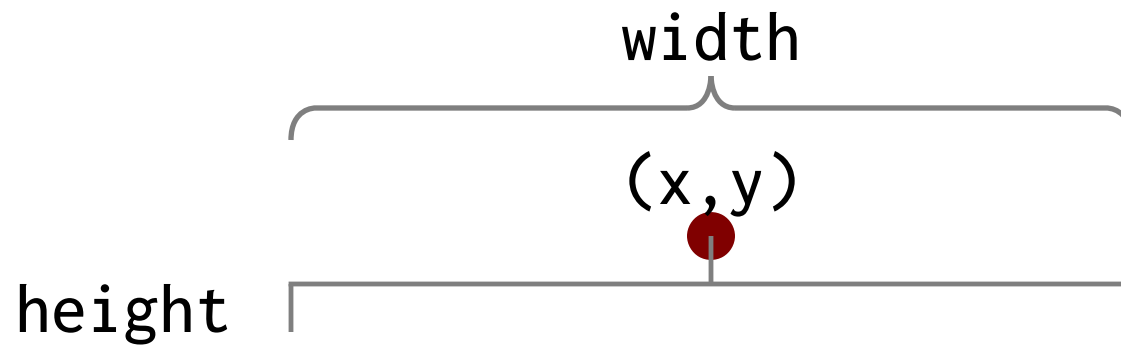
lbrace 20 25 2 20



rbracket 50 25 4 20 1



rbracket 80 25 6 20 1 "maroon"



ubracket *x y width height lw color op*



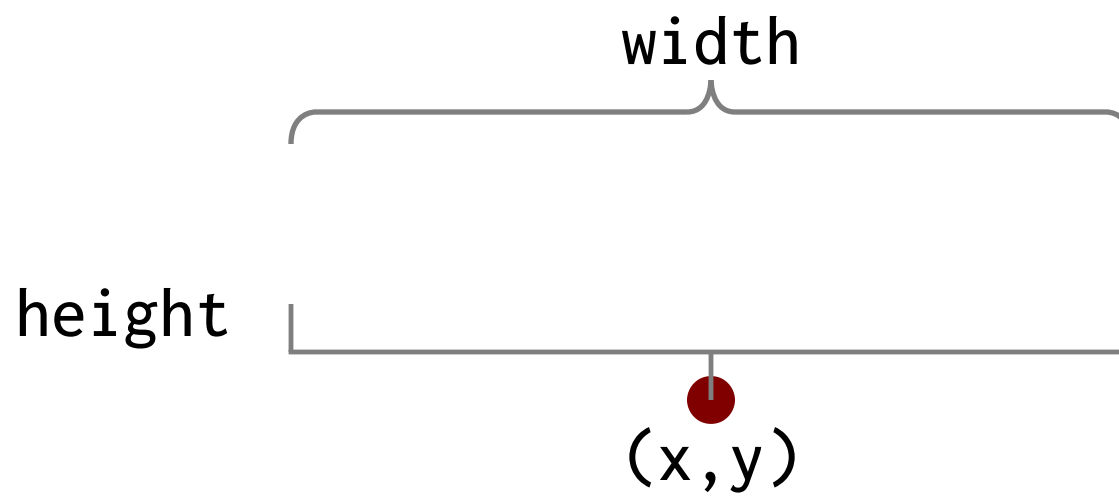
`ubracket 20 25 20 4`



`ubracket 50 25 20 4 0.3`



`ubracket 80 25 20 4 1 "maroon"`



dbracket *x* *y* *width* *height* *lw* *color* *op*



dbracket 20 25 20 4



dbracket 50 25 20 4 0.3



dbracket 80 25 20 4 1 "maroon"

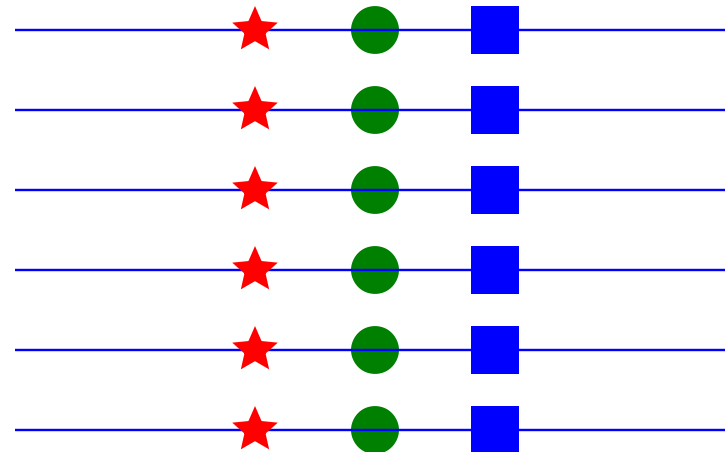
Loop, If, Built-ins

description	keyword	arguments
Loop	<code>for v=</code>	<code>begin end [increment] ... efor</code>
Conditional	<code>if</code>	<code>condition ... [else] ... eif</code>
Polar coordinate (x)	<code>x=polarx</code>	<code>x y radius angle</code>
Polar coordinate (y)	<code>y=polary</code>	<code>x y radius angle</code>
Polar coordinates	<code>value=polar</code>	<code>x y radius angle</code>
Geo coordinates	<code>value=geocoord</code>	<code>geo: URL or lat long pair</code>
Area	<code>value=area</code>	<code>expression</code>
Formatted text	<code>value=format</code>	<code>fmt expression or up to 5 args</code>
Substring	<code>value=substr</code>	<code>string begin end</code>
Random number	<code>value=random</code>	<code>min max</code>
Value mapping	<code>value=vmap</code>	<code>data min1 max1 min2 max2</code>
Define function	<code>def</code>	<code>name arg1 ... argn ... edef</code>
Import function	<code>import</code>	<code>"file"</code>
In-line data	<code>data</code>	<code>"file" ["plain"] ... edata</code>
Objects on a grid	<code>grid</code>	<code>"file" x y hspace vspace edge</code>
Rulers	<code>ruler/georuler</code>	<code>increment [color]</code>

```
for v=begin end [increment]  
...items to repeat using v  
efor
```

```
for v=begin end increment ...efor
```

```
for v=10 35 5  
  hline 50 v 30 0.1 "blue"  
  star 60 v 5 1 0.4 "red"  
  circle 65 v 2 "green"  
  square 70 v 2 "blue"  
efor
```



if condition

...statements when true

else

...statements when false

endif

} else block
may be omitted

if condition ... else ... endif

Condition

Description

if a == b

if a equals b

if a != b

if a not equal to b

if a > b

if a less than b

if a < b

if a greater than b

if a >= b

if a greater than or equal to b

if a <= b

if a less than or equal to b

if a >< b c

if a is between b and c

a=60

b=3

if a > b

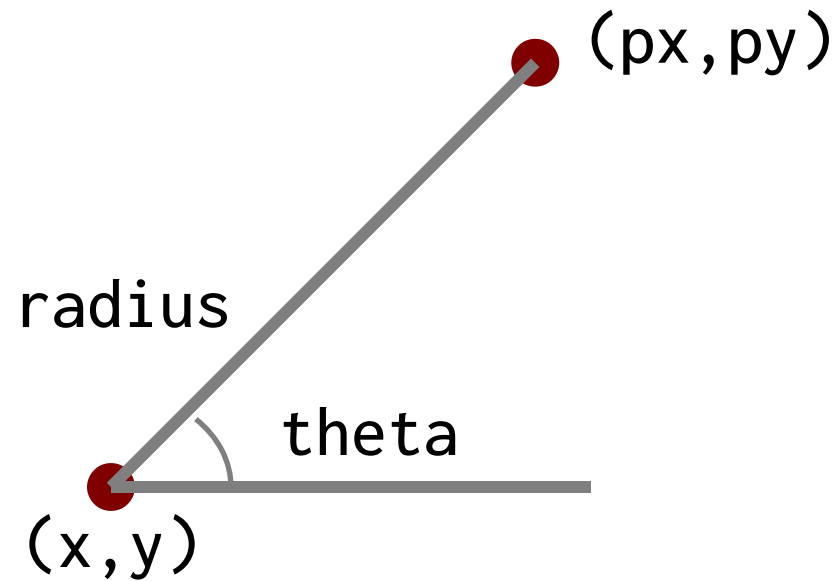
text "hello" a b 2.5 "sans" "red"

else

text "bye" a b 2.5 "sans" "blue"

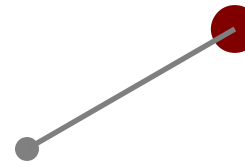
endif

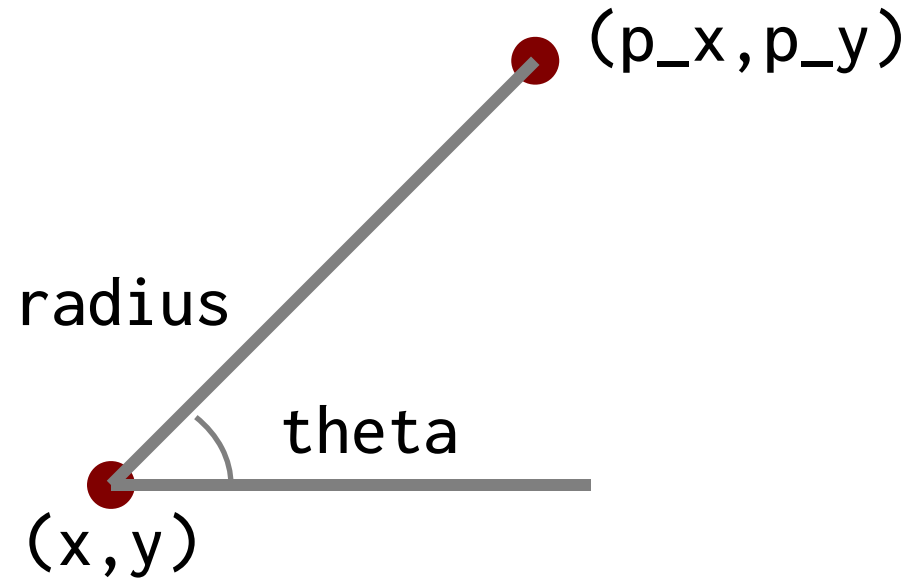
hello



```
px=polarx x y radius theta  
py=polary x y radius theta
```

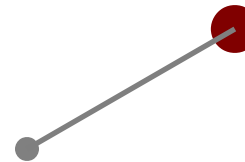
```
cpx=60  
cpy=20  
px1=polarx cpx cpy 10 30  
py1=polary cpx cpy 10 30  
line cpx cpy px1 py1  
circle cpx cpy 1 "gray"  
circle px1 py1 2 "maroon"
```





`p=polar x y radius theta`

```
cpx=60  
cpy=20  
point=polar cpx cpy 10 30  
line cpx cpy point_x point_y  
circle cpx cpy 1 "gray"  
circle point_x point_y 2 "maroon"
```



v=123.45

a=area v



area



original value

value=**area** expression

```
m1=100
```

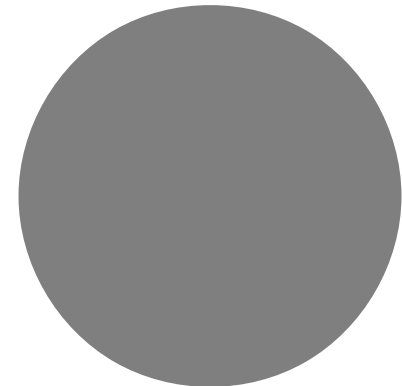
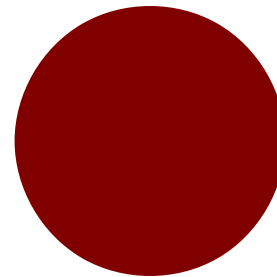
```
m2=200
```

```
a1=area m1
```

```
a2=area m2
```

```
circle 60 20 a1 "maroon"
```

```
circle 80 20 a2
```



x=3.14159

y=2.0

title=format "Value=%.2f" x*y
Value=6.28 format string expression

value=**format** fmt expression or up to 5 args

v1=100.3

v2=200.234

pi=3.1415926

title=format "%.2f Million (USD)" v1

subtitle=format "Total value: %.2f" v1+v2

args=format "Multiple args (%v,%v,%.3f)" 80 10 pi

ctext title 80 30 4 "sans" "maroon"

ctext subtitle 80 20 3 "sans" "gray"

ctext args 80 10 2 "mono"

100.30 Million (USD)

Total value: 300.53

Multiple args (80,10,3.142)

```
s="hello, world"
h=substr s 3 8
      ↑           ↑ ↑
"lo, wo"       begin end
```

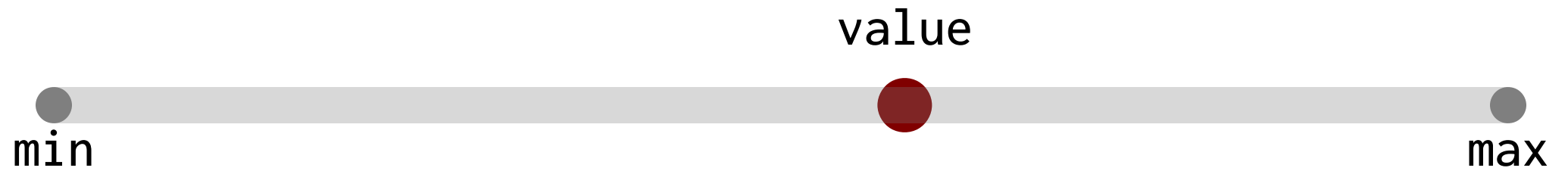
value=**substr** string begin end

```
now="Now is the time for all good men"
s1=substr now 0 14
s2=substr now 16 18
s3=substr now 24 end
ctext s1 70 34 3
ctext s2 70 24 3
ctext s3 70 15 3
```

Now is the time

for

good men



`value=random min max`



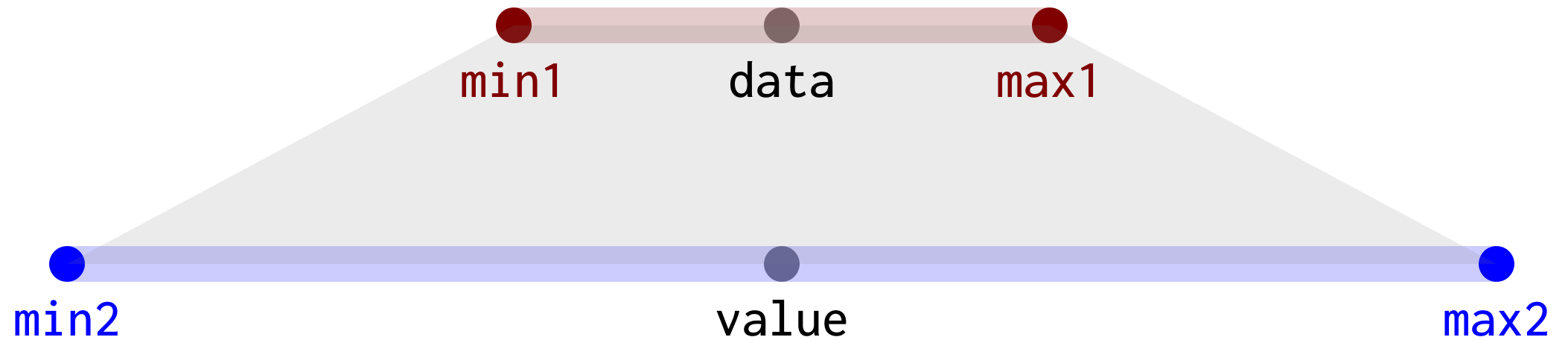
```
rx1=random 5 30  
ry1=random 15 35  
circle rx1 ry1 3 "maroon"
```



```
rx2=random 40 60  
ry2=random 15 35  
circle rx2 ry2 3 "green"
```



```
rx1=random 75 95  
ry1=random 15 35  
circle rx3 ry3 3 "blue"
```



`value=vmap data min1 max1 min2 max2`

```
yrmin=1776
yrmax=2021
smin=60
smax=90
vp=vmap 1945 yrmin yrmax smin smax
line smin 20 smax 20 0.5 "gray" 20
circle smin 20 1
circle smax 20 1
circle vp 20 2 "maroon"
```



```
import "doit.dsh"
```

contents of "doit.dsh"

```
def doit fx fy fs ft  
  ctext ft fx fy fs "serif" "purple"  
edef
```



```
doit 50 20 2.5 "hello"
```

```
fx=50  
fy=20  
fs=2.5  
ft="hello"  
ctxt ft fx fy fs "serif" "purple"
```

```
import "file"
```

calling the function
call again

```
doit 50 30 5 "calling the function"  
doit 50 20 4 "call again"
```

```
data "file.d" ← data file
first 20
second 100
third 200
edata
```

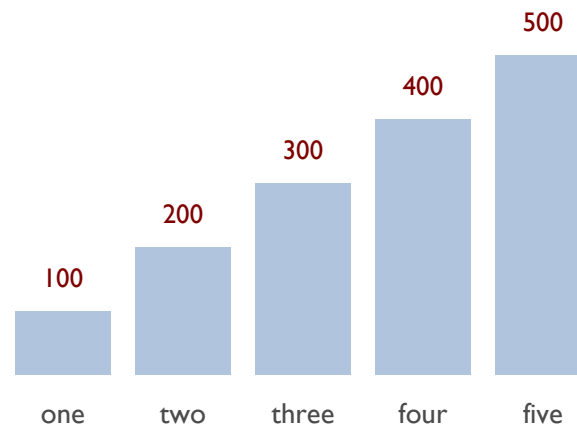
} data values

```
data "file" ... edata
```

```
data "test.d"
  one 100
  two 200
  three 300
  four 400
  five 500
```

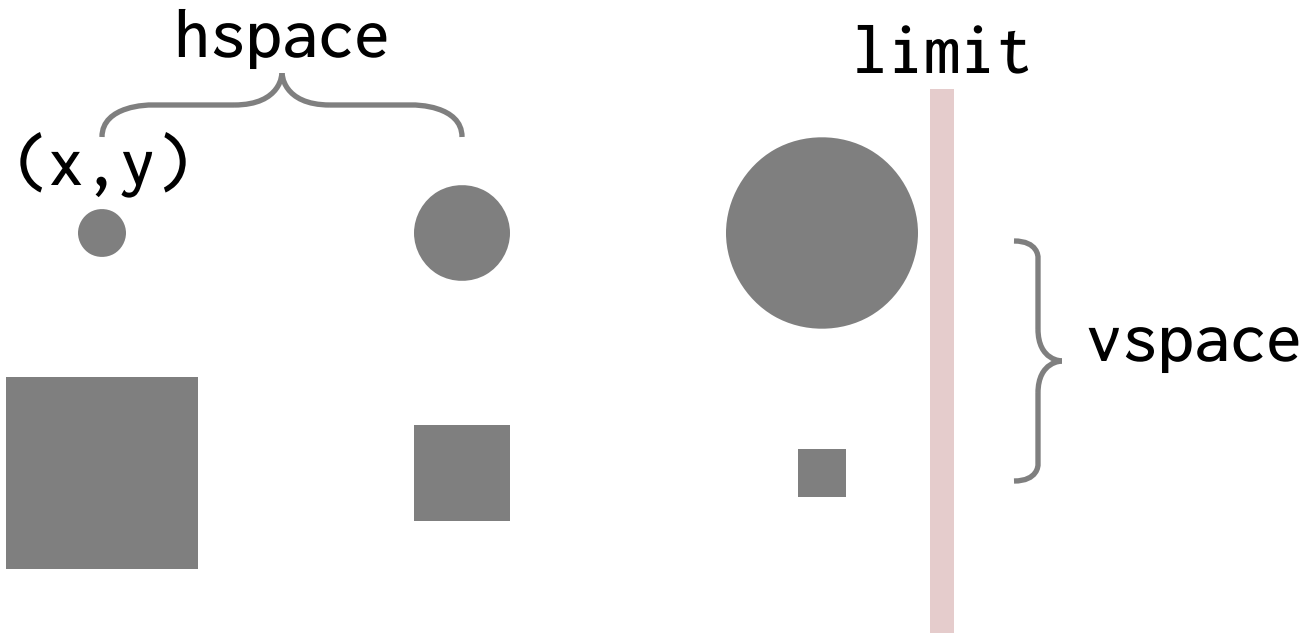
```
edata
```

```
dchart -bar -left 50 -bottom 15 -right 70 -top 35 "test.d"
```



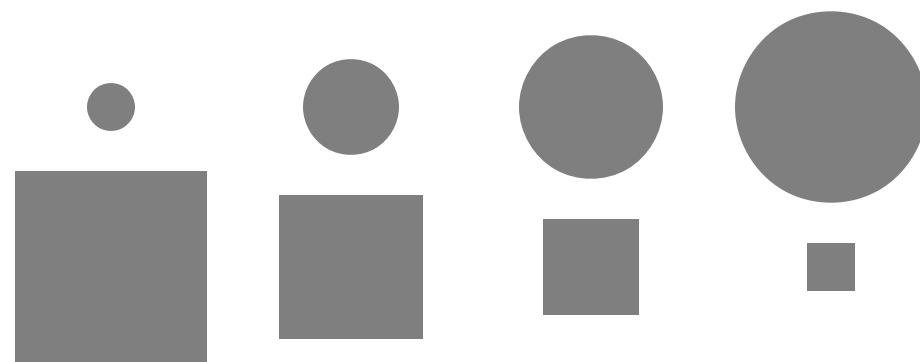
file

```
circle x y 2  
circle x y 4  
circle x y 8  
square x y 8  
square x y 4  
square x y 2
```



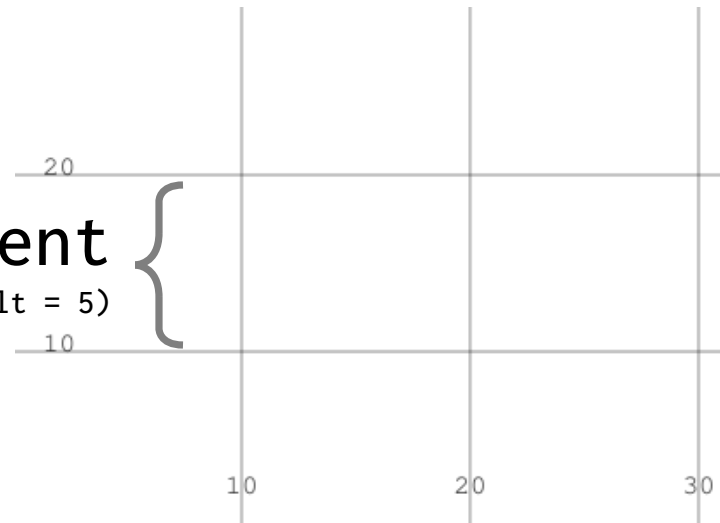
grid "file" x y hspace vspace limit

```
circle x y 2  
circle x y 4  
circle x y 6  
circle x y 8  
square x y 8  
square x y 6  
square x y 4  
square x y 2
```

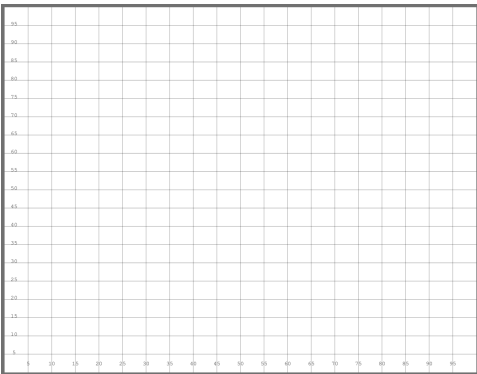


grid "code/grid-ex.dsh" 35 33 10 10 65

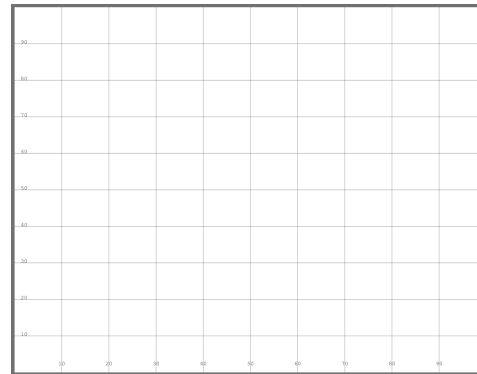
increment
(default = 5)



ruler/georuler increment color



ruler



ruler 10



ruler 2 "red"

Math Functions

description	keyword	mandatory
Cosine	cosine	number or expression
Sine	sine	number or expression
Square Root	sqrt	number or expression
Tangent	tangent	number or expression

a=3.14159265359

b=2*a

x=cosine 4 ← x = -1

y=cosine a ← y = -0.65364

n=cosine a+b ← n = -1

value=cosine number or expression

a=10

b=71

x=sin 4 ← x = -0.75680

y=sin a ← y = -0.5440

n=sin a+b ← n = -0.62989

value=sine number or expression

a=10

b=71

x=sqrt 4

← x = 2

y=sqrt a

← y = 3.1622776

n=sqrt a+b

← n = 9

value=**sqrt** number or expression

```
a_squared=10*10
```

```
b_squared=20*20
```

```
c=sqrt a_squared + b_squared
```

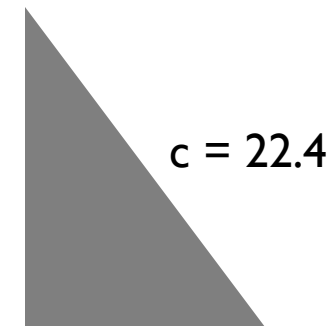
```
w=70+10
```

```
h=15+20
```

```
polygon "70 w 70" "15 15 h"
```

```
clabel=format "c = %.1f" c
```

```
text clabel 76 25 2
```



a=10

b=71

x=tangent 4 ← x = 1.1578213

y=tangent a ← y = 0.6483608

n=tangent a+b ← n = -0.8109944

value=tangent number or expression

Charts

description	keyword	mandatory	optional
General Charts	dchart	options "file"	
Chart Legends	legend	"text" x y size font color	
Chart Bounding Box	chartbbox	left	right top bottom
Area Chart	areachart	"file"	color lcolor vcolor
Bar Chart	barchart	"file"	color lcolor vcolor
Dot Chart	dotchart	"file"	color lcolor vcolor
Horizontal Bar Chart	hbarchart	"file"	color lcolor vcolor
Line Chart	linechart	"file"	color lcolor vcolor
Word Bar Chart	wbarchart	"file"	color lcolor vcolor
Slope Chart	slopechart	"file"	color lcolor vcolor
Proportional Map	pmap	"file"	size
Pie Chart	pie	"file"	size
Donut Chart	donut	"file"	width size
Proportional Grid	pgrid	"file"	size
Waffle Chart	waffle	"file"	size
Fan Chart	fanchart	"file"	size
Bowtie Chart	bowtie	"file"	size

Chart Variables

description	variable	default value
Chart Top	chartTop	80
Chart Bottom	chartBottom	30
Chart Left	chartLeft	10
Chart Right	chartRight	90
Chart text size	chartTextSize	1.5
Chart line width	chartLineWidth	0.2
Chart bar width	chartBarWidth	0 (autoscale)
Chart line spacing	chartLineSpacing	2.4
Chart Volume Opacity	chartVolOp	50
X-axis label interval	chartXLabel	1 (1: all, 0: none, every nth label)
Y-axis range	chartYRange	" " (min,max,interval)
CSV Columns	chartCSVCols	" "
Data Format	chartDataFmt	%.1f
Show % Value	chartPercent	0 (0: off, 1: on)
Show data values	chartVal	1 (0: off, 1: on)
Show Y-axis grid	chartGrid	0 (0: off, 1: on)
Show title	chartTitle	1 (0: off, 1: on)
Read CSV	chartReadCSV	0 (0: off, 1: on)
Show last X-axis label	chartXLast	0 (0: off, 1: on)

dchart options: chart types

option	default	description
-bar	true	bar chart
-wbar	false	word bar chart
-hbar	false	horizontal bar chart
-donut	false	donut chart
-dot	false	dot chart
-lego	false	lego chart
-line	false	line chart
-pgrid	false	proportional grid
-pmap	false	proportional map
-bowtie	false	bowtie chart
-fan	false	fan chart
-radial	false	radial chart
-scatter	false	scatter chart
-slope	false	slope chart
-vol	false	volume (area) chart

dchart options: elements

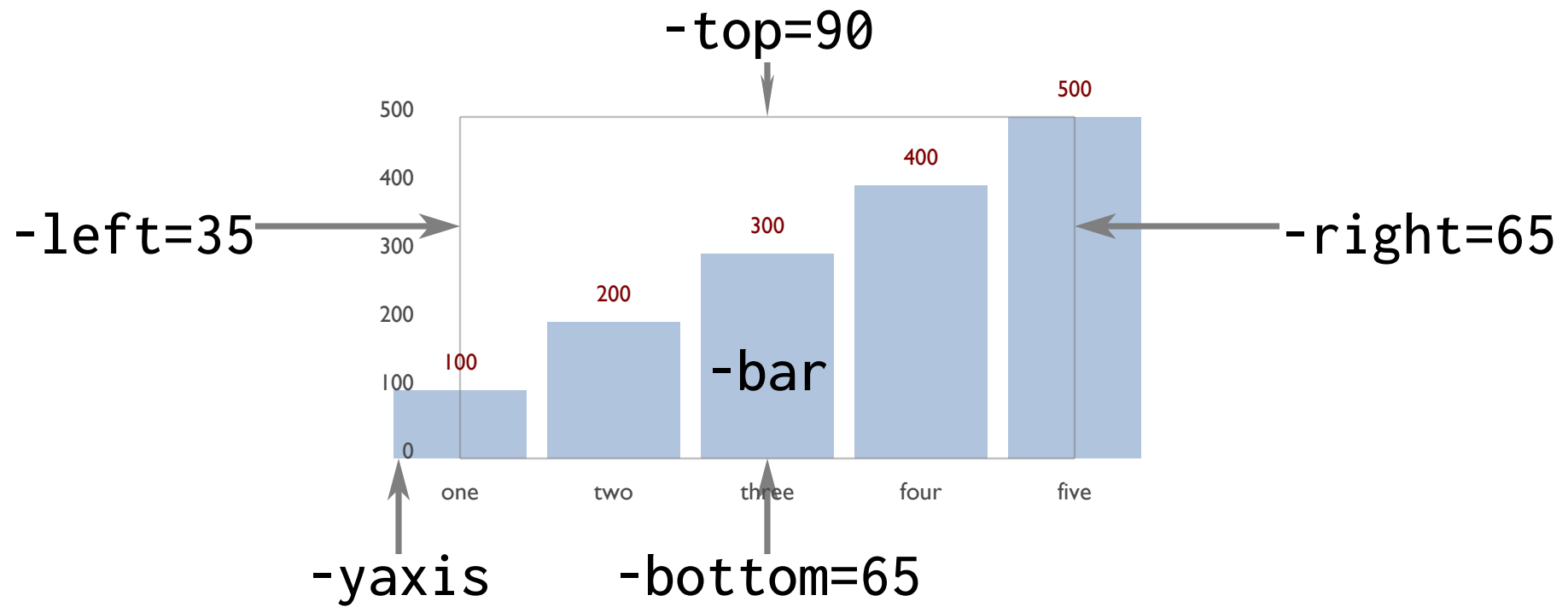
option	default	description
-csv	false	read CSV files
-frame	false	show a colored frame
-fulldeck	true	generate full deck markup
-grid	false	show gridlines on the y axis
-note	true	show annotations
-pct	false	show computed percentage
-rline	false	show a regression line
-solidpmap	false	show solid pmap colors
-spokes	false	show spokes in radial chart
-title	true	show the title
-val	true	show values
-xlast	false	show the last x label
-xstagger	false	stagger x axis labels
-yaxis	false	show a y axis
-chartitle	override title in data	specify the title
-datacond	low,high,color	conditional data colors
-hline	value,label	label horizontal line at value
-valpos	t=top, b=bottom, m=middle	value position
-xlabel	default=1, 0 to suppress	x axis label interval
-yrange	min,max.step	specify the y axis label range

dchart options: measures and attributes

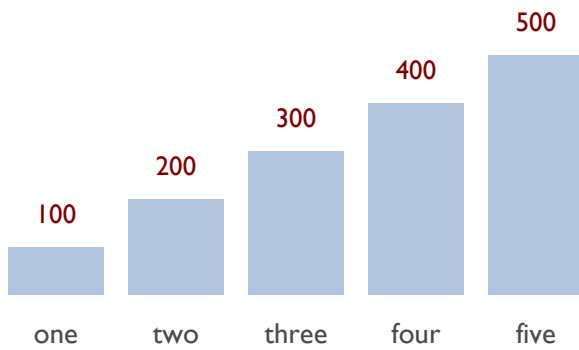
option	default	description
-bgcolor	white	background color
-barwidth	computed from data size	barwidth
-color	lightsteelblue	data color
-csvcol	label1,label2	specify csv columns
-datafmt	%.1f	data format for values
-dmin	false	use data minimum, not zero
-framecolor	rgb(127,127,127)	frame color
-lcolor	rgb(75,75,75)	label color
-linewidth	0.2	linewidth
-ls	2.4	linespacing
-noteloc	c=center, r=right, l=left	annotation location
-pmlen	20	pmap label length
-psize	30	diameter of the donut
-pwidth	3	width of the donut or pmap
-rlcolor	rgb(127,0,0)	regression line color
-textsize	1.5	text size
-xlabrot	0	xlabel rotation (deg.)
-vcolor	rgb(127,0,0)	value color
-volop	50	volume opacity %

dchart options: position and scaling

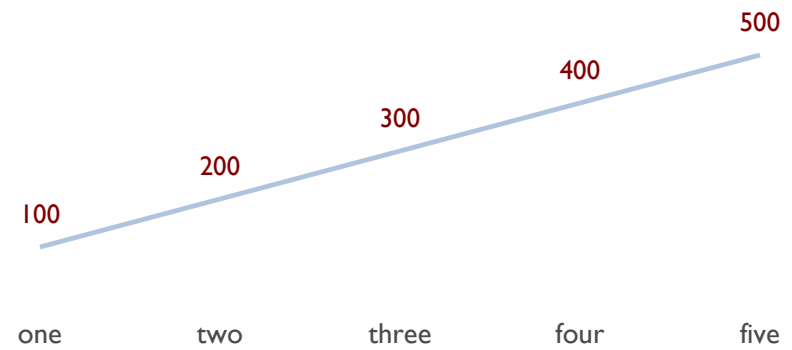
option	default	description
-top	80	top of the chart
-bottom	30	bottom of the chart
-left	20	left margin
-right	80	right margin
-min	data min	set the minimum data value
-max	data max	set the maximum data value
-bounds	""	set left,right,top,bottom



dchart options "file"



```
dchart -left=10 -right=30 -top=35 -bottom=20 "test.d"
```



```
dchart -left=55 -right=85 -top=35 -bottom=20 -bar=f -line "test.d"
```



■ My text
(x, y)

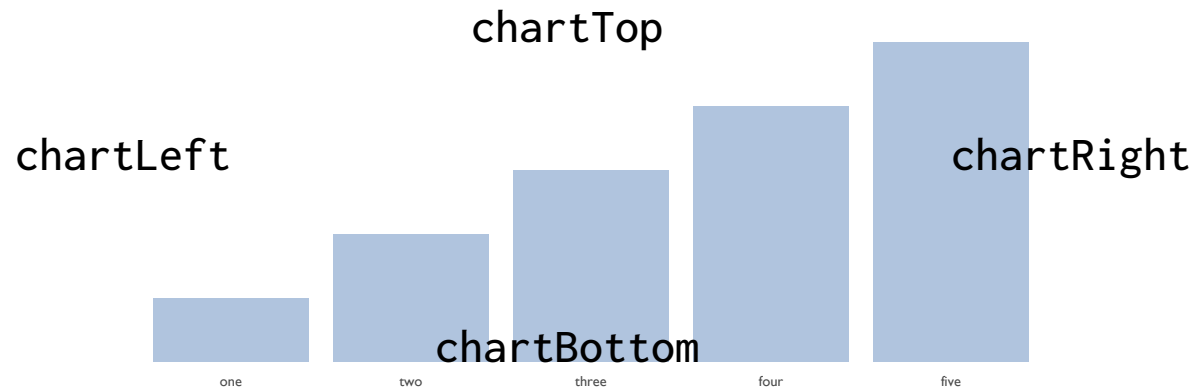
legend "text" x y fontsize font color

■ Item on the chart

■ Thing

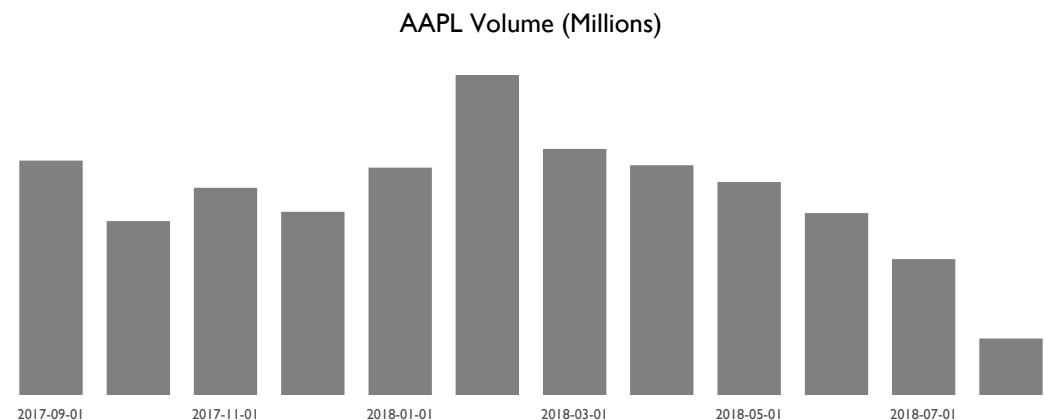
```
legend "Item on the chart" 20 30 3 "sans" "red"
```

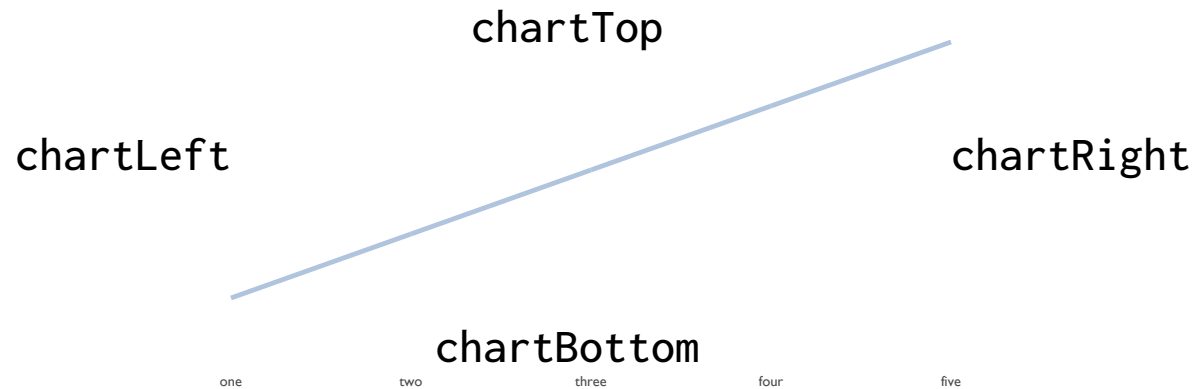
```
legend "Thing" 70 30 2 "serif" "blue"
```



barchart "file" color lcolor vcolor

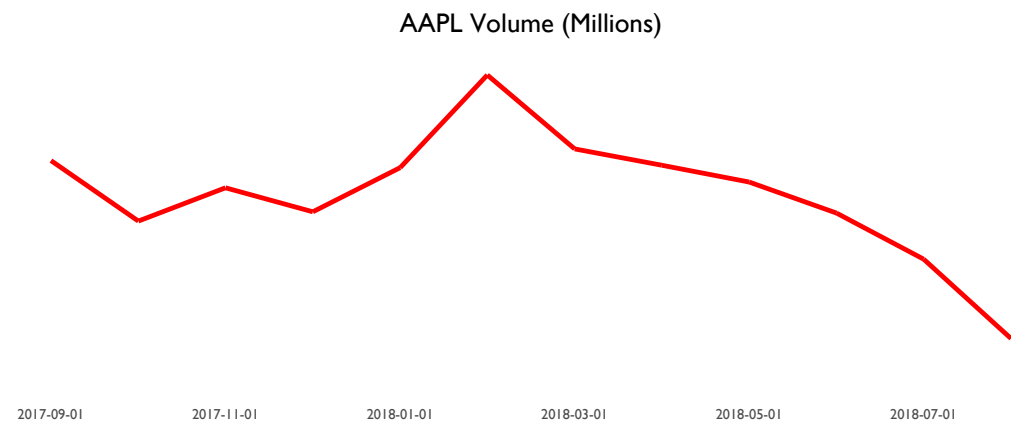
```
chartLeft=50
chartRight=90
chartTop=30
chartBottom=10
chartXLabel=2
barchart "AAPL.d" "gray"
```

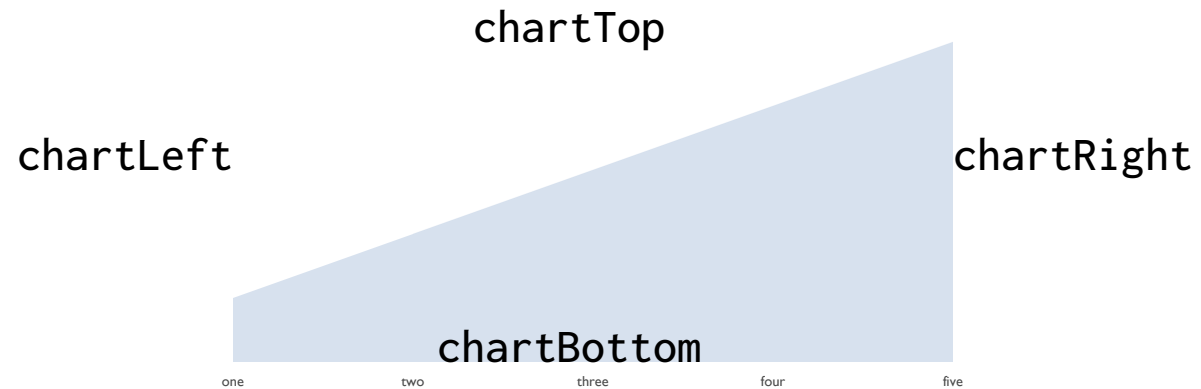




`linechart "file" color lcolor vcolor`

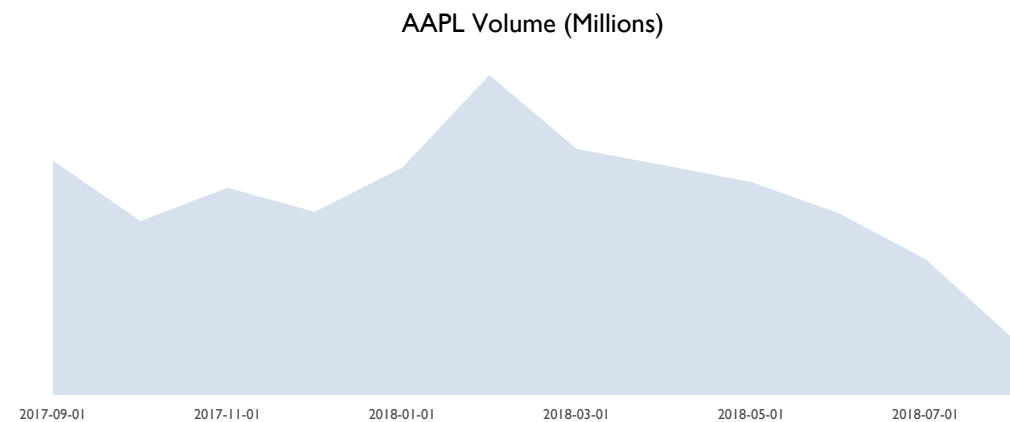
```
chartLeft=50
chartRight=90
chartTop=30
chartBottom=10
chartXLabel=2
linechart "AAPL.d" "red"
```

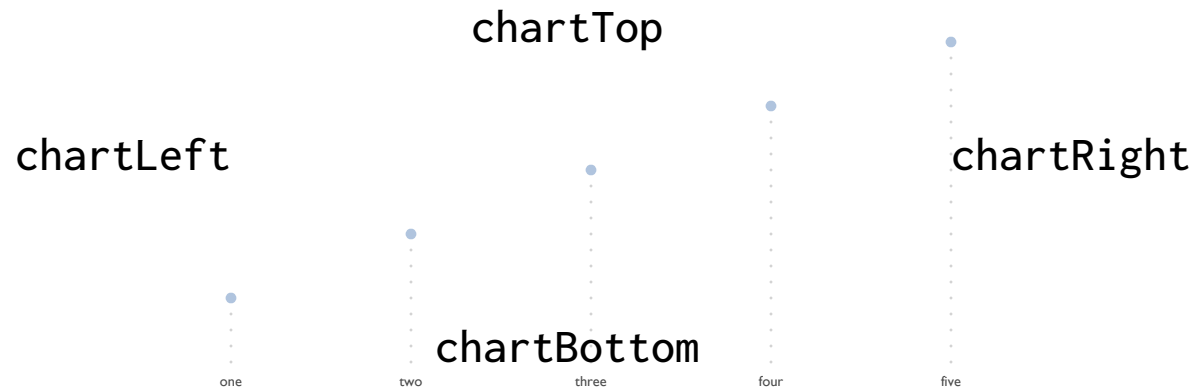




areachart "file" color lcolor vcolor

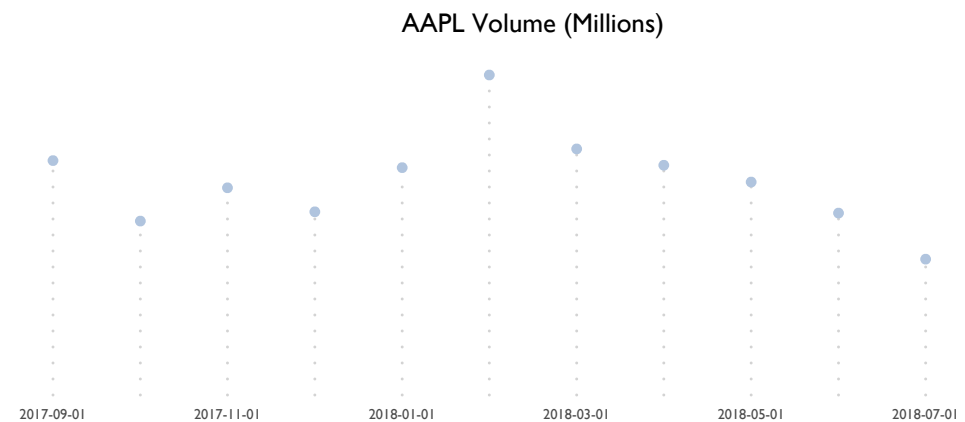
```
chartLeft=50
chartRight=90
chartTop=30
chartBottom=10
chartXLabel=2
areachart "AAPL.d"
```

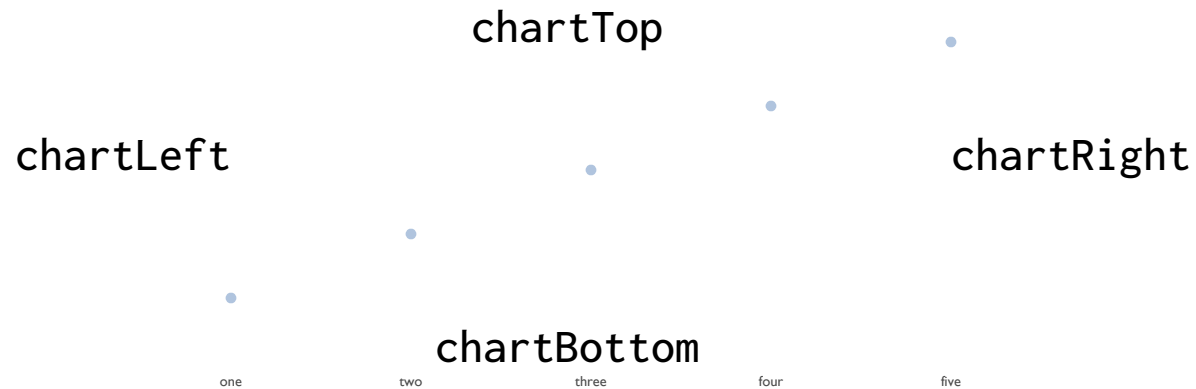




`dotchart "file" color lcolor vcolor`

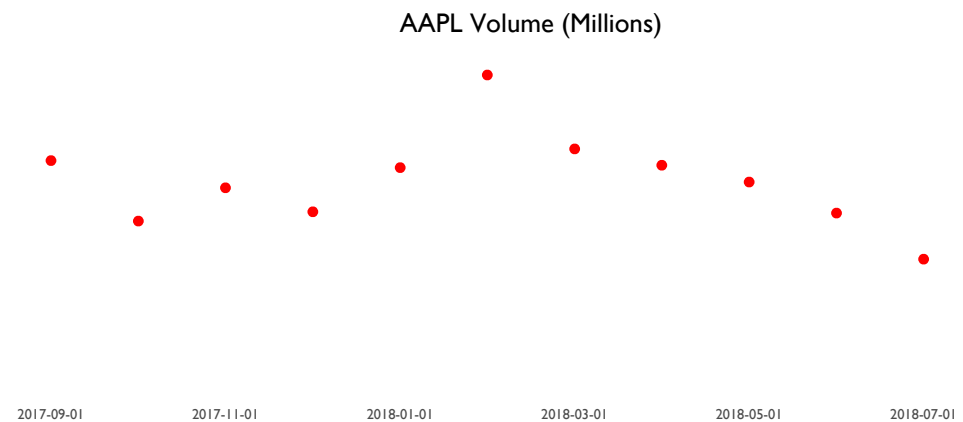
```
chartLeft=50
chartRight=90
chartTop=30
chartBottom=10
chartXLabel=2
dotchart "AAPL.d"
```

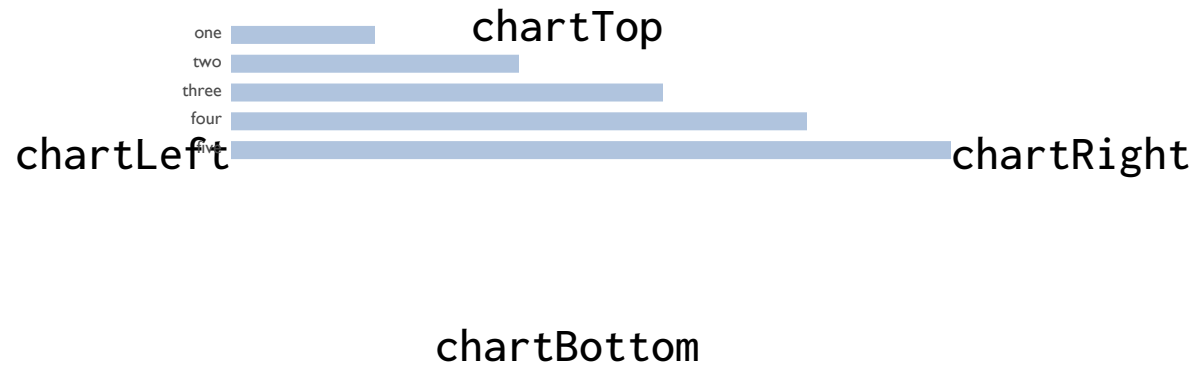




scatterchart "file" color lcolor vcolor

```
chartLeft=50
chartRight=90
chartTop=30
chartBottom=10
chartXLabel=2
scatterchart "AAPL.d" "red"
```

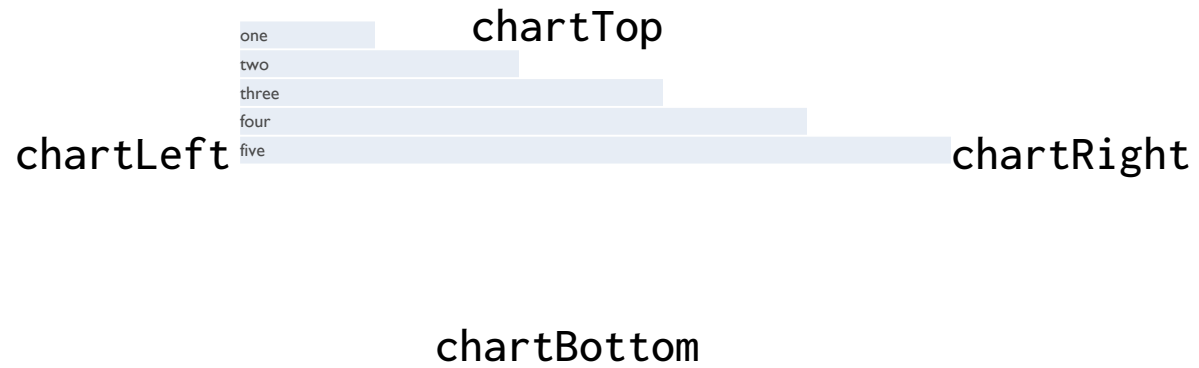




hbarchart "file" color lcolor vcolor

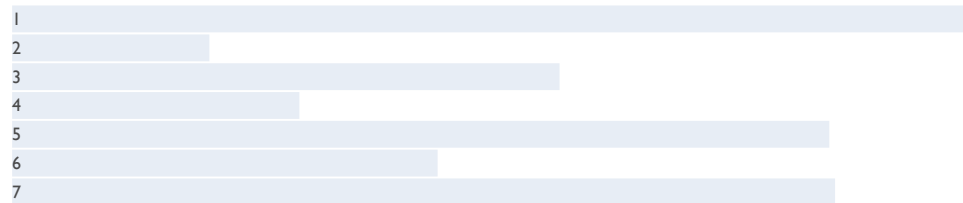
```
chartLeft=50
chartRight=90
chartTop=30
chartBottom=10
chartXLabel=2
hbarchart "rand.d" "maroon"
```

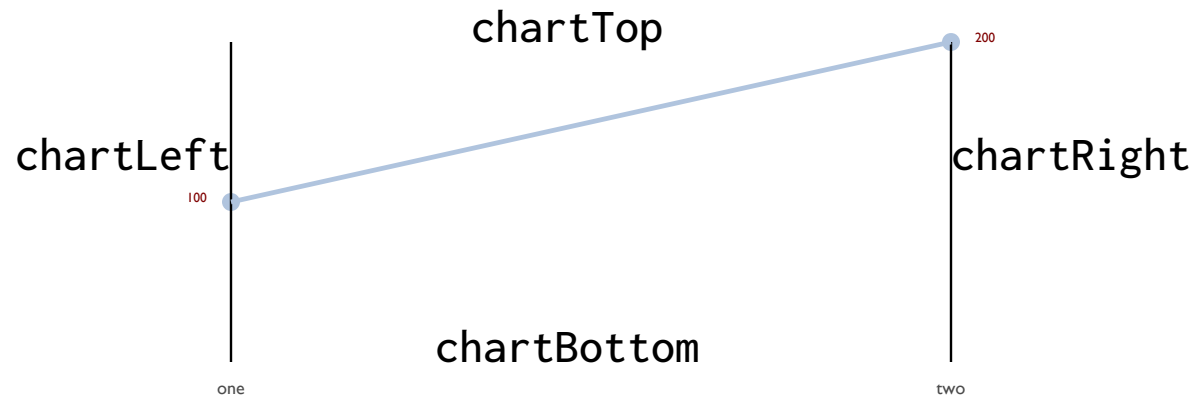




wbarchart "file" color lcolor vcolor

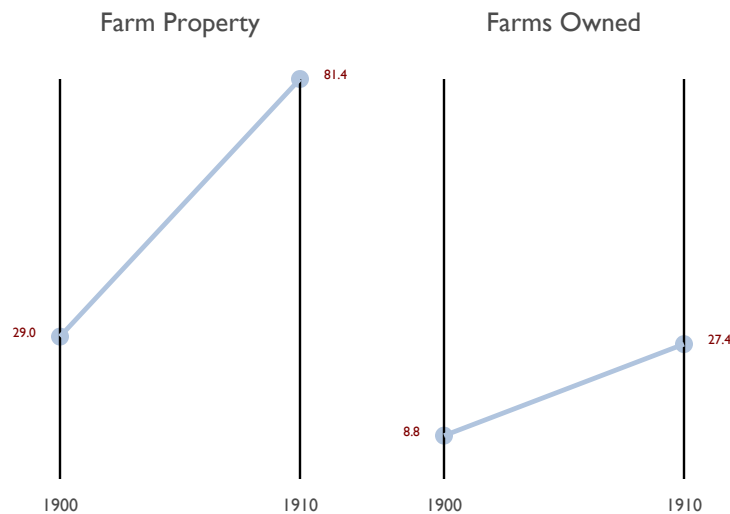
```
chartLeft=50
chartRight=90
chartTop=30
chartBottom=10
chartXLabel=2
wbarchart "rand.d"
```



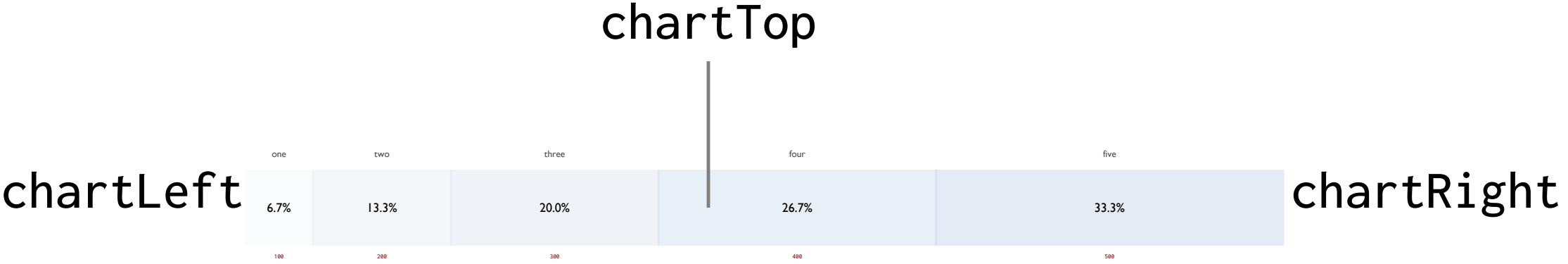


`slopechart "file" color lcolor vcolor`

```
chartLeft=40
chartRight=50
chartTop=35
chartBottom=10
slopechart "slopex.d"
```

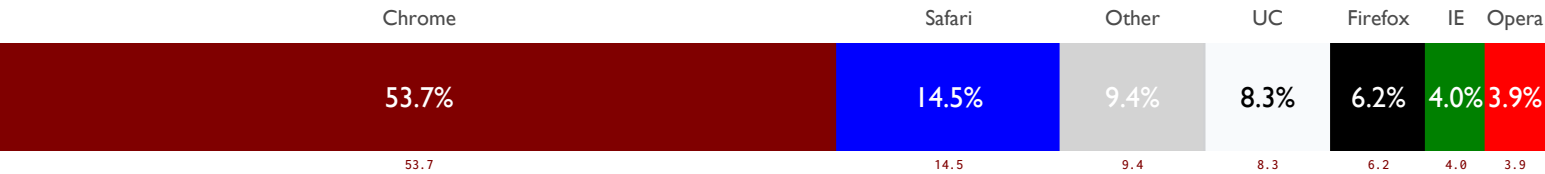


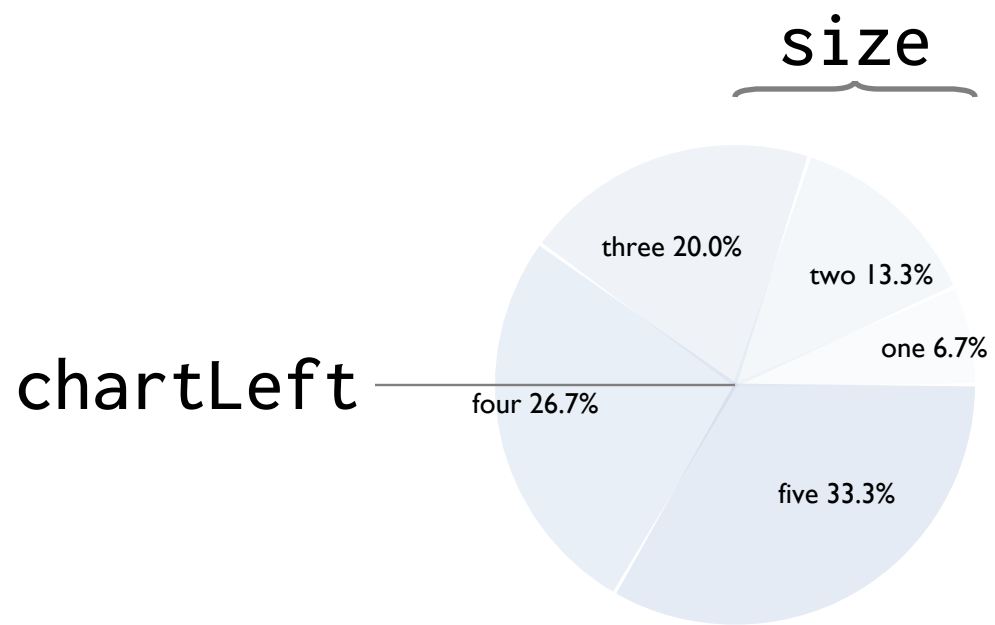
1900	28.99	Farm Property
1910	81.42	
1900	8.83	Farms Owned
1910	27.45	



pmap "file" size

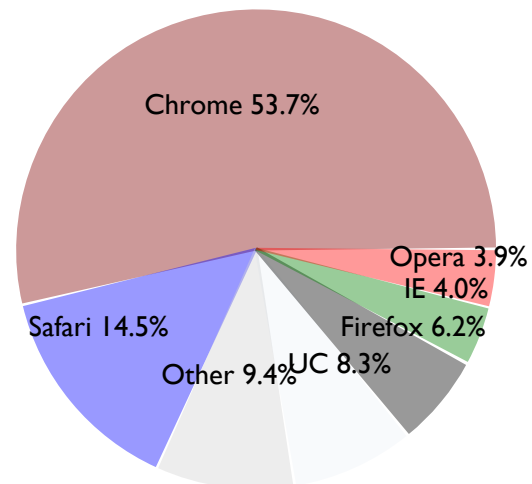
chartLeft=25
chartTop=25
chartRight=90
chartTitle=0
chartTextSize=1.2
pmap "browser.d"



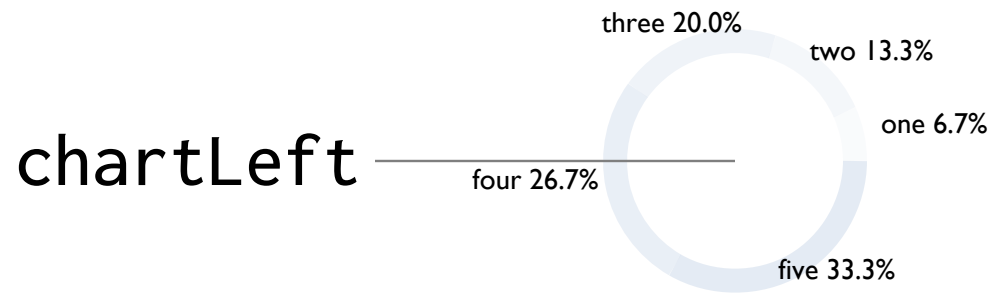


pie "file" size

```
chartLeft=50  
chartTop=25  
chartRight=95  
chartTextSize=1.2  
chartTitle=0  
pie "browser.d" 10
```

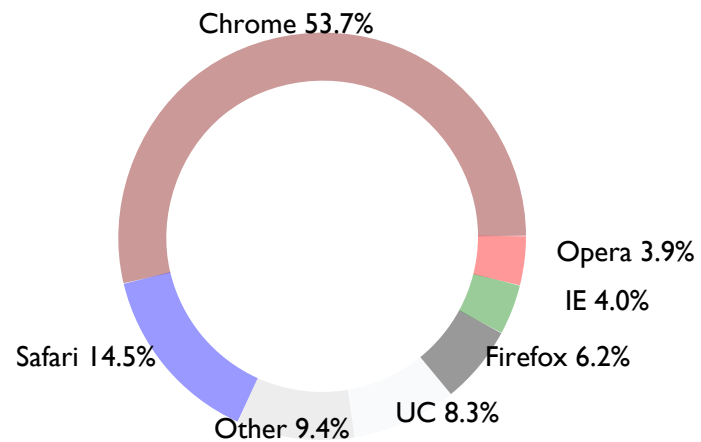


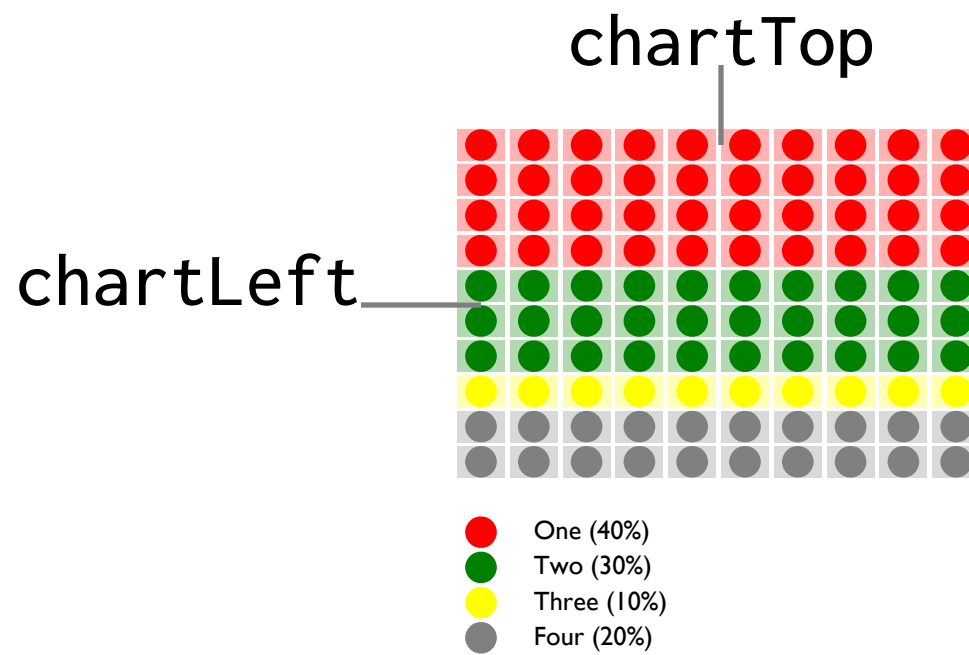
size



donut "file" size width

```
chartLeft=50
chartTop=30
chartRight=60
chartTextSize=1.2
chartTitle=0
donut "browser.d" 2 15
```

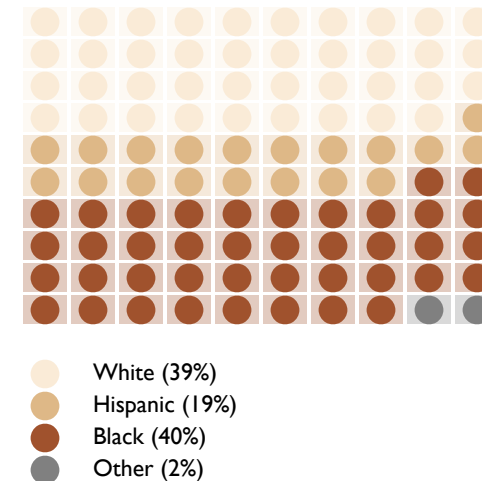


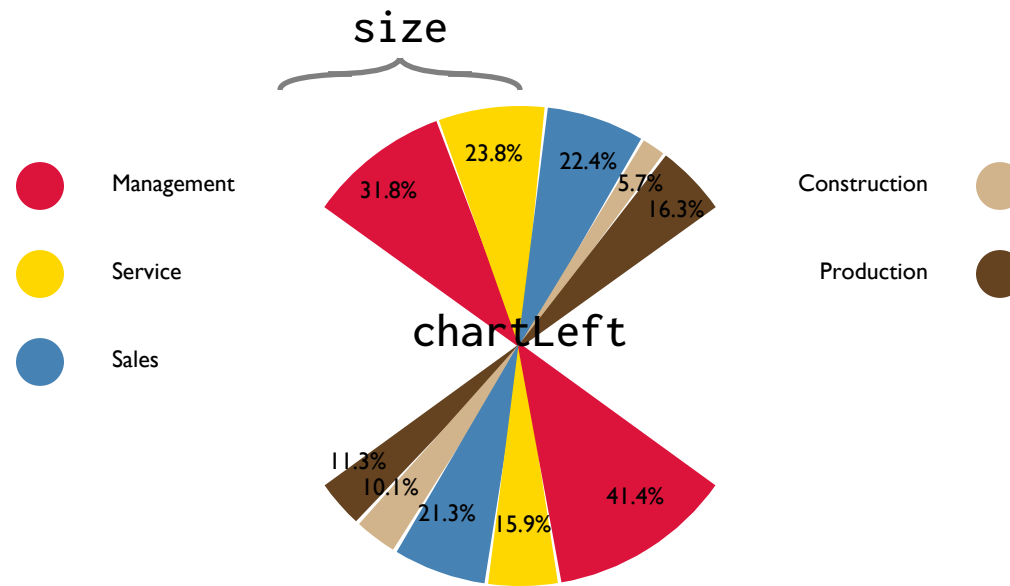


lego "file"

```
chartLeft=65
chartTop=35
chartTextSize=2
lego "incarcer.d"
```

```
# US Incarceration Rate
White      39      antiquewhite
Hispanic   19      burlywood
Black      40      sienna
Other       2      gray
```





fanchart "file" size

chartLeft=50

chartTop=80

chartVal=1

chartTextSize=1

fanchart "occupation.d" 10

Occupations of African-Americans and Whites (2019)

Management 31.8 crimson

Service 23.8 gold

Sales 22.4 steelblue

Construction 5.7 tan

Production 16.3 rgb(101,67,33)

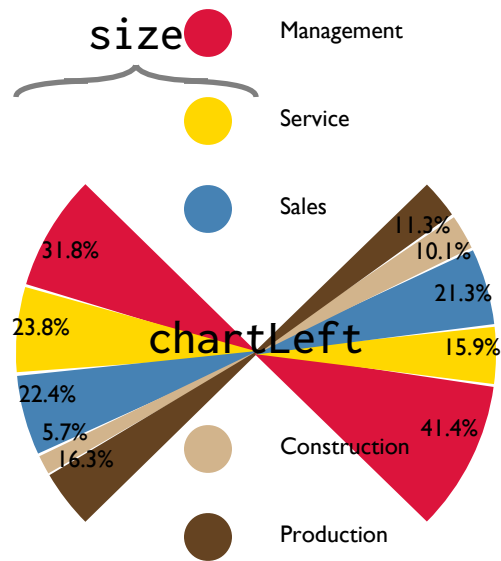
Management 41.4 crimson

Service 15.9 gold

Sales 21.3 steelblue

Construction 10.1 tan

Production 11.3 rgb(101,67,33)



bowtie "file" size

```
chartLeft=50
chartTop=80
chartVal=1
chartTextSize=1
bowtie "occupation.d" 10
```

Occupations of African-Americans and Whites (2019)

```
Management 31.8 crimson
Service 23.8 gold
Sales 22.4 steelblue
Construction 5.7 tan
Production 16.3 rgb(101,67,33)
Management 41.4 crimson
Service 15.9 gold
Sales 21.3 steelblue
Construction 10.1 tan
Production 11.3 rgb(101,67,33)
```

Geographic Functions

description	keyword	mandatory	optional
Geographic Regions	georegion	"file"	color op
Geographic Borders	geoborder geoline	"file"	lw color op
Text labels	geolabel	"loc"	size font color op
Dot markers	geomark	"loc"	size color op
Text with markers	geoloc	"loc"	align size font color op
Place images	geoimage	"loc" w h	
Lines between points	geopath	"p1" "p2"	lw color op
Arcs between points	geoarc	"p1" "p2"	lw color op
Lines between points	geopathfile	"file"	lw color op gap
Geographic boundaries	geobbox geobound	latmin or "p1" "p2"	latmax longmin longmax
Geographic canvas	geocanvas	xmin	xmax ymin ymax

"file" refer to Shapefile (.shp), GeoJSON (.json or .geojson) or Keyhole Markup Language (.kml) files from [opendatasoft](#)

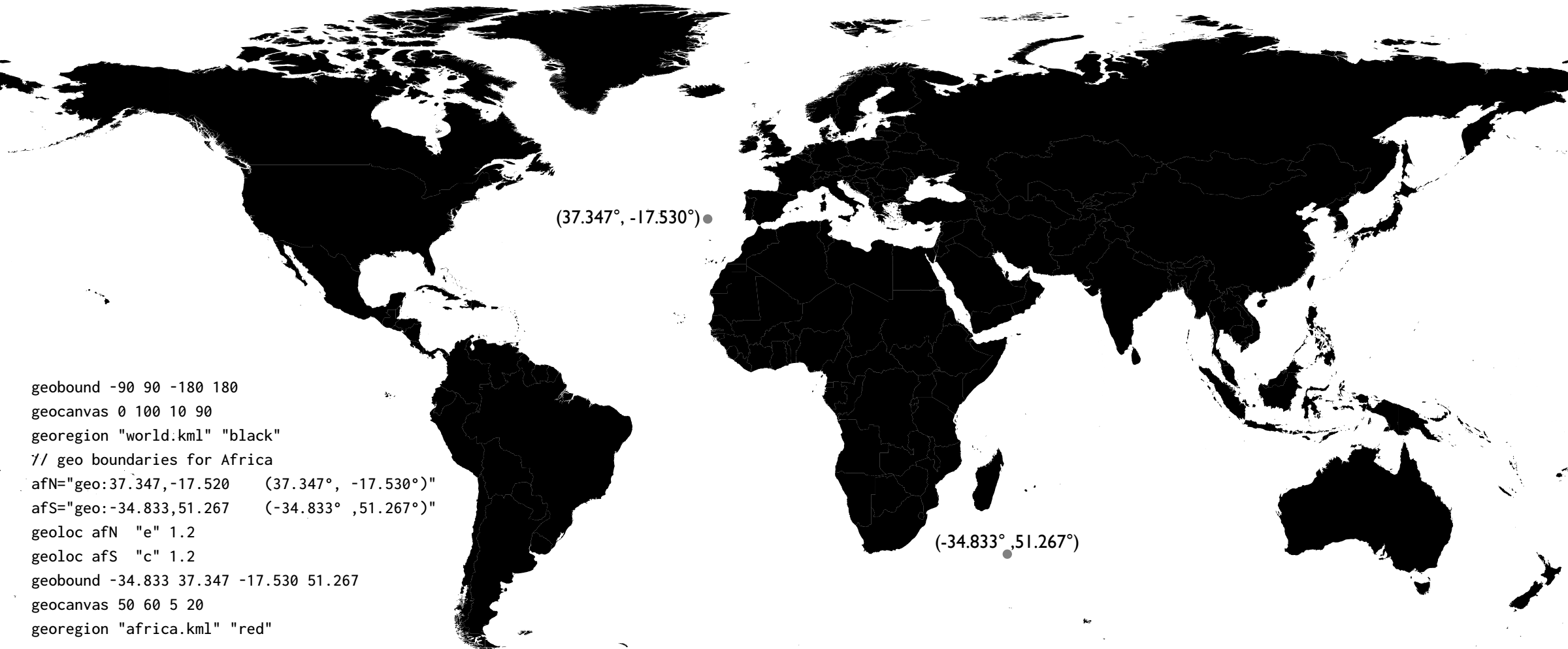
"loc" refers to a geo URI string ("geo:lat,long"), or a filename containing multiple locations

"p1" and "p2" are geo URIs

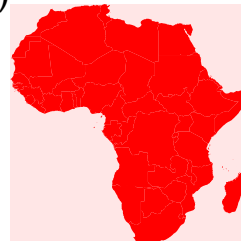
Geo Variables

description	variable	default value
Minimum latitude	geoLatMin	-90
Maximum latitude	geoLatMax	90
Minimum longitude	geoLongMin	-180
Maximum longitude	geoLongMax	180
Geo canvas x minimum	geoXmin	0
Geo canvas x maximum	geoXmax	100
Geo canvas y minimum	geoYmin	0
Geo canvas y maximum	geoYmax	100

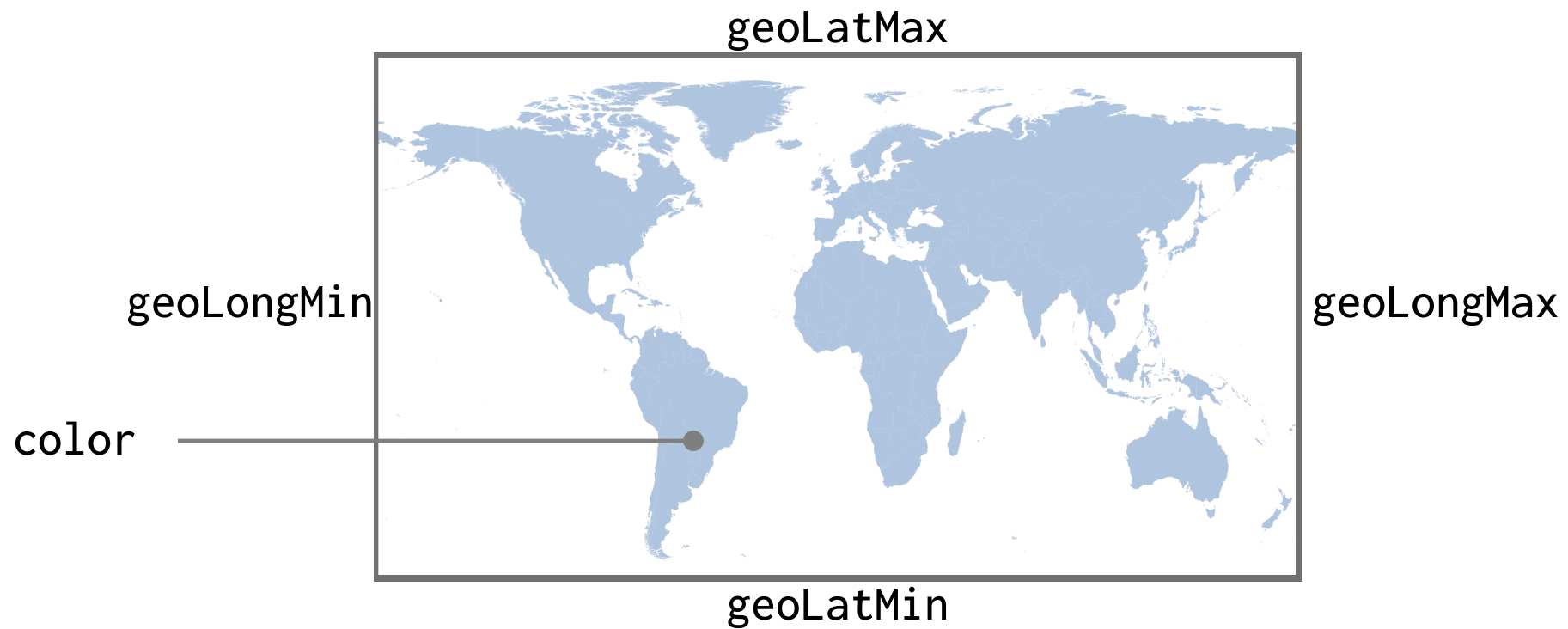
Geographic Scaling



(50,20)



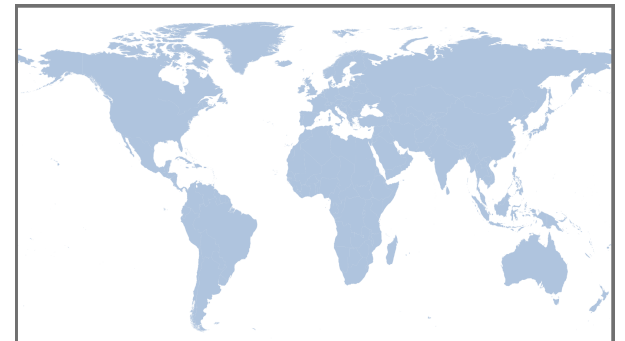
(60,5)

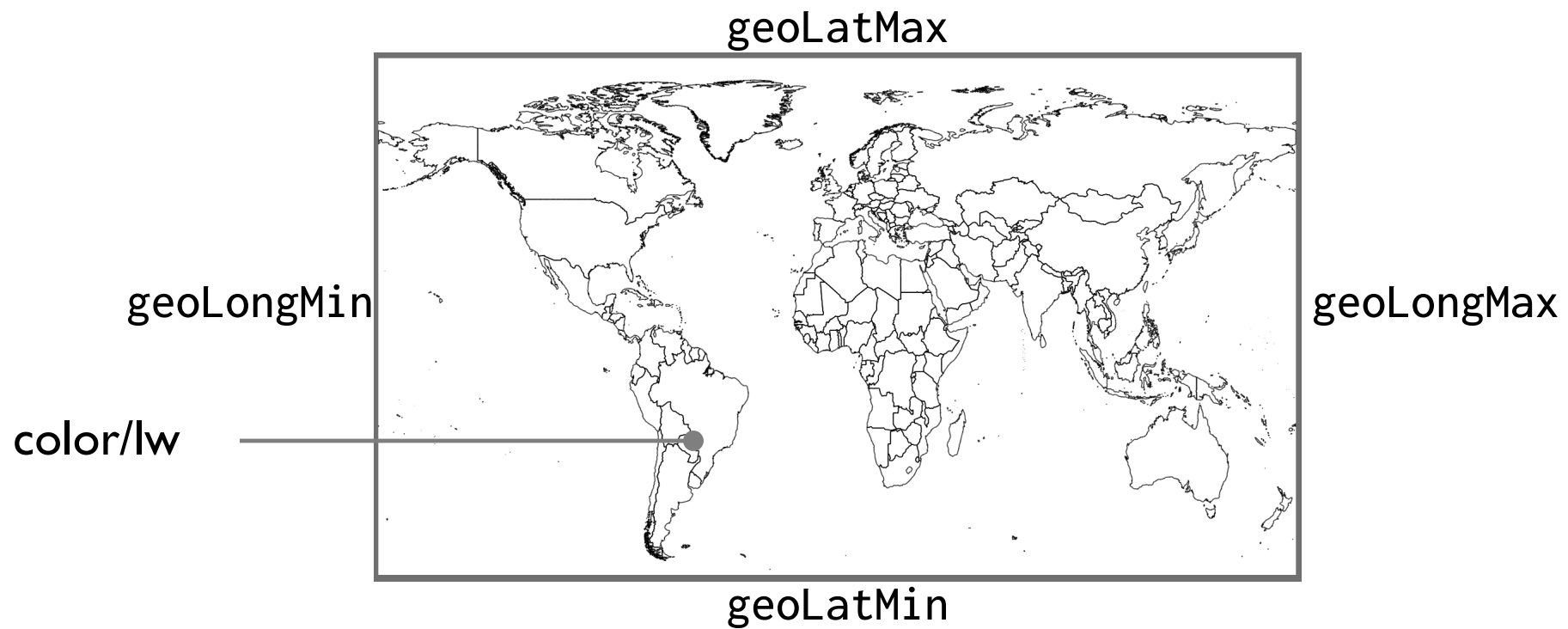


`georegion "file" color op`

`geoLatMin=0-60`
`geoLatMax=90`
`geoLongMin=0-180`
`geoLongMax=180`

`georegion "world.kml" "white"`

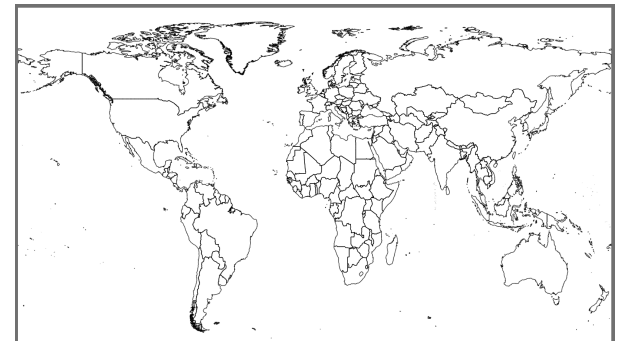




`geoborder "file" lw color`

`geoLatMin=0-60`
`geoLatMax=90`
`geoLongMin=0-180`
`geoLongMax=180`

`geoborder "world.kml" 0.1 "black"`

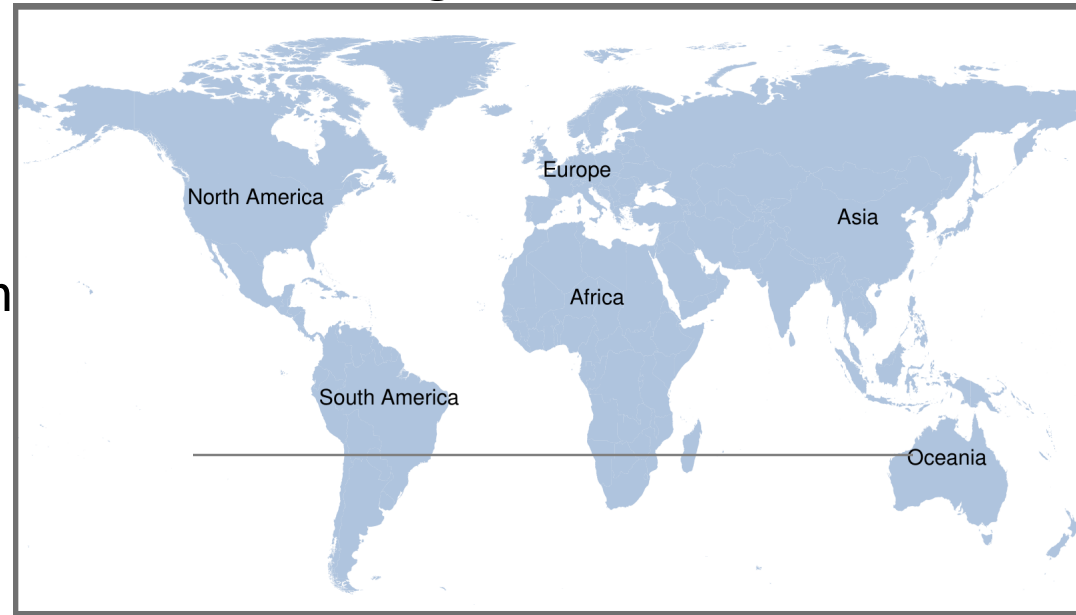


geoLatMax

geoLongMin

geoLongMax

geo:lat,long<tab>Label



geoLatMin

geolabel "loc" size color op

geoLatMin=0-60

geoLatMax=90

geoLongMin=0-180

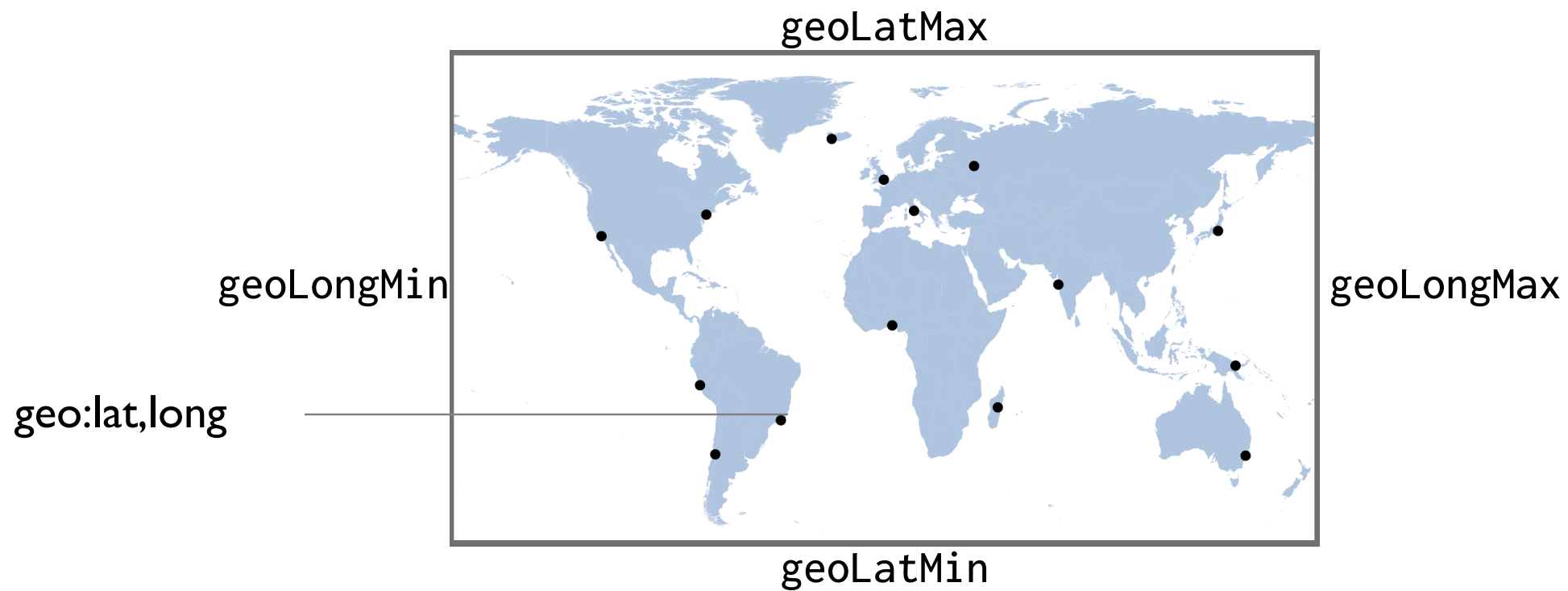
geoLongMax=180

geo:-10,-55	South America
geo:40,-100	North America
geo:46.8,8.3	Europe
geo:15,15	Africa
geo:35,103	Asia
geo:-25,133	Oceania

georegion "world.kml" "white"

geolabel "continents.d" 2 "sans" "black"





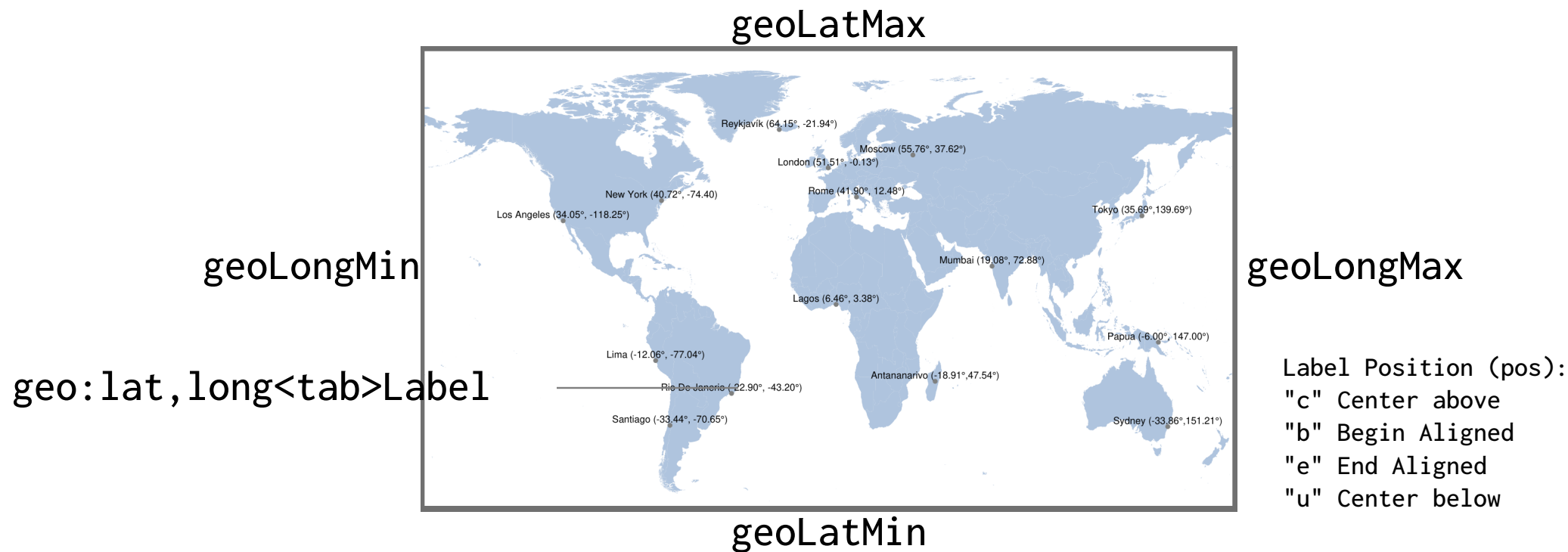
`geomark "loc" size color op`

`geoLatMin=0-60`
`geoLatMax=90`
`geoLongMin=0-180`
`geoLongMax=180`

<code>geo:41.8967,12.4822000</code>	Rome (41.90°, 12.48°)
<code>geo:-18.91368,47.53610</code>	Antananarivo (-18.91°, 47.54°)
<code>geo:-33.8559,151.20670</code>	Sydney (-33.86°, 151.21°)
<code>geo:40.7167,-74.400000</code>	New York (40.72°, -74.40°)
<code>...</code>	
<code>geo:34.0500,-118.25000</code>	Los Angeles (34.05°, -118.25°)

`georegion "world.kml" "white"`
`geolabel "cities.d" 2 "sans" "black"`



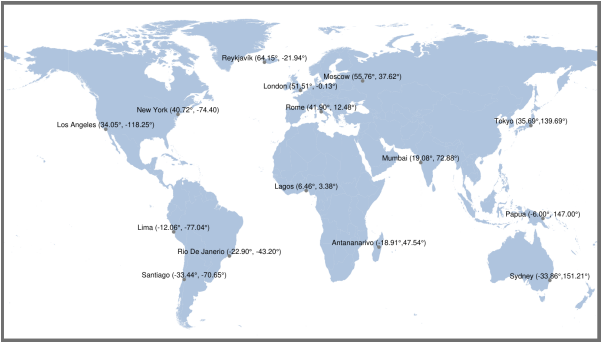


geoloc "loc" pos size font color op

geoLatMin=0-60
geoLatMax=90
geoLongMin=0-180
geoLongMax=180

geo:41.8967,12.4822000	Rome (41.90°, 12.48°)
geo:-18.91368,47.53610	Antananarivo (-18.91°,47.54°)
geo:-33.8559,151.20670	Sydney (-33.86°,151.21°)
geo:40.7167,-74.400000	New York (40.72°, -74.40)
...	
geo:34.0500,-118.25000	Los Angeles (34.05°, -118.25°)

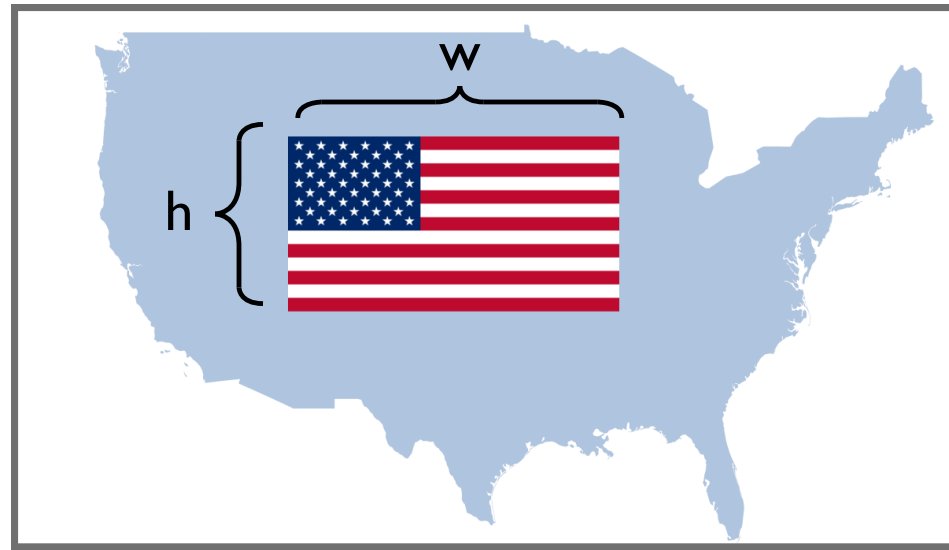
georegion "world.kml" "white"
geoloc "cities.d" "c" ts



geoLatMax

geoLongMin

geoLongMax



geo:lat,long<tab>imgfile

geoLatMin

geoimage "loc" w h

geoLatMin=25

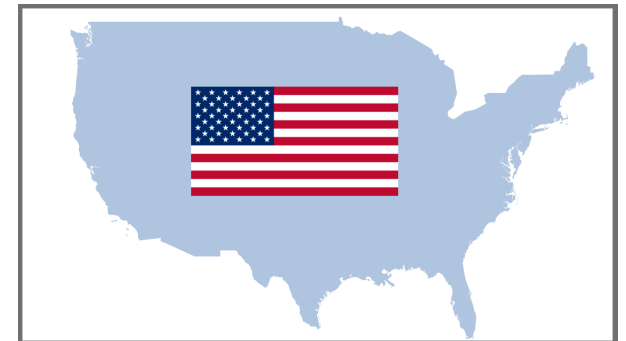
geoLatMax=50

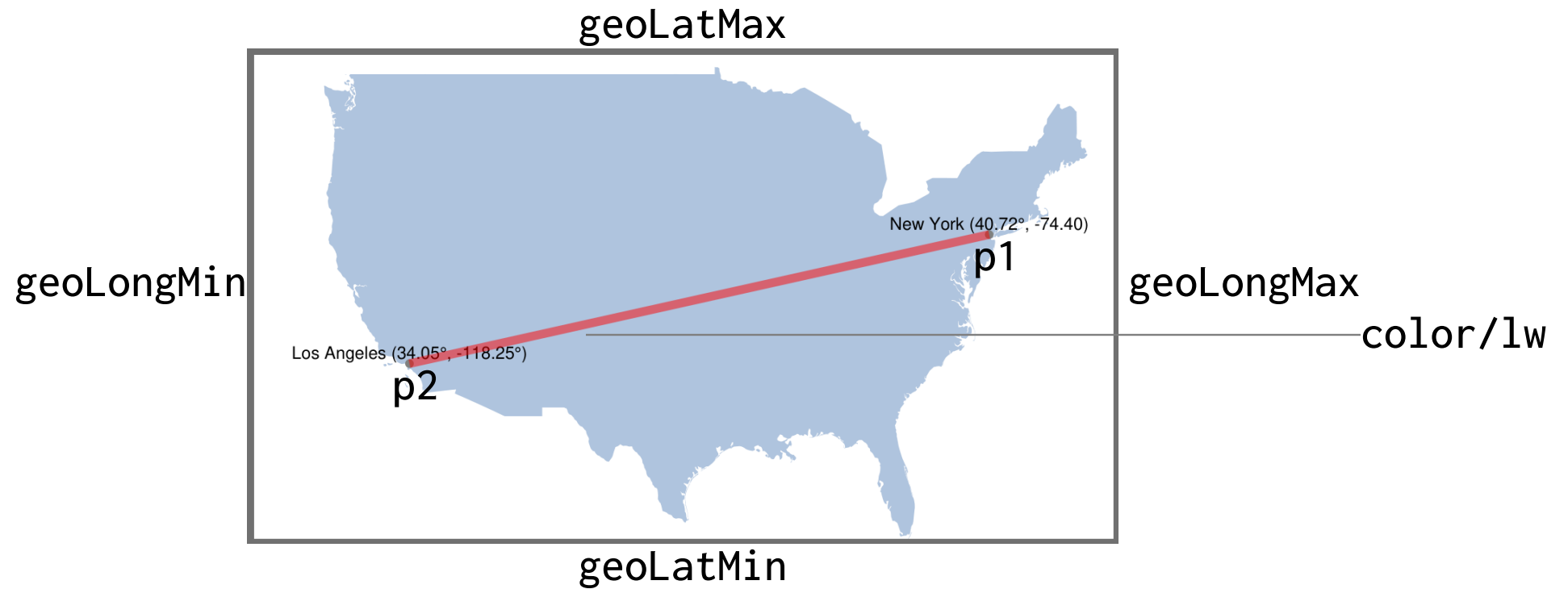
geoLongMin=0-130

geoLongMax=0-65

georegion "usa.kml" "lightsteelblue"

geoimage "geo:40,-100 usa.png" 35 0



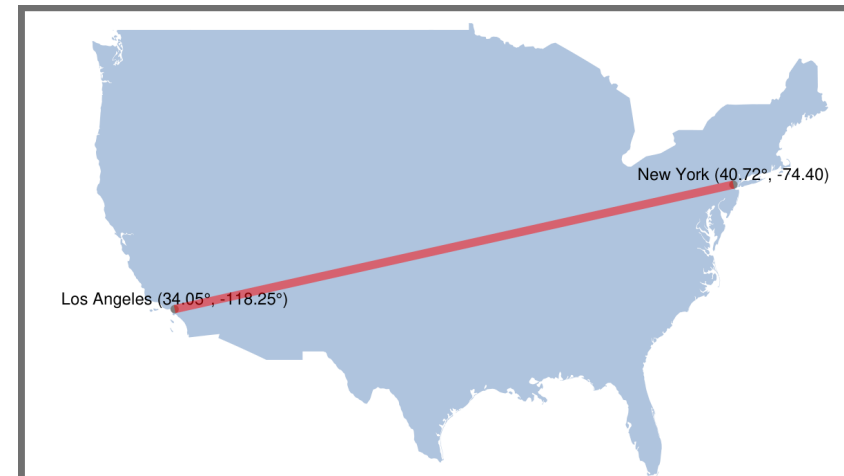


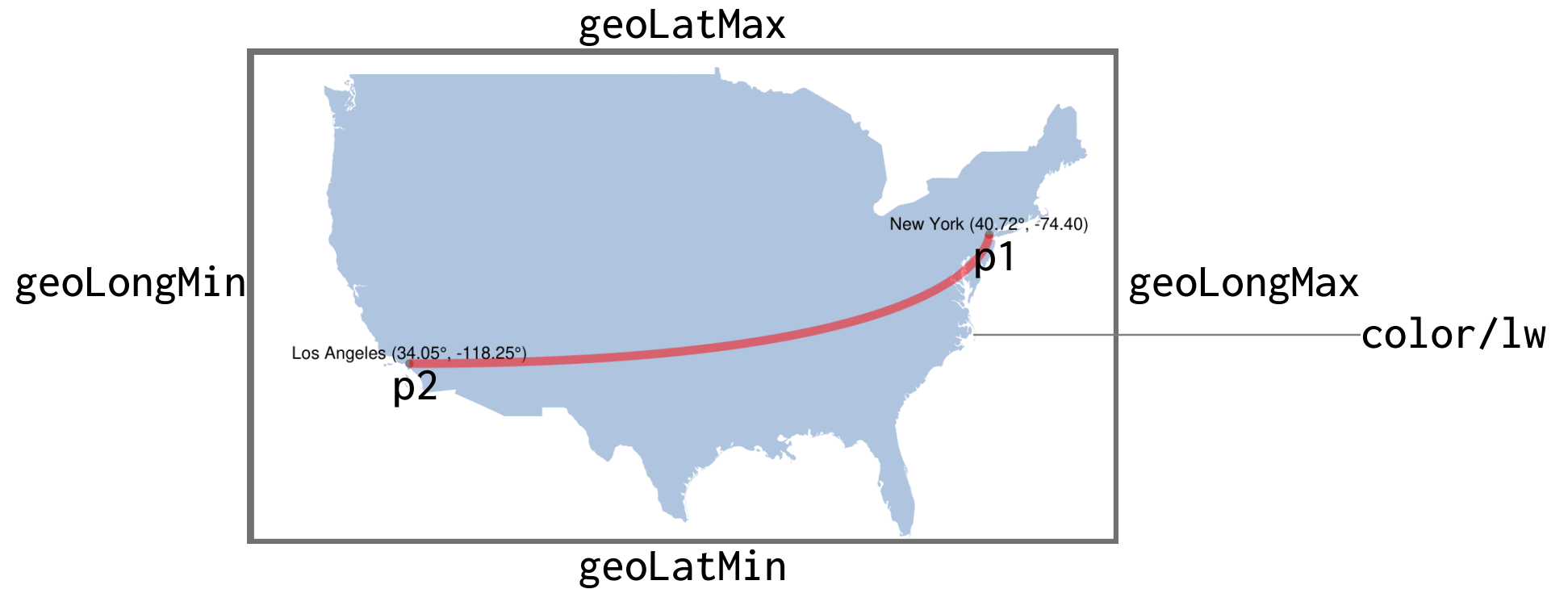
geopath "p1" "p2" lw lw color op

```

geoLatMin=25
geoLatMax=50
geoLongMin=0-130
geoLongMax=0-65
nyc="geo:+40.7167,-74.4000    New York (40.72°, -74.40)"
los="geo:+34.05000,-118.250    Los Angeles (34.05°, -118.25°)"

georegion "usa.kml" "lightsteelblue"
geoloc    nyc  "c" 2
geoloc    los  "c" 2
geopath   nyc los 1 "red" 50
  
```

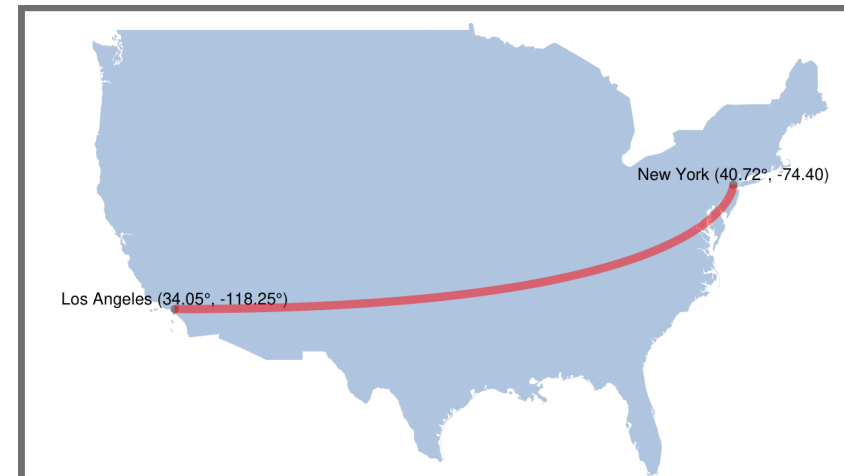


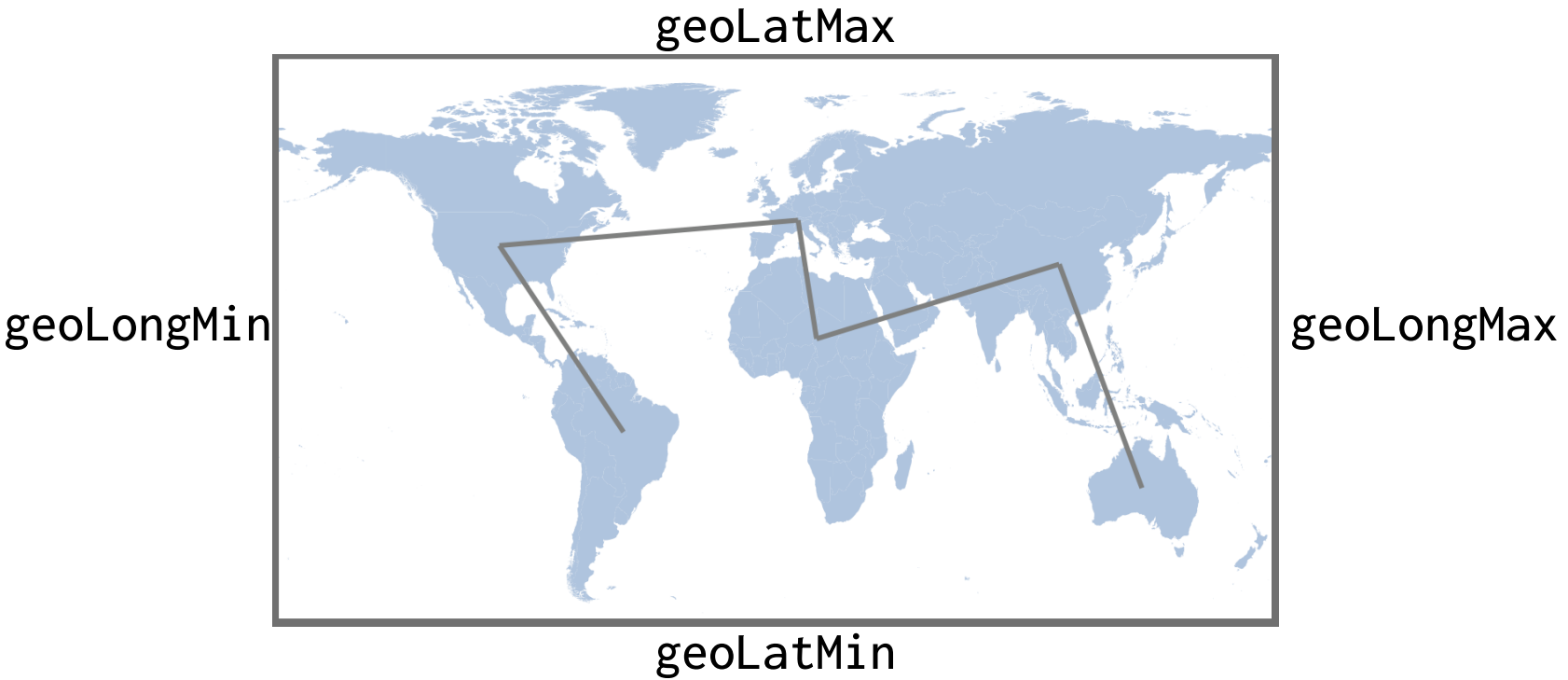


```
geoarc "p1" "p2" lw color op
```

```
geoLatMin=25
geolatmax=50
geoLongMin=0-130
geoLongMax=0-65
nyc="geo:+40.7167,-74.4000    New York (40.72°, -74.40°)"
los="geo:+34.05000,-118.250    Los Angeles (34.05°, -118.25°)"
```

```
georegion "usa.kml" "lightsteelblue"
geoloc    nyc "c" 2
geoloc    los "c" 2
geoarc     nyc los 1 "red" 50
```





geopathfile "file" lw color op gap

geoLatMin=25
geoLatMax=50
geoLongMin=0-130
geoLongMax=0-65

geo:40.712778,-74.006111	New York
geo:39.952778,-75.163611	Philadelphia
geo:39.768611,-86.158056	Indianapolis
geo:38.627222,-90.197778	St. Louis
geo:35.468611,-97.521389	Oklahoma City
geo:35.084444,-106.650278	Albuquerque
geo:36.167222,-115.148611	Las Vegas
geo:34.050000,-118.250000	Los Angeles

georegion "usa.kml" "lightsteelblue"
geoloc "roadtrip.d" "c" 2
geopathfile "roadtrip.d" 0.5 "red" 25



Keyword Reference

	Keyword	Arguments	Description
Structure	deck		Begin a deck; end with "edeck"
	def	name args...	Define a function; end with "edef"
	for	x=begin end [increment]	Begin loop; end with "efor"
	if	condition...[else]...eif	Conditional; one of: a==b, a!=b, a>b, a<b, a>=b, a<=b, a<>b c
	import	"file"	import function found in a file
	include	"file"	Include the contents of a file
	slide	[bgcolor] [fgcolor]	Begin a slide; end with "eslide"
	edeck		End the deck
	edef		End the definition
	efor		End the for loop
	eif		End the conditional
	else		Begin the else clause
Utility	canvas	width height	Define with dimensions of the canvas
	content	"scheme://file"	Embed content
	dump	[name]	Dump variables
	grid	"file" x y hspace vspace edge	Define a content grid
	ruler	[increment] [color]	draw a (x,y) ruler
Graphics	acircle	x y w [color] [opacity]	Circle with size based on area
	arc	x y w h a1 a2 [lw] [color] [opacity]	Ellipical arc centered at (x,y), dimensions (w,h) between angles a1 and a2
	circle	x y w [color] [opacity]	Circle centered at (x,y), diameter w
	curve	bx by cx cy ex ey [lw] [color] [opacity]	Quadratic Bezier Curve begin (bx,by), control (cx, cy), end (ex,ey)
	ellipse	x y w h [color] [opacity]	Ellipse centered at (x,y), dimension (w,h)
	hline	x y w [lw] [color] [opacity]	Horizontal line begin at (x,y), length w
	line	x1 y1 x2 y2 [lw] [color] [opacity]	Line between (x1,y1) and (x2,y2)
	dline	x1 y1 x2 y2 [lw] [gap] [color] [opacity]	Dotted line between (x1,y1) and (x2,y2), size lw
	pill	x y w h [color]	Pill shape beginning at (x,y), dimensions (w,h)
	polygon	"x1 x2 x3...." "y1 y2 y3..." [color] [opacity]	Polygon with specified x, y coordinates
	polyline	"x1 x2 x3...." "y1 y2 y3..." [lw] [color] [opacity]	Polyline with specified x, y coordinates
	rect	x y w h [color] [opacity]	Rectangle centered at (x,y), dimensions (w,h)
	rrect	x y w h r [color] [opacity]	Rounded rectangle centered at (x,y), dimensions (w,h), corner radius r
	square	x y w [color] [opacity]	Square centered at (x,y), size w
	star	x y sides inner outer [color] [opacity]	Star centered at (x,y), with sides, innner and outer sizes
	vline	x y h [lw] [color] [opacity]	Vertical line beginning at (x,y), h high

	Keyword	Arguments	Description
Text	arctext	"string" x y radius a1 a2 fontsize [font] [color] [opacity] [link]	Text on an arc, at fontsize, center (x,y), radius r, between a1. a2
	btext	"string" x y fontsize [font] [color] [opacity] [link]	Text beginning at (x,y), at fontsize
	ctext	"string" x y fontsize [font] [color] [opacity] [link]	Centered text beginning at (x,y), at fontsize
	etext	"string" x y fontsize [font] [color] [opacity] [link]	End-aligned text at (x,y), at fontsize
	rtext	"string" x y angle fontsize [font] [color] [opacity] [link]	Rotated text centered at x,y), at angle and fontsize
	text	"string" x y fontsize [font] [color] [opacity] [link]	Text beginning at (x,y), at fontsize
	para	x y w fontsize [font] [color] [opacity] [link]...epara	Paragraphs beginning at (x,y), at fontsize with width w
	textblock	"string" x y w fontsize [font] [color] [opacity] [link]	Block of text beginning at (x,y), at fontsize, with width w
	textblockfile	"file" x y w fontsize [font] [color] [opacity] [link]	Block of text read for a file, beginning at (x,y), at fontsize, with width w
	textcode	"file" x y w fontsize [font] [color] [opacity]	Lines of code, read from a file, upper right corner at (x,y), margin at w
	textfile	"file" x y fontsize [font] [color] [opacity] [spacing]	Contents of a text file upper right corner at (x,y)
Lists	blist	x y fontsize [font] [color] [opacity] [spacing]	Bulleted list starting at (x,y), at fontsize
	clist	x y fontsize [font] [color] [opacity] [spacing]	Centered list starting at (x,y), at fontsize
	list	x y fontsize [font] [color] [opacity] [spacing]	List starting at (x,y), at fontsize
	nlist	x y fontsize [font] [color] [opacity] [spacing]	Numbered list starting at (x,y), at fontsize
	li	"item" [font] [color] [opacity]	List item
	elist		End the list
Images	cimage	"file" "caption" x y w h [scale] [link] capsize	Captioned image; center (x,y), dimensions (w,h) (h=0, w is % of canvas width)
	image	"file" x y w h [scale] [link]	Image center at (x,y), dimensions (w,h) (h=0, w is % of canvas width)
Braces/ Brackets	dbrace	x y w bw bh [lw] [color] [opacity]	Downward pointing brace
	dbracket	x y w h [lw] [color] [opacity]	Downward pointing bracket
	lbrace	x y h bw bh [lw] [color] [opacity]	Left pointing brace
	lbracket	x y w h [lw] [color] [opacity]	Left pointing bracket
	rbrace	x y h bw bh [lw] [color] [opacity]	Right pointing brace
	rbracket	x y w h [lw] [color] [opacity]	Right pointing bracket
	ubrace	x y w bw bh [lw] [color] [opacity]	Upward facing brace
	ubracket	x y w h [lw] [color] [opacity]	Upward facing bracket

	Keyword	Arguments	Description
Arrows	arrow	x1 y1 x2 y2 [lw] [aw] [ah] [color] [opacity]	Arrow starting at (x1,y1), ending at (x2,y2), aw=width, ah=height
	dcarrow	bx by bx xy ex ey [lw] [aw] [ah] [color] [opacity]	Downward curved arrow; curve specified by (bx,by), (cx,cy), (ex,ey)
	lcarrow	bx by bx xy ex ey [lw] [aw] [ah] [color] [opacity]	Left curved arrow; curve specified by (bx,by), (cx,cy), (ex,ey)
	rcarrow	bx by bx xy ex ey [lw] [aw] [ah] [color] [opacity]	Right curved arrow; curve specified by (bx,by), (cx,cy), (ex,ey)
	ucarrow	bx by bx xy ex ey [lw] [aw] [ah] [color] [opacity]	Upward curved arrow; curve specified by (bx,by), (cx,cy), (ex,ey)
Built-ins	x=area	expression	Assign an area
	x=format	"fmt" expr... (up to 5)	Assign formatting to expressions
	x=polar	x y radius angle	Assign polar coordinate centered at (x,y) at radius and angle (0-360)
	x=polarx	x y radius angle	Assign X-polar coordinate centered at (x,y) at radius and angle (0-360)
	x=polary	x y radius angle	Assign Y-polar coordinate centered at (x,y) at radius and angle (0-360)
	x=random	min max	Assign a random number between two values
	x=substr	"string" begin end	Assign a substring
	x=vmap	data min1 max1 min2 max2	Assign a value mapped to two ranges
	x=geocoord	geo URL or lat long	Map geo to canvas coordinates
Math	x=cosine	expression	Assign the cosine of expression
	x=sine	expression	Assign the sine of expression
	x=sqrt	expression	Assign the square root of expression
	x=tangent	expression	Assign the tangent of expression
Geographic	geoarc	"p1" "p2" [lw] [color] [opacity]	Draw arcs between points
	geoborder	"file" [lw] [color] [opacity]	Reads KML data from the specified file and renders the map borders
	geoimage	"loc" width height	Place an image at a geographical location
	geolabel	"loc" [size] [font] [color] [opacity]	Reads data from the specified file or location and renders the map labels
	geoloc	"loc" [align] [size] [font] [color] [opacity]	Reads data from the specified file or location and a make map point and labels
	geomark	"loc" [size] [color] [opacity]	Reads data from the specified file or location and renders map points
	geopath	"p1" "p2" [lw] [color] [opacity]	Draw line between points
	geopathfile	"file" [lw] [color] [opacity]	Reads data from the specified file and a make lines between points
	georegion	"file" [color] [opacity] [gap]	Reads KML data from the specified file and renders the map regions
	geobound	latmin [latmax] [longmin] [longmax]	Sets overall geographic bounds
	geocanvas	xmin [xmax] [ymin] [ymax]	Sets geographic canvas
	georuler	[increment] [color]	Draw a geographic ruler

	Keyword	Arguments	Description
Charts	dchart	options... file	Chart with specified options
	legend	"string" x y fontsize font color	Chart legend
	chartbbox	left [right] [top] [bottom]	Chart bounding box
	areachart	"file" [color] [lcolor] [vcolor]	Area chart with specified data
	barchart	"file" [color] [lcolor] [vcolor]	Bar chart with specified data
	dotchart	"file" [color] [lcolor] [vcolor]	Dot chart with specified data
	hbarchart	"file" [color] [lcolor] [vcolor]	Horizontal bar chart with specified data
	linechart	"file" [color] [lcolor] [vcolor]	Line chart with specified data
	wbarchart	"file" [color] [lcolor] [vcolor]	Word bar chart with specified data
	slopechart	"file" [color] [lcolor] [vcolor]	Slope chart with specified data
	pmap	"file" [size]	Proportional map chart with specified data
	pie	"file" [size]	Pie chart with specified data
	donut	"file" [width] [size]	Donut chart with specified data
	pgrid	"file" [size]	Proportional grid chart with specified data
	waffle	"file" [size]	Waffle chart with specified data
	fanchart	"file" [size]	Fan chart with specified data
	bowtie	"file" [size]	Bowtie chart with specified data